NEET 2018

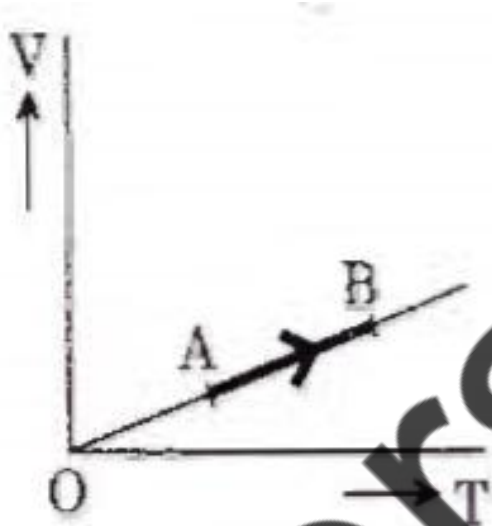
OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTION

All questions with detailed solutions of Physics, Chemistry and Biology

CAREERS 360

Physics

Q. 1 The volume (V) of a monatomic gas varies with its temperature (T), as shown in the graph. The ratio of work done by the gas, to the heat absorbed by it, when it undergoes a change from state A to state B, is



Option 1:
$\frac{1}{3}$

Option 2:
$\frac{2}{3}$

Option 3:
$\frac{2}{5}$

Option 4:
$\frac{2}{7}$

careers360

Correct Answer:

$\frac{2}{5}$

Solution:

As we have learned

First law in isobaric process -

$$\Delta U = n\,C_v \Delta T$$

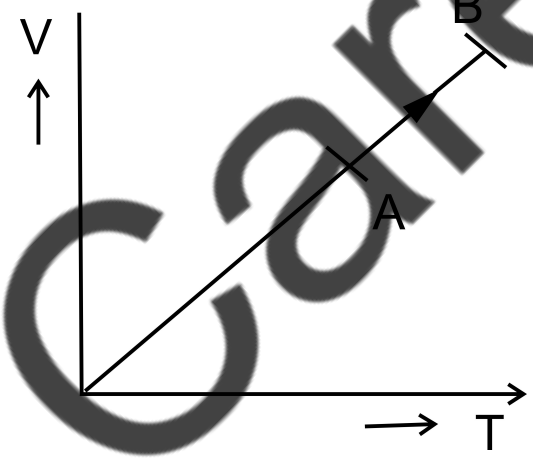
$$= n\frac{R}{\gamma - 1}\Delta T$$

- wherein

$$\Delta Q = \Delta U + W$$

$$= n\frac{\gamma\, R}{\gamma - 1} \cdot \Delta T$$

$$\Delta Q = nC_p\,\Delta T$$



$$V \propto T$$

$$V = KT$$

$$PV = nRT$$

Careers360

or $p = \frac{nRT}{KT} = \frac{nR}{K}$

work done =

$$= P(\Delta V) = \frac{nR}{K}(K\Delta T) = nR\Delta T$$

heat absorbed = $nC_p\Delta T$

$$= n(\frac{5R}{2})\Delta T = \frac{5}{2}(nR\Delta T)$$

ratio of work to heat

$$\frac{nR\Delta T}{5/2nR\Delta T} = 2/5$$

Q. 2 The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is :

Option 1:
12.5 cm

Option 2:
8.0 cm

Option 3:
13.2 cm

Option 4:
16.0 cm

Correct Answer:
13.2 cm

Careers360

Solution:

The frequency in open organ pipe -$\nu = n \cdot \frac{V}{2l}$

$n = 1, 2, 3.....$

wherein

$V =$ velocity of sound wave

$l =$ length of pipe

$n =$ number of overtones

Let the length of the open pipe be l

=> v_1 of open pipe = v_3 of closed pipe

$$\frac{v}{2l} = 3 \times \frac{v}{4 \times 20cm} \Rightarrow l = 13.2cm$$

Hence, the answer is the option (3).

Q. 3 At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere?

(Given: Mass of oxygen molecule (in kg) =. 2.76 x 10^{-26} kg
Boltzmann's constant K_B = 1.38 x 10^{-23} J K^{-1})

Option 1:
5.016 X 10^4 K

Option 2:
8.360 X 10^4 K

Option 3:
2.508 X 10^4 K

Careers360

Option 4:
1.254 X 10^4 K

Correct Answer:
8.360 X 10^4 K

Solution:
Root mean square velocity -

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$= \sqrt{\frac{3P}{\rho}}$$

wherein

R = Universal gas constant

M = molar mass

P = pressure due to gas

ρ = density

$$v_{escape} = v_{rms} = \sqrt{\left(\frac{3K_BT}{M_{O_2}}\right)} = 11.2km/s$$

$$T = \frac{M_{O_2}}{3K_B} \times (11.2 \times 10^3)^2$$

$$= \frac{2.76 \times 10^{-26}}{3 \times 1.38 \times 10^{-23}} \times 125.44 \times 10^6 = 8.36 \times 10^4 K$$

Hence, the answer is the option (2).

Q. 4 The efficiency of an ideal heat engine working

between the freezing point and boiling point **of** water, is

Careers360

Option 1:
6.25%

Option 2:
20%

Option 3:
26.8%

Option 4:
12.5%

Correct Answer:
26.8%

Solution:
As we have learned

Efficiency of a carnot cycle -

$$\eta = \frac{W}{Q_1 - Q_2} = 1 - \frac{T_2}{T_1}$$

$T_1 \, and \, T_2$ are in kelvin

- wherein

$T_1 =$ Source temperature

$T_2 =$ Sink Temperature

$(T_1 > T_2)$

$$n = 1 - \frac{T_2}{T_1} \times 100\% = (1 - \frac{273}{373}) \times 100\%$$

$$= \frac{100}{373} * 100\% = 26.8\%$$

Careers360

Q. 5 A carbon resistor of $(47 \pm 4.7)\mathrm{k}\Omega$ is to be marked with rings of different colours for its identification. The colour code sequence will be:

Option 1:
Yellow — Green — Violet — Gold

Option 2:
Yellow — Violet — Orange — Silver

Option 3:
Violet — Yellow — Orange — Silver

Option 4:
Green — Orange — Violet — Gold

Correct Answer:
Yellow — Violet — Orange — Silver

Solution:
As we have learned

To remember the sequence of colour code -

B B ROY Great **B**ritain **V**ery **G**ood **W**ife

Careers360

- wherein

Letters as an aid to memory	Colour	Figure (A, B)	Multiplier (C)
B	Black	0	10^0
B	Brown	1	10^1
R	Red	2	10^2
O	Orange	3	10^3
Y	Yellow	4	10^4
G	Green	5	10^5
B	Blue	6	10^6
V	Violet	7	10^7
G	Grey	8	10^8
W	White	9	10^9

$$R = (47 \pm 4.7) \times 10^3$$

$$= 47 * 10^3 \pm 10\%\Omega$$

As per code : 3- orange

4-yellow

7-violet

10%- silver

Careers360

Q. 6 A set of 'n' equal resistors, of value 'R' each, are connected in series to a battery of emf 'E' and internal resistance 'R'. The current drawn is I. Now, the 'n' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes 10I. The value of 'n' is

Option 1:
20

Option 2:
11

Option 3:
10

Option 4:
9

Correct Answer:
10

Solution:
As we have learned

Current given by the cell -

$$i=\frac{E}{R+r}$$

- wherein

$R-$ External resistance

$r-$ internal resistance

Careers360

In series Grouping -

$$R_{eq} = R_1 + R_2 + R_3 + \cdots + R_n$$

- wherein

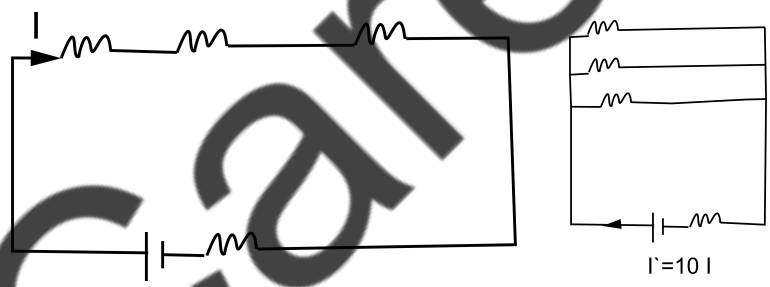
R_{eq} — Equivalent Resistance

In parallel Grouping -

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

-

Case I Case II



$$I = \frac{E}{R + nR} \quad \text{And} \quad 10I = \frac{E}{R/n + R}$$

$$10I = \frac{nE}{R + nR} = nI$$

$$n = 10$$

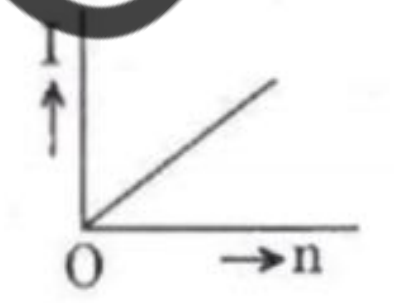
Careers360

Q. 7 A battery consists of a variable number 'n' of

identical cells (having internal resistance 'r' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n ?

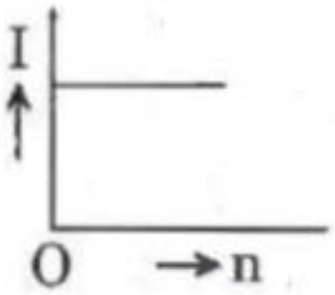
Option 1:

Option 2:

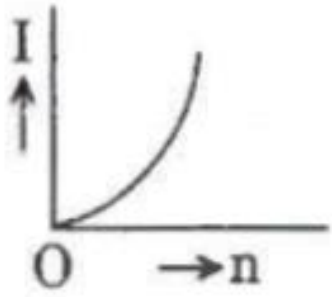


Careers360

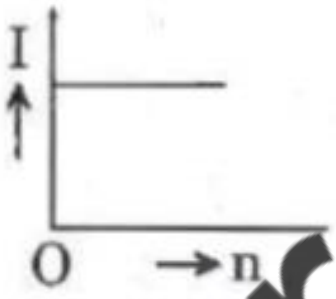
Option 3:



Option 4:



Correct Answer:



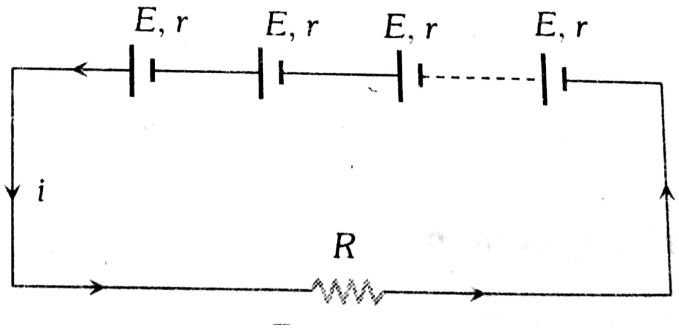
Solution:
As we have learnt

Series Grouping -

In series grouping anode of one cell is connected to cathode of other cell

- wherein

Careers360



Main current / current from each cell -

$$i = \frac{nE}{R + nr}$$

- wherein

n — identical cells which connected in series

$$I = \frac{nE}{nR} = \frac{E}{r} = constant$$

Q. 8 Unpolarised light is is incident from air on a plane surface of a material of refractive index μ. At a particular angle of incidence 'i', it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is correct for this situation ?

Option 1:

$$i = sin^{-1}\left(\frac{1}{\mu}\right)$$

Option 2:

Reflected light is polarised with its electric vector perpendicular to the plane of incidence

Option 3:

Reflected light is polarised with its electric vector parallel to the plane of incidence

Careers360

Option 4:

$$i = tan^{-1}\left(\frac{1}{\mu}\right)$$

Correct Answer:

Reflected light is polarised with its electric vector perpendicular to the plane of incidence

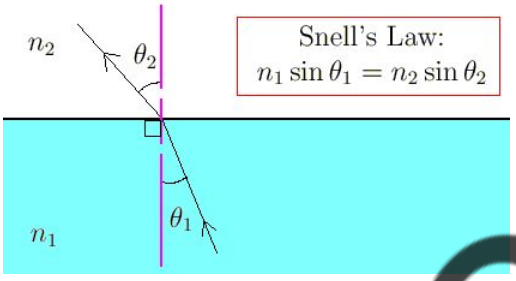
Solution:

As we have learned

Relation between angle of incidence and angle of refaction -

$$\mu_1 \sin i = \mu_2 \sin r$$

- wherein

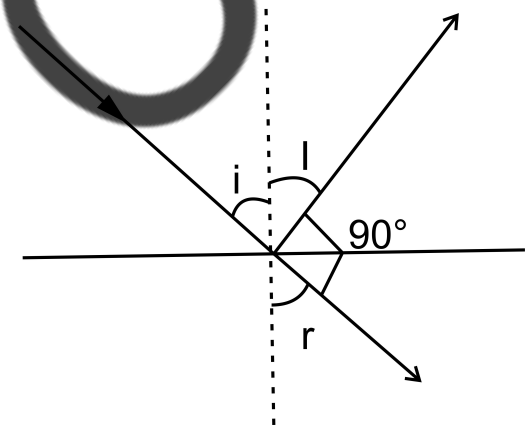


$\mu_1 =$ refractive index of medium of incidence.

$\mu_2 =$ refractive index of medium where rays is refracted.

$i =$ angle of incidence.

$r =$ angle of refraction.



Careers360

Apply snells law

$\sin i / \sin r = \mu \rightarrow (1)$

also $i + r = 90^\circ$

$\Rightarrow r = 90 - i$

$\Rightarrow \frac{\sin i}{\sin(90 - i)} = \mu$

or $\tan i = \mu$

$i = \tan^{-1} \mu$

also reflected light is polarized with its electric field perpendicular to plane of incidence

Careers360

Q. 9 In Young's double-slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is 5896 $\AA^0$ and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20°. To increase the fringe angular width to 0.21° (with the same λ and D) the separation between the slits needs to be changed to

Option 1:
2.1 mm

Option 2:
1.9 mm

Option 3:
1.8 mm

Option 4:
1.7 mm

Correct Answer:
1.9 mm

Solution:
Since in YDSE, angular width is given by

Angular Width A.W.=$\frac{\beta}{D} = \frac{\lambda}{d}$

For a given wavelength,

A.W.1d1=A.W.2d2

0.20×2=0.21×d2

d2=1.9mm

Hence, the answer is the option (2).

Q. 10 An astronomical refracting telescope will have large angular magnification and high angular resolution when it has an objective lens of

Option 1:
large focal length and large diameter

Option 2:
large focal length and small diameter

Option 3:
small focal length and large diameter

Careers360

Option 4:
small focal length and small diameter

Correct Answer:
large focal length and large diameter

Solution:
As we have learned

Astronomical Telescope -$m = \frac{-f_o}{f_e}\left(1+\frac{f_e}{D}\right)$

wherein

f_o = focal length of objective

f_e = focal length of the eyepiece

for large angular magnification and high angular resolution, the objective lens of the refracting telescope should have a large diameter and large focal length

$M = \frac{f_0}{f_e}$ and resolving power $= \frac{d}{1.22\lambda}$

Hence, the answer is the option (1).

Q. 11 The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom is

Option 1:
2 : -1

Option 2:
-1 : 1

Option 3:
1:1

Careers360

Option 4:
1 : -2

Correct Answer:
-1 : 1

Solution:
As we have learned

Relation between K.E., P.E. & total energy

$$K.E. = (Total\ energy)$$
$$= \frac{1}{2}(Potential\ energy)$$

wherein

Potential Energy and total energy are negative. While kinetic energy is positive

The kinetic energy of an electron in a Bohr orbit of the hydrogen atom, K=−E
E= total energy
Hence required ratio is 1:−1

Hence, the answer is the option (2).

Q. 12 An electron of mass m with an initial velocity $\vec{V} = V_o\hat{i}\,(V_o > 0)$

enters an electric field $\vec{E} = -E_o\hat{i}\,(E_o = constant > 0)$

at t = 0. If λ_o is its de-Broglie wavelength initially , then its de-Broglie wavelength at time t is :

Option 1:
$\lambda_o t$

Option 2:
$\lambda_o\left(1 + \frac{eE_o}{mV_o}t\right)$

Careers360

Option 3:

$$\frac{\lambda_o}{\left(1+\frac{eE_o}{mV_o}t\right)}$$

Option 4:

λ_o

Correct Answer:

$$\frac{\lambda_o}{\left(1+\frac{eE_o}{mV_o}t\right)}$$

Solution:

De - Broglie wavelength -

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

wherein

$h = plank's\ constant$

$m = mass\ of\ particle$

$v = speed\ of\ the\ particle$

$E = Kinetic\ energy\ of\ particle$

$\vec{V} = v_0\hat{i}$

$\vec{E} = E_0\hat{i}$

$\vec{F} = -e\vec{E} = eE_0\hat{i}$

$$v = v_0 + \frac{eE_0}{m}t$$

$$p(t) = m(v_0 + \frac{eE_0t}{m})$$

Careers360

$$\lambda = \frac{h}{p(t)} = \frac{h}{mv_0(1+\frac{eE_0t}{mv_0})} = \frac{\lambda_0}{1+\frac{eE_0t}{mv_0}}$$

Hence, the answer is the option (3).

Q. 13 . For a radioactive material, half-life is 10 minutes. If initially there are 600 number of nuclei, the time taken (in minutes) for the disintegration of 450 nuclei is

Option 1:
30

Option 2:
10

Option 3:
20

Option 4:
15

Correct Answer:
20

Solution:
As we have learned

Number of nuclei in terms of half life -

$$N = \frac{N_0}{2^{t/t_{1/2}}}$$

- wherein

Very useful to determine number of nuclei in terms of half life

Careers360

$t_{1/2} = 10 minutes$

$N_0 = 600$

$$N = 150 = \frac{N_0}{2^{\frac{t}{t_{1/2}}}} = \frac{600}{2^{\frac{t}{t_{1/2}}}}$$

$$\Rightarrow \frac{t}{t_{1/2}} = 2$$

or $t = 2t_{1/2} = 20 minutes$

Q. 14 When the light of frequency $2v_0$ (where v_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5v_0$, the maximum velocity of electrons emitted from the same plate is v_2 . The ratio of v_1 to v_2 is

Option 1:
4:1

Option 2:
1:4

Option 3:
1:2

Careers360

Option 4:
2:1

Correct Answer:
1:2

Solution:
As we have learned

Conservation of energy -

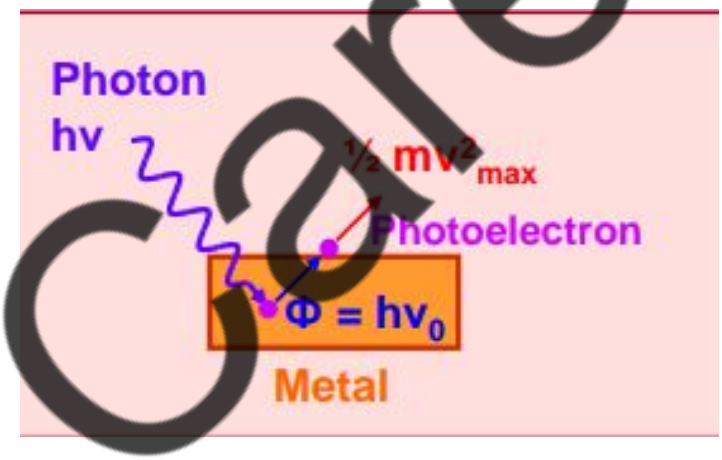
$$h\nu = \phi_0 + \frac{1}{2}mv_{max}^2$$

$$h\nu = h\nu_0 + \frac{1}{2}mv_{max}^2$$

$$h\left(\nu - \nu_0\right) = \frac{1}{2}mv_{max}^2$$

$where, h - Plank's constant\ \nu - Frequency\ \nu_0 - threshold\ frequency\ \phi_0 - work function$

- wherein



From einstein equation

$$1/2mv^2 = hv - \phi = h(v - v_0)$$

Careers360

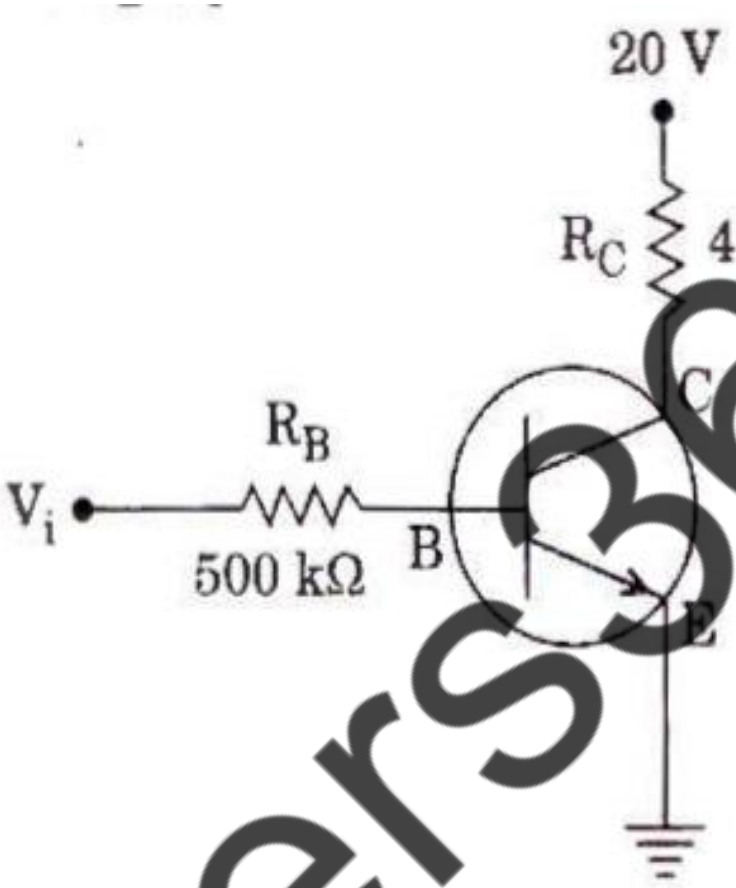
Initially $1/2mv_1^2 = h(2v_0 - v_0) = hv_0$

finally $1/2mv_2^2 = h(5v_0 - v_0) = 4hv_0$

$\Rightarrow \frac{v_2^2}{v_1^2} = 4$ or $\Rightarrow \frac{v_1}{v_2} = 1/2$

Careers360

Q. 15 In the circuit shown in the figure, the input voltage V_i is 20 V, V_{BE} = 0 and V_{CE} = 0. The values of I_B, I_C and β are given by



Option 1:
$I_B = 20\mu A, I_C = 5mA, \beta = 250$

Option 2:
$I_B = 25\mu A, I_C = 5mA, \beta = 200$

Option 3:
$I_B = 40\mu A, I_C = 10mA, \beta = 250$

Option 4:
$I_B = 40\mu A, I_C = 5mA, \beta = 125$

Correct Answer:
$I_B = 40\mu A, I_C = 5mA, \beta = 125$

Solution:

Careers360

As we have learned

Relation between α and β -

$$\beta = \frac{\alpha}{1-\alpha}$$

- wherein

$$\alpha = \frac{I_C}{I_E}$$

$\beta = \frac{I_C}{I_B}$ (current gain)

$$I_B = \frac{V_i - V_B}{R_B} = \frac{20-0}{500k\Omega} = 40\mu A$$

$$I_C = \frac{20-0}{4k\Omega} = 5mA$$

$$\beta = \frac{I_C}{I_B} = 125$$

Careers360

Q. 16 In a p-n junction diode, change in temperature due to heating -

Option 1:
does not affect resistance of p-n junction

Option 2:
affects only forward resistance

Option 3:
affects only reverse resistance

Option 4:
affects the overall V - I characteristics of p-n junction

Correct Answer:
affects the overall V - I characteristics of p-n junction

Solution:
Number of electrons or holes

$$n_e = n_\lambda = AT^{3/2}e^{-Eg/2KT}$$

wherein

E_g = Energy gap

K = Boltzmann Constant

T = Temperature in kelvin

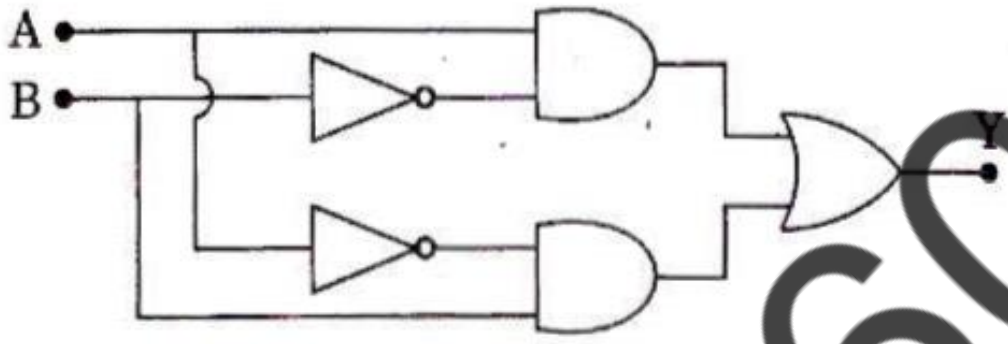
On increasing temperature, the number of currents carried increases

Temperature change changes the concentration of the e-p pair and hence affects the overall V- characteristics of the PN junction.

Hence, the answer is the option (4).

Careers360

Q. 17 In the combination of the following gates the output Y can be written in terms of inputs A and B as



Option 1:

$\overline{A.B} + A.B$

Option 2:

$\bar{A}.B + A.\bar{B}$

Option 3:

$\overline{A.B}$

Option 4:

$\overline{A + B}$

Correct Answer:

$\bar{A}.B + A.\bar{B}$

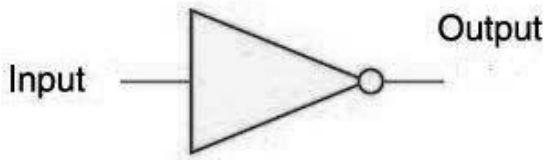
Solution:

As we have learned

NOT gate -

$Y = \bar{A}$

Careers360



- wherein

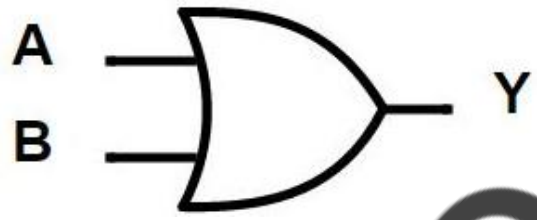
A is input

Y is out put

OR gate -

Relation between input and output

$$Y = A + B$$



- wherein

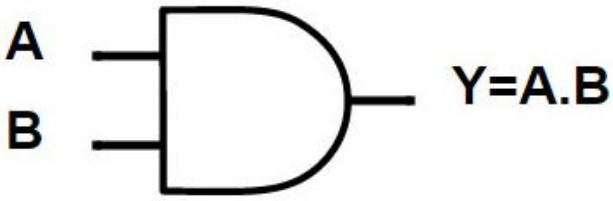
A and B are input

Y is out put

AND Gate -

$$Y = A \cdot B$$

Careers360



- wherein

A and B are input

Y is out put

$$Y = A\bar{B} + \bar{A}B$$

Q. 18 An em wave is propagating in a medium with a velocity $\vec{V} = V\hat{i}$. The instantaneous oscillating electric field of this em wave is along the +y axis. Then the direction of the oscillating magnetic field of the em wave will be along

Option 1:
— y direction

Option 2:
+ z direction

Option 3:
— z direction

Option 4:
— x direction

Careers360

Correct Answer:
+ z direction

Solution:

$\vec{S} = V\hat{i}$

$\vec{E} = E\hat{j}$

$\hat{i} = \hat{j} \times \hat{k}$

$\vec{B} = B\hat{k}, +z - direction$

Hence, the answer is the option (2).

Q. 19 The refractive index of the material of a prism is $\sqrt{2}$ and the angle of the prism is 30°. One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is

Option 1:
30°

Option 2:
45°

Option 3:
60°

Option 4:
zero

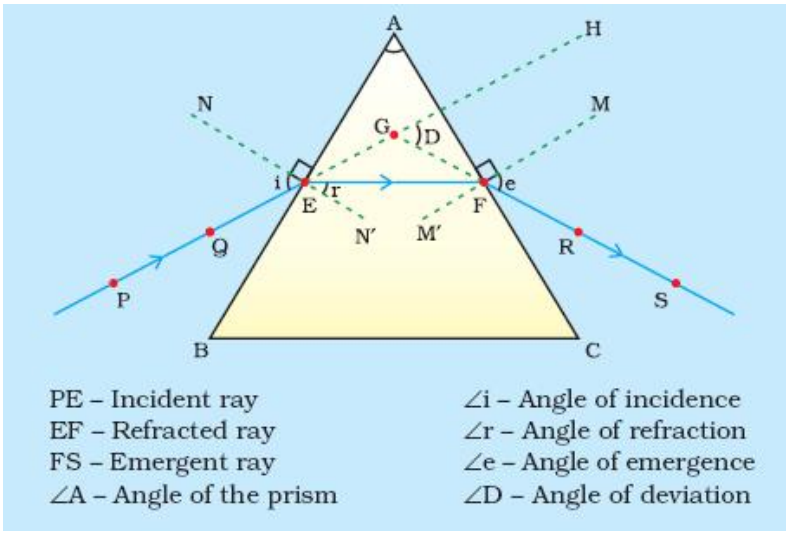
Correct Answer:
45°

Solution:
As we have learned

Careers360

Angle of deviation through prism -

$$\delta = i + e - A$$

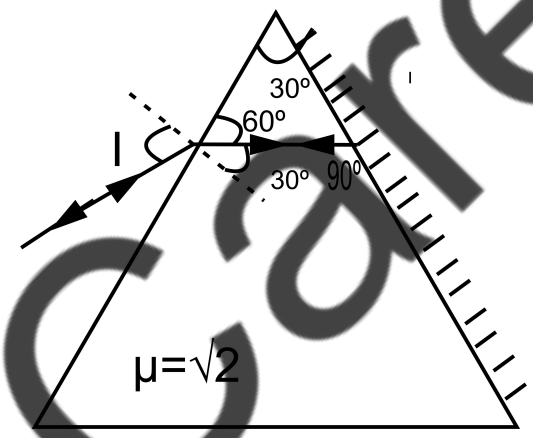


- wherein

$i =$ angle of incidence

$e =$ angle of emergence

$A =$ angle of prism



$r = 30^\circ$

$\sin i = \mu \sin 30^\circ$

$\sqrt{2} \cdot 1/2 = 1/\sqrt{2} \Rightarrow i = 45^\circ$

Careers360

Q. 20 An object is placed at a distance of 40 cm from a concave mirror of a focal length of 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be

Option 1:
30 cm towards the mirror

Option 2:
36 cm away from the mirror

Option 3:
30 cm away from the mirror

Option 4:
36 cm towards the mirror

Correct Answer:
36 cm away from the mirror

Solution:
As we have learned

Mirror Formula

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

wherein

$u =$ Object distance from pole of mirror.

$v =$ Image distance from pole of mirror.

Careers360

$f =$ focal length of the mirror.

$u_1 = -40cm$

$f = -15cm$

$$\frac{1}{v_1} = \frac{1}{f_1} - \frac{1}{u_1} = \frac{-1}{15} + \frac{1}{40} = \frac{-8+3}{120} = \frac{-5}{120}$$

$v_1 = -24cm$

2nd case

$v_2 = -20cm$

$$\frac{1}{v_2} = \frac{1}{f} - \frac{1}{u_2} = \frac{-1}{15} + \frac{1}{20} = \frac{-4+3}{60}$$

$v_2 = -60cm$

The image gets displaced by 36 cm, away from the mirror

Hence, the answer is the option (2).

Q. 21 The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance

Option 1:
1.389 H

Option 2:
138.88 H

Option 3:
0.138 H

Careers360

Option 4:
13.89 H

Correct Answer:
13.89 H

Solution:
As we have learned

Energy stored in the magnetic field of the inductor

$$U = \frac{1}{2}LI^2$$

$$v = 1/2Li^2$$

$$L = \frac{2v}{i^2} = \frac{2 \times 25 \times 10^{-3}}{36 \times 10^{-4}} = \frac{500}{36}H$$

$$L = 13.89H$$

Hence, the answer is the option (4).

Q. 22 .An electron falls from rest through a vertical distance h in a uniform and vertically upward directed electric field E. The direction of electric field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in it through the same vertical distance h. The time of fall of the electron, in comparison to the time of fall of the proton is

Option 1:
10 times greater

Option 2:
5 times greater

Option 3:
smaller

Careers360

Option 4:
equal

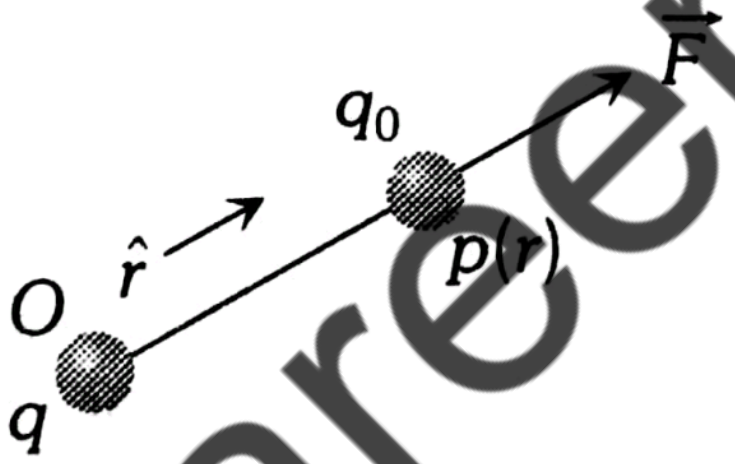
Correct Answer:
smaller

Solution:
As we have learned

Electric Field Intensity -

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{kQ}{r^2}$$

- wherein



FOR ELECTRON

$$a_e = \frac{eE}{m_e}$$

$$t_e = \sqrt{(\frac{2h}{a})} = \sqrt{\frac{2hm_e}{eE}}$$

FOR PROTON :

Careers360

$$a_P = \frac{eE}{m_p} \Rightarrow t_p = \sqrt{\frac{2hm_p}{eE}}$$

$$t_e/t_p = \sqrt{\frac{m_e}{m_p}} < 1$$

$\therefore$ electron will take less time

Q. 23 The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

Option 1:
proportional to the square root of the distance between the plates.

Option 2:
linearly proportional to the distance between the plates.

Option 3:
independent of the distance between the plates.

Option 4:
inversely proportional to the distance between the plates.

Correct Answer:
independent of the distance between the plates.

Solution:
As we have learned

The force between Parallel Plates' Capacitor

Careers360

$$F = \frac{\sigma^2 A}{2\epsilon_0} = \frac{Q^2}{2\epsilon_0 A} = \frac{CV^2}{2d}$$

wherein

$\sigma - Surface\ charge\ density.$

$$F = \frac{\sigma^2 A}{2\epsilon_0}$$

F is independent of the distance between the plates

Hence, the answer is the option (3).

Q. 24 A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is

Option 1:
350 m/s

Option 2:
339 m/s

Option 3:
330 m/s

Option 4:
300 m/s

Correct Answer:
339 m/s

Solution:
As we have learned

Careers360

Fundamental frequency with end correction -

$\nu_0 = \frac{V}{4\,(l+e)}$ (one end open)

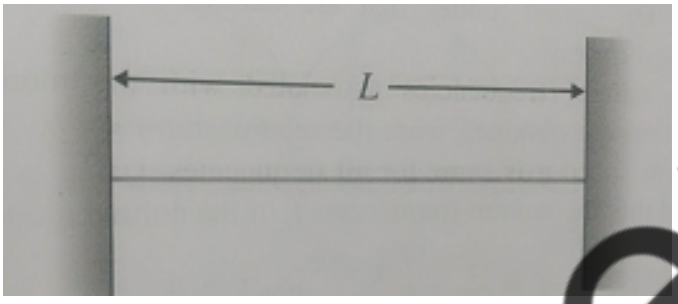
$\nu_0 = \frac{V}{2\,(l+2e)}$ (Both end open)

e = end correction

Velocity of transverse wave using sonometer -

$v = n_0\lambda$

$= 2\,n_0\,(l_2 - l_1)$



- wherein

where $\frac{\lambda}{2} = l_2 - l_1$

$l_2\ and\ l_1$ are the two successive resonance length.

$V = 2f(l_2 - l_1)$

$V = 2 * 320 * (0.73 - 0.20)$

$V = 339.2m/s$

Careers360

Q. 25 A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is

Option 1:
2 s

Option 2:
π s

Option 3:
2π s

Option 4:
1 s

Correct Answer:
π s

Solution:
As we had learnt in

Equation of S.H.M.

$$a = -\frac{d^2x}{dt^2} = -w^2x$$

$$w = \sqrt{\frac{k}{m}}$$

wherein

$$x = A\sin(wt + \delta)$$

$$a = w^2x \Rightarrow 20w^2 \times 5 \Rightarrow w = 2rad/sec$$

Careers360

$$T = \frac{2\pi}{w} sec = \pi \ seconds$$

Hence, the answer is the option (2).

Q. 26 A metallic rod of mass per unit length 0.5 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 30^o with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. The current flowing in the rod to keep it stationary is :

Option 1:
14.76 A

Option 2:
5.98 A

Option 3:
7.14 A

Option 4:
11.32 A

Correct Answer:
11.32 A

Solution:
As we have learned

Force on a current-carrying conductor

$$\underset{dF}{\rightarrow} = i(\underset{dl}{\rightarrow} \times \underset{B}{\rightarrow})$$

wherein

idl=current element

Careers360

To keep the rod stationary

$$F = i(\vec{l} \times \vec{B}) = mg \sin \Theta$$

or

$$F = \vec{l} \times \vec{B} = (g \sin \Theta)\frac{m}{i} \rightarrow (1)$$

the angle between the rod and the magnetic field is 90-Θ

i.e $ilB \cos \Theta = mg \sin \theta$

$\Rightarrow ilB = mg \tan \theta$ or $i = \left(\frac{m}{l}\right)\frac{g \tan \theta}{B}$

$$i = \left(\frac{0.5 \times 10 \times \frac{1}{\sqrt{3}}}{0.25}\right) = \frac{20}{\sqrt{3}} = 11.3 amp$$

Hence, the answer is the option (4).

Q. 27 A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from

Option 1:
the lattice structure of the material of the rod

Option 2:
the magnetic field

Option 3:
the current source

Option 4:
the induced electric field due to the changing magnetic field

Careers360

Correct Answer:
the current source

Solution:
As we have learned

Diamagnetic Substance -

B<B_0, B/B_0<1 or $\frac{\mu}{\mu_0} <, \mu_r < 1$

Relative permeability of diamagnetic substance.

-

when current source is switched on, magnetic field sets up between poles on electromagnet

Diamagnetic material due to its tendency to move from stronger to weaker field , is thus repelled out

Careers360

Q. 28 An inductor 20 mH, a capacitor 100 μF and a resistor 50 Ω are connected in series across a source of emf V = 10 sin 314 t. The power loss in the circuit is

Option 1:
2.74 W

Option 2:
0.43 W

Option 3:
0.79 W

Option 4:
1.13 W

Correct Answer:
0.79 W

Solution:
As we have learned

Average power

$$P_{av} = i_{rms}^{'2} R = \frac{V_{rms}^2 R}{Z^2}$$

power loss= $v_{rms} I_{rms} \cos\phi$

$$\left(\frac{v_{rms}}{Z}\right)(V_{rms})\frac{R}{Z}$$

$$\left(\frac{v_{rms}}{Z}\right)^2 R$$

$$X_l = WL = 314 * 20 * 10^{-3}\Omega = 6.28\Omega$$

$$X_c = 1/WC = (1/314 * 10^{-4})\Omega = 31.84\Omega$$

$$Z = \sqrt{R^2 + (X_c - X_l)^2} = 56\Omega$$

$$p = \frac{\frac{10^2}{\sqrt{2}^2}}{56^2} \times 50 = 0.79w$$

Hence, the answer is the option (3).

Careers360

Q. 29 The current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is

Option 1:
250 Ω

Option 2:
25 Ω

Option 3:
40 Ω

Option 4:
500 Ω

Correct Answer:
250 Ω

Solution:
As we have learned

Current sensitivity

$$S_i = \frac{\alpha}{i} = \frac{NBA}{C}$$

Voltage sensitivity -

$$S_v = \frac{\alpha}{iR} = \frac{S_i}{R} = \frac{NBA}{RC}$$

wherein

R-resistance

Ohm's Law -Current flowing through the conductor is directly proportional to the Potential difference across two ends.

Careers360

wherein

unit - $\frac{m^2}{volt \cdot sec}$\

$V \alpha I$

$V = IR$

R– Electric Resistance

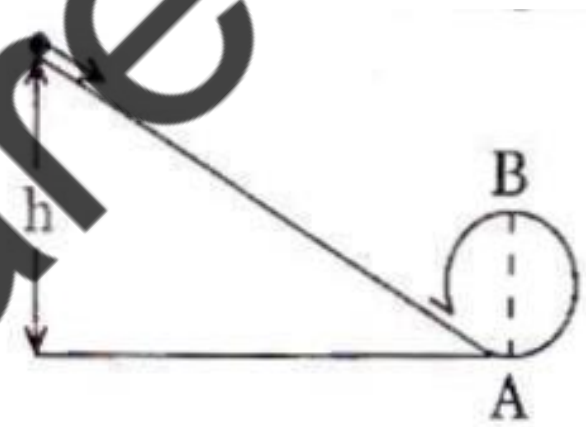
Current sensitivity $\theta/i = 5div/mA = 5000div/A$

voltage sensitivity $\theta/v = 20div/V$

$\frac{\frac{\theta}{i}}{\frac{\theta}{v}} = v/i = \frac{5000}{20} = 250\Omega$

Hence, the answer is the option (1).

Q. 30 A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter AB = D. The height h is equal to



Option 1:

$\frac{7}{5}D$

Careers360

Option 2:
D

Option 3:
$\frac{3}{2}D$

Option 4:
$\frac{5}{4}D$

Correct Answer:
$\frac{5}{4}D$

Solution:
As we have learned

If only conservative forces act on a system, total mechnical energy remains constant -

$$K+U=E\,(constant)$$

$$\Delta K+\Delta U=0$$

$$\Delta K=-\Delta U$$

-

For particles to complete circle

$$v=\sqrt{5gR}$$

also $v=\sqrt{2gh}\Rightarrow\sqrt{2gh}=\sqrt{5g\frac{D}{2}}$

$$h=\frac{5D}{4}$$

Careers360

Q. 31 Three objects, A : (a solid sphere), B : (a thin circular disk) and C : (a circular ring), each have the same mass M and radius R. They all spin with the same angular speed ω about their own symmetry axes. The amount of work (W) required to bring them to rest, would satisfy the relation

Option 1:
$W_B > W_A > W_C$

Option 2:
$W_A > W_B > W_C$

Option 3:
$W_C > W_B > W_A$

Option 4:
$W_A > W_C > W_B$

Correct Answer:
$W_C > W_B > W_A$

Solution:
As we have learned

The kinetic energy of rotation

$$K = \frac{1}{2} I w^2$$

wherein

I = moment of inertia about the axis of rotation

w = angular velocity

Work energy theorem states that work done =change in kinetic energy $= \frac{1}{2} I W^2$

$$I_c > I_B > I_A$$

Careers360

$\therefore K \cdot E_c > K \cdot E_B > K \cdot E_A$

$W_c > W_B > W_A$

Hence, the answer is the option (3).

Q. 32 A moving block having mass m, collides with another stationary block having a mass 4m. The lighter block comes to rest after collision. When the initial velocity of the lighter block is v, then the value of the coefficient of restitution (e) will be :

Option 1:
0.8

Option 2:
0.25

Option 3:
0.5

Option 4:
0.4

Correct Answer:
0.25

Solution:
Coefficient of Restitution (e)

$$e = \frac{v_2 - v_1}{u_2 - u_1}$$

wherein

The ratio of relative velocity after a collision to relative velocity before the collision.

Let the velocity of 4m mass be v after collision

Careers360

then $mv = (4m)\, v$

$$\Rightarrow V = \frac{V}{4}$$

$$e = \frac{velocity of separation}{velocity of approach} = \frac{\frac{v}{4}}{v} = \frac{1}{4}$$

e=0.25

Hence, the answer is the option (2).

Q. 33 Which one of the following statements is incorrect?

Option 1:
Frictional force opposes the relative motion.

Option 2:
Limiting value of static friction is directly proportional to normal reaction.

Option 3:
Rolling friction is smaller than sliding friction.

Option 4:
Coefficient of sliding friction has dimensions of length.

Correct Answer:
Coefficient of sliding friction has dimensions of length.

Solution:
Kinetic or Dynamic Friction -

$$f_K \ \alpha \ R$$

Careers360

$f_K = \mu_K R$

$f_K =$ kinetic friction

$\mu_K =$ coefficient of kinetic friction

R = reaction

- wherein

$f_K < F_l$

$\therefore \ \mu_K < \mu_s$

$\mu_K =$ depends on the nature of the surface in contact.

The coefficient of sliding friction is dimensionless.

Hence the answer is Option (4).

Q. 34 A toy car with charge q moves on a frictionless horizontal plane surface under the influence of a uniform electric field $\vec{E}$. Due to the force q$\vec{E}$, its velocity increases from 0 to 6 m/s in one second duration. At that instant the direction of the field is reversed. The car continues to move for two more seconds under the influence of this field. The average velocity and the average speed of the toy car between 0 to 3 seconds are
respectively

Option 1:
1 m/s, 3.5 m/s

Option 2:
1 m/s , 3 m/s

Careers360

Option 3:
2 m/s , 4 m/s

Option 4:
1.5 m/s , 3 m/s

Correct Answer:
1 m/s , 3 m/s

Solution:
As we have learned

2nd equation or Position- time equation -

$$s = ut + \frac{1}{2}at^2$$

$s \rightarrow$ Displacement

$u \rightarrow$Initial velocity

$a \rightarrow$acceleration

$t \rightarrow$ time

-

$0 < t < 1s,$ velocity increases from 0 to 6 m/s

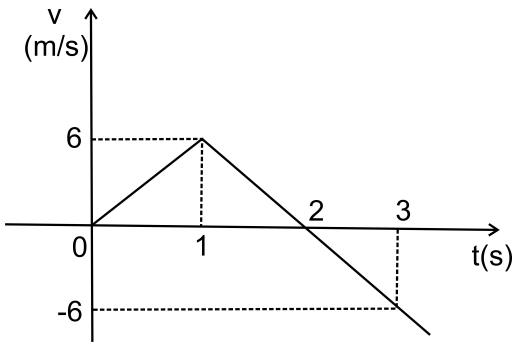
$1 < t < 2s,$ velocity decreases from 6 to 0 m/s

but car continues to move forword

$2 < t < 3s,$ since acceleration is same

$\Rightarrow$ car's velocity increases from 0 to -6 m/s

Careers360



Distance travelled in first second

$$= \frac{1}{2}at^2 = \frac{1}{2} \times \left(\frac{v-u}{t}\right) t^2$$

$$= \frac{1}{2} \times 6 \times 1 = 3m$$

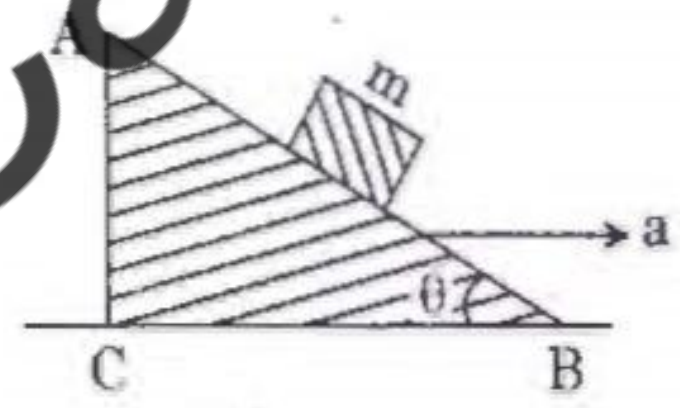
Total displacement = 3m

$\therefore$ Average velocity = 3/3 =1 m/s

Total Distance =(3+3+3)m =9m

$\therefore$ Average speed = $\therefore \frac{9m}{3s} = 3m/s$

Q. 35 A block of mass m is placed on a smooth inclined wedge ABC of inclination $\ominus$ as shown in the figure. The wedge is given an acceleration 'a' towards the right. The relation between a and $\ominus$ for the block to remain stationary on the wedge is



Careers360

Option 1:
$a = gcos\Theta$

Option 2:
$a = \frac{g}{sin\Theta}$

Option 3:
$a = \frac{g}{cosec\Theta}$

Option 4:
$a = gtan\Theta$

Correct Answer:
$a = gtan\Theta$

Solution:
As we have learned

Equilibrium of Concurrent Forces -

If all the forces working on a body are acting on the same point then they are said to be concurrent.

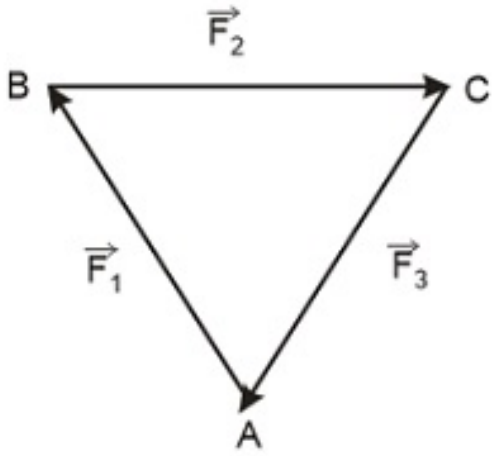
- wherein

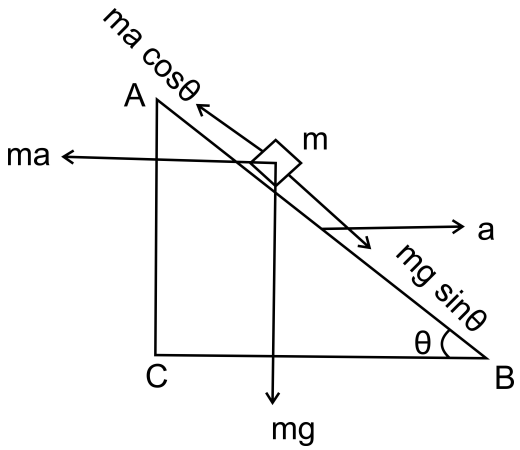
$\sum \vec{F_{net}} = 0$

$or \sum \vec{F}_x = 0, \sum \vec{F}_y = 0, \sum \vec{F}_z = 0$

* Three forces will be in equilibrium if they are represented by three sides of triangle taken in order

Careers360





For Block to be stationary net force along the plane =0

$\Rightarrow mg\sin\theta = ma\cos\theta \Rightarrow a = g\tan\theta$

Q. 36 The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at (2, 0, -3) , about the point (2, -2, -2) is given by

Option 1:
$-7\hat{i} - 8\hat{j} - 4\hat{k}$

Option 2:
$-4\hat{i} - \hat{j} - 8\hat{k}$

Option 3:
$-8\hat{i} - 4\hat{j} - 7\hat{k}$

Careers360

Option 4:

$-7\hat{i}-4\hat{j}-8\hat{k}$

Correct Answer:

$-7\hat{i}-4\hat{j}-8\hat{k}$

Solution:

Torque -

$$\underset{\tau}{\rightarrow}=\underset{r}{\rightarrow}\times\underset{F}{\rightarrow}$$

- wherein

This can be calculated by using either $\tau = r_1 F \; or \; \tau = r \cdot F_1$

r_1 = perpendicular distance from the origin to the line of force.

F_1 = component of force perpendicular to line joining force.

$$\vec{r}=\vec{r_2}-\vec{r_1}$$

$$=\left(2\hat{i}+(-3\hat{k})\right)-(2\hat{i}-2\hat{j}-2\hat{k})$$

$$=-\hat{k}+2\hat{j}$$

$$\therefore \tau=\vec{r}\times\vec{F}$$

$$=\left(2\hat{j}-\hat{k}\right)\times(4\hat{i}+5\hat{j}-6\hat{k})$$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 2 & -1\\ 4 & 5 & -6\end{vmatrix}=\hat{i}(-12+5)-\hat{j}(0+4)+\hat{k}(0-8)$$

$$-7\hat{i}-4\hat{j}-8\hat{k}$$

Hence, the answer is the Option (4).

Careers360

Q. 37 A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If the screw gauge has a zero error of — 0.004 cm, the correct diameter of the ball is :

Option 1:
0.053 cm

Option 2:
0.525 cm

Option 3:
0.521 cm

Option 4:
0.529 cm

Correct Answer:
0.529 cm

Solution:
Reading of screw gauge = MSR+VSR × LC+ZERO ERROR

$= 0.5 + 25 \times 0.001 + 0.004$

$= 0.529cm$

Hence, the answer is the option (4).

Careers360

Q. 38 A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere ?

Option 1:
Rotational kinetic energy

Option 2:
Moment of inertia

Option 3:
Angular velocity

Option 4:
Angular momentum

Correct Answer:
Angular momentum

Solution:
As we have learned

Rotational angular momentum -

$$\vec{L} = I\vec{w}$$

- wherein

I = Moment of inertia about fixed axis of rotation

w = angular velocity of rotation

Law of conservation of angular moment -

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

- wherein

If net torque is zero

i.e. $\frac{d\vec{L}}{dt} = 0$

Careers360

$\vec{L} = constant$

angular momentum is conserved only when external torque is zero .

when we have incraese radius of sphere keeping the mass same then moment of inertia increases

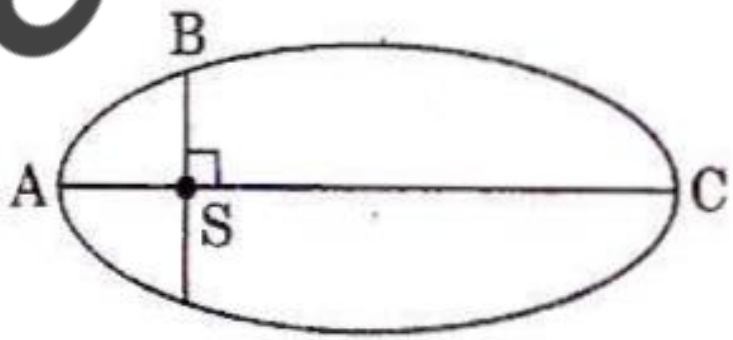
since there is no torque acting since it is rotating freely , angular momemntum remains constant

$L = Iw$

L = constant , I increases $\Rightarrow$ w decreases

Careers360

Q. 39 The kinetic energies of a planet in an elliptical orbit about the Sun, at positions A, B and C are K_A, K_B and K_C, respectively. AC is the major axis and SB is perpendicular to AC at the position of the Sun S as shown in the figure. Then



Option 1:

$K_B < K_A < K_C$

Option 2:

$K_A > K_B > K_C$

Option 3:

$K_A < K_B < K_C$

Option 4:

$K_B > K_A > K_C$

Correct Answer:

$K_A > K_B > K_C$

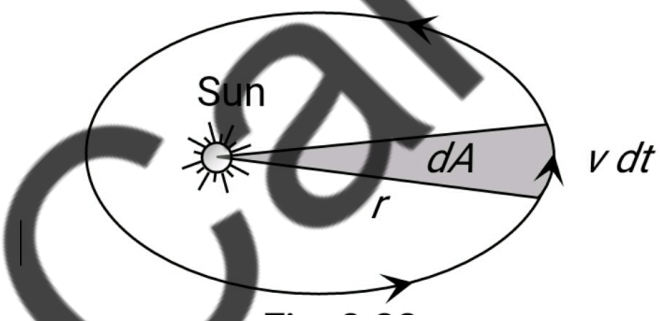
Solution:

As we have learned

Kepler's 2nd law -

Area of velocity = $\frac{dA}{dt}$

$$\frac{dA}{dt} = \frac{1}{2}\frac{(r)\,(V\,dt)}{dt} = \frac{1}{2}rV$$



$\frac{dA}{dt} \rightarrow$ Areal velocity

$dA \rightarrow$ small area traced

- wherein

Simiar to Law of conservation of momentum

Careers360

$$\frac{dA}{dt} = \frac{L}{2m}$$

$L \rightarrow$ Angular momentum

Known as Law of Area

$$V_A > V_B > V_C (vr = constant)$$

$$K_A > K_B > K_C$$

Q. 40 If the mass of the Sun were ten times smaller and the universal gravitational constant were ten times larger in magnitude, which of the following is not correct ?

Option 1:
Time period of a simple pendulum on the Earth would decrease.

Option 2:
Walking on the ground would become more difficult.

Option 3:
Raindrops will fall faster.

Option 4:
'g' on the Earth will not change.

Correct Answer:
'g' on the Earth will not change.

Careers360

Solution:
As we have learned

Acceleration due to gravity (g) -

Force extended by earth on a body is gravity.

Formula: $g = \frac{GM}{R^2},$

$g = \frac{4}{3}\pi\rho GR$

$g \rightarrow$ gravity

$\rho \rightarrow$ density of earth

$R \rightarrow$ Radius of earth

- wherein

It's average value is $9.8\ m/s^2$ or $981cm/sec^2$ or $32feet/s^2$ on the surface of earth

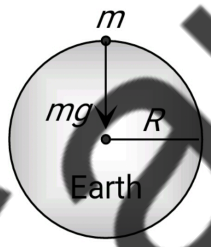
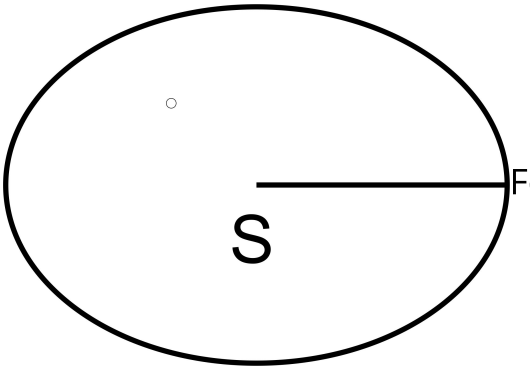


Fig. 8.4

Careers360



For earth $\frac{G'm_eM's}{r^2} = \frac{m_ev^2}{r}$

$$v^2 = \frac{G'M's}{r}$$

$$g = \frac{G'M'e}{R^2}$$

$$G' = 10g, M's = 10Ms$$

$$g' = 10g$$

rain drop will fall faster walking on the ground become more difficult

Time period of simple pendulum $(2\pi\sqrt{l/g})$ will decrease

Hence 'g' on the Earth will not change.

Careers360

Q. 41 A solid sphere is in a rolling motion. In rolling motion, a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is :

Option 1:
10 : 7

Option 2:
5 : 7

Option 3:
7 : 10

Option 4:
2 : 5

Correct Answer:
5 : 7

Solution:
The kinetic energy of a body in combined rotation and translation -

$$K.E = \frac{1}{2} I_{cm} w^2 + \frac{1}{2} m {v_0}^2$$

where

$\frac{1}{2} m {v_0}^2$ = Translational kinetic energy of the centre of mass

$\frac{1}{2} I_{cm} w^2$ = Rotational kinetic energy about the centre of mass

$$k_t = 1/2 m {v_c m}^2$$

$$k_r = 1/2 I w^2 = 1/2 (2/5 m r^2)(v/r)^2 = 1/5 m v^2$$

$$\frac{kt}{kt + k_r} = \frac{\frac{1}{2} m {V_c m}^2}{(\frac{1}{2} + \frac{1}{5}) m v^2} = 5/7$$

Hence the answer is Option (2).

Careers360

Q. 42 A small sphere of radius `r' falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity is proportional to -

Option 1:
r^5

Option 2:
r^2

Option 3:
r^3

Option 4:
r^4

Correct Answer:
r^5

Solution:
The net force on the body -

$$F_B + F_v = W$$

$$\rightarrow 6\pi\eta r v + \frac{4}{3}\pi r^3 \sigma g = \frac{4}{3}\pi r^3 \rho g$$

$$\rightarrow 6\pi\eta r v = \frac{4}{3}\pi r^3 g\left(\rho - \sigma\right)$$

$$\rightarrow v_t = \frac{2}{9}\frac{r^2\left(\rho - \sigma\right)}{\eta}g$$

where

Careers360

$F_B - Buoyant\ force$

$F_v - viscous\ force$

$w - weight$

$\rho \rightarrow density\ of\ ball$

$\sigma \rightarrow density\ of\ water$

$V_T = terminal\ velocity$

Rate of heat production $= \vec{F} \cdot \vec{V}$

$= (6\pi\eta r v) \cdot v$

$= 6\pi\eta(rv^2)$

$v \propto r^2$

$\therefore$ rate of heat

$\therefore \propto r^5$

Hence the answer is option (1).

Q. 43 The power radiated by a black body is P and it radiates maximum energy at wavelength, λ_o. If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_o$, the power radiated becomes nP. The value of 'n' is

Option 1:

$\frac{256}{81}$

Careers360

Option 2:

$\frac{4}{3}$

Option 3:

$\frac{3}{4}$

Option 4:

$\frac{81}{256}$

Correct Answer:

$\frac{256}{81}$

Solution:

$$\lambda T = costant$$

$$\lambda_0 T_1 = \frac{3}{4}\lambda_0 T_2 \Rightarrow T_2 = \frac{4}{3}T_1$$

$$p \propto T_1^4$$

$$np \propto (\frac{4}{3}T_1)^4 = (256/81)T_1^4$$

$$np/p = (256/81)$$ or n= 256/81

Hence the correct option is (1).

Q. 44 Two wires are made of the same material and have the same volume, The first wire has a cross-sectional area of A and the second wire has a cross-sectional area 3A. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount?

Careers360

Option 1:
4 F

Option 2:
6 F

Option 3:
9 F

Option 4:
F

Correct Answer:
9 F

Solution:
Young Modulus -

The ratio of normal stress to longitudinal strain

it denoted by Y

$$Y = \frac{Normal\ stress}{longitudnal\ strain}$$

- wherein

$$Y = \frac{F/A}{\Delta l/L}$$

F - applied force

A - Area

Δl - Change in length

l - original length

$$F/A = Y\frac{\Delta l}{l}$$

Careers360

$$F = AY\frac{\Delta l}{l}$$

in the second case,

$$F' = A'Y\frac{\Delta l}{l'}$$

volume = constant

$$Al = const. \Rightarrow Al = 3Al'$$

or $l' = l/3$

$$F' = Y\frac{3A\Delta l}{l/3} = 9(\frac{YA\Delta l}{l}) = 9F$$

Hence, the answer is the option (3).

Q. 45 A sample of 0.1 g of water at 100°C and normal pressure (1.013 x 10^5 Nm^{-2}) requires 54 cal of heat energy to convert to steam at 100°C. If the volume of the steam produced is 167.1 cc, the change in internal energy of the sample is:

Option 1:
42.2 J

Option 2:
209.8 J

Option 3:
104.3 J

Option 4:
84.5 J

Careers360

Correct Answer:
209.8 J

Solution:

$$dQ = dU + dW$$

$$Heat\ Supplied = 54cal = 54 \times 4.2\ J = 226.8\ J$$

$$Work\ Done = P\Delta V\ = 1.01 \times 10^5 \times 167.1 \times 10^{-6} J$$

$$= 168.8 \times 10^{-1} = 16.92\ J$$

$$\therefore \quad \Delta U = 226.8 - 16.92 J \simeq 209.88 J$$

Hence the answer is Option (2).

Chemistry

Q. 1 The correct order of N-compounds in its decreasing order of oxidation states is:

Option 1:
HNO_3, NH_4Cl, NO, N_2

Option 2:
HNO_3, NO, NH_4C1, N_2

Option 3:
HNO_3, NO, N_2, NH_4C1

Option 4:
NH_4C1, N_2, NO, HNO_3

Careers360

Correct Answer:
HNO_3, NO, N_2, NH_4C1

Solution:
The oxidation state of N in the given compounds is -

$$\overset{+5}{H}\overset{+2}{N}O_3, \overset{0}{N}O, \overset{-3}{N}H_4Cl$$

Hence, the answer is the option (3).

Q. 2 Which one of the following elements is unable to form MF_6^{3-} ion?

Option 1:
B

Option 2:
Al

Option 3:
Ga

Option 4:
In

Correct Answer:
B

Solution:
Boron belongs to the Second Period and is not able to expand its octet as it does not have vacant d-orbitals.

Thus, Boron will not be able to form a hexaco-ordinated complex.

Hence, the correct answer is an option (1).

Careers360

Q. 3 Considering the Ellingham diagram, which of the following metals can be used to reduce alumina?

Option 1:
Mg

Option 2:
Zn

Option 3:
Fe

Option 4:
Cu

Correct Answer:
Mg

Solution:
Ellingham Diagram - Ellingham diagram normally consists of plots of $\triangle fG^{\circ}$ Vs T for the formation of oxides of elements.

$$2xM(s) + O_2(g) \rightarrow 2M_xO(s)$$

Mg has more negative $\triangle G$ value than alumina. The metal which has more $-\triangle G$ value can reduce the metal oxide which has less $-\triangle G$ value.

Hence, option 1 is correct.

Q. 4 The correct order of atomic radii in group 13 elements is

Option 1:
$B < Ga < Al < Tl < In$

Careers360

Option 2:

$B < Al < Ga < In < Tl$

Option 3:

$B < AI < In < Ga < Tl$

Option 4:

B < Ga < Al < In < Tl

Correct Answer:

B < Ga < Al < In < Tl

Solution:

The size of elements increases down the group, but in the case of group 13 size of Gallium (period - 4) is smaller than Aluminum (period - 3). This is because of the poor shielding effect of the d- electrons and the effective nuclear charge experienced by outermost electrons increases and size decreases slightly.

Hence, the answer is the option (4).

Q. 5 Which of the following statements is not true for halogens ?

Option 1:

All but fluorine show positive oxidation states.

Option 2:

All are oxidizing agents.

Option 3:

All form monobasic oxyacids.

Option 4:

Chlorine has the highest electron-gain enthalpy.

Correct Answer:

All but fluorine show positive oxidation states.

Careers360

Solution:
Acidic strength $\propto$ electronegativity of the central atom. F being the most electronegative element, shows only a -1 oxidation state.

Hence, the answer is the option (1).

Q. 6 In the structure of ClF_3, the number of lone pairs of electrons on central atom 'Cl' is

Option 1:
four

Option 2:
two

Option 3:
one

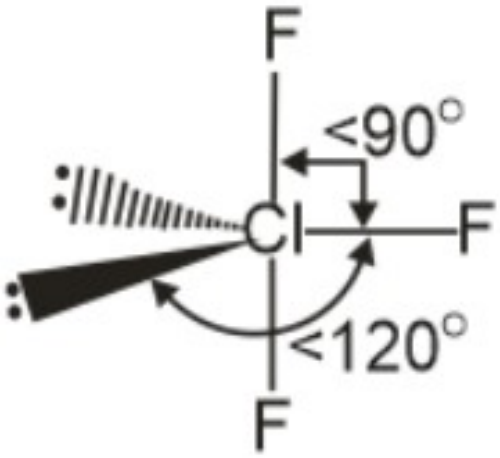
Option 4:
three

Correct Answer:
two

Solution:

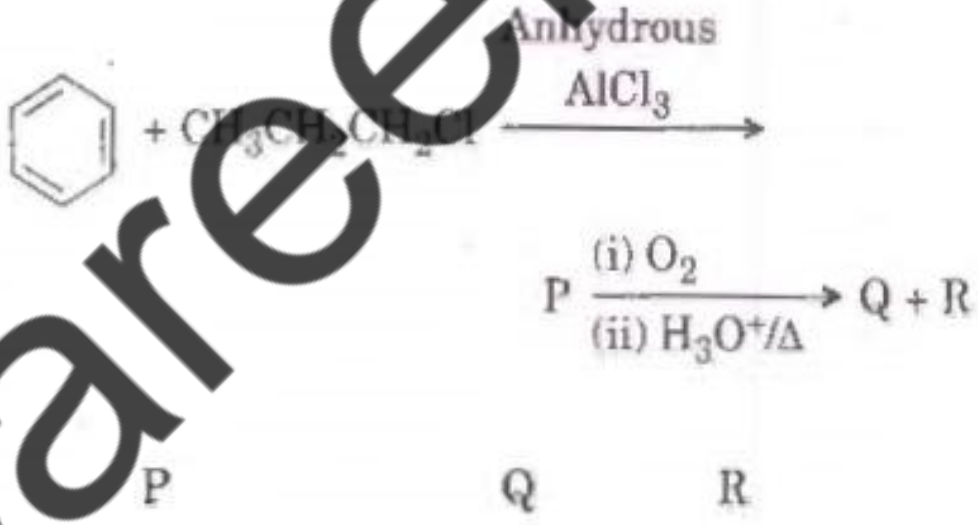
In ClF_3 there are two lone pairs in Chlorine at equitorial position .

Careers360



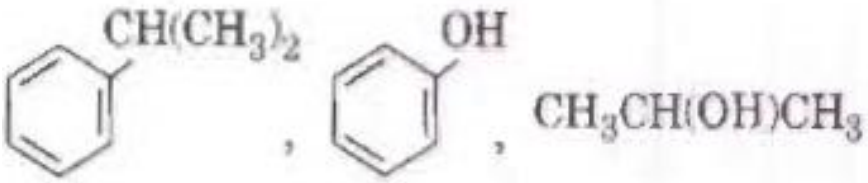
Hence, the option number (2) is correct.

Q. 7 Identify the major products P, Q and R in the following sequence of reactions :

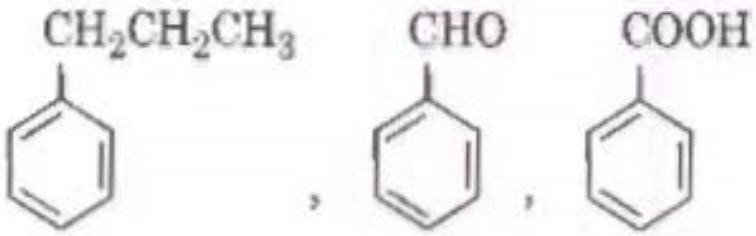


Careers360

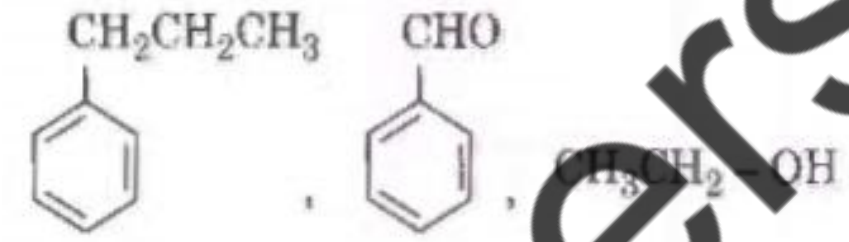
Option 1:



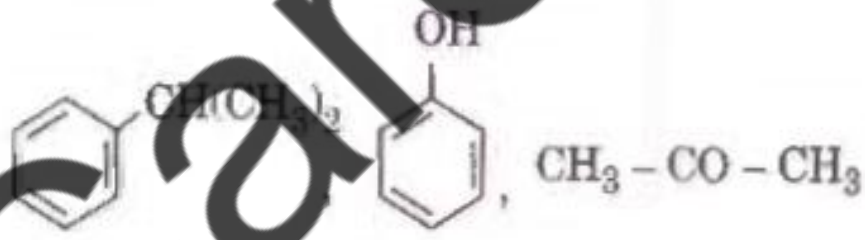
Option 2:



Option 3:

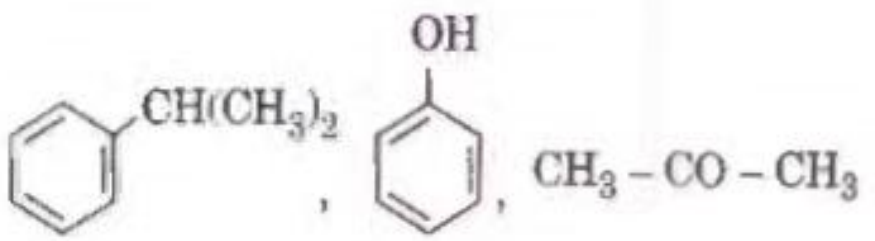


Option 4:



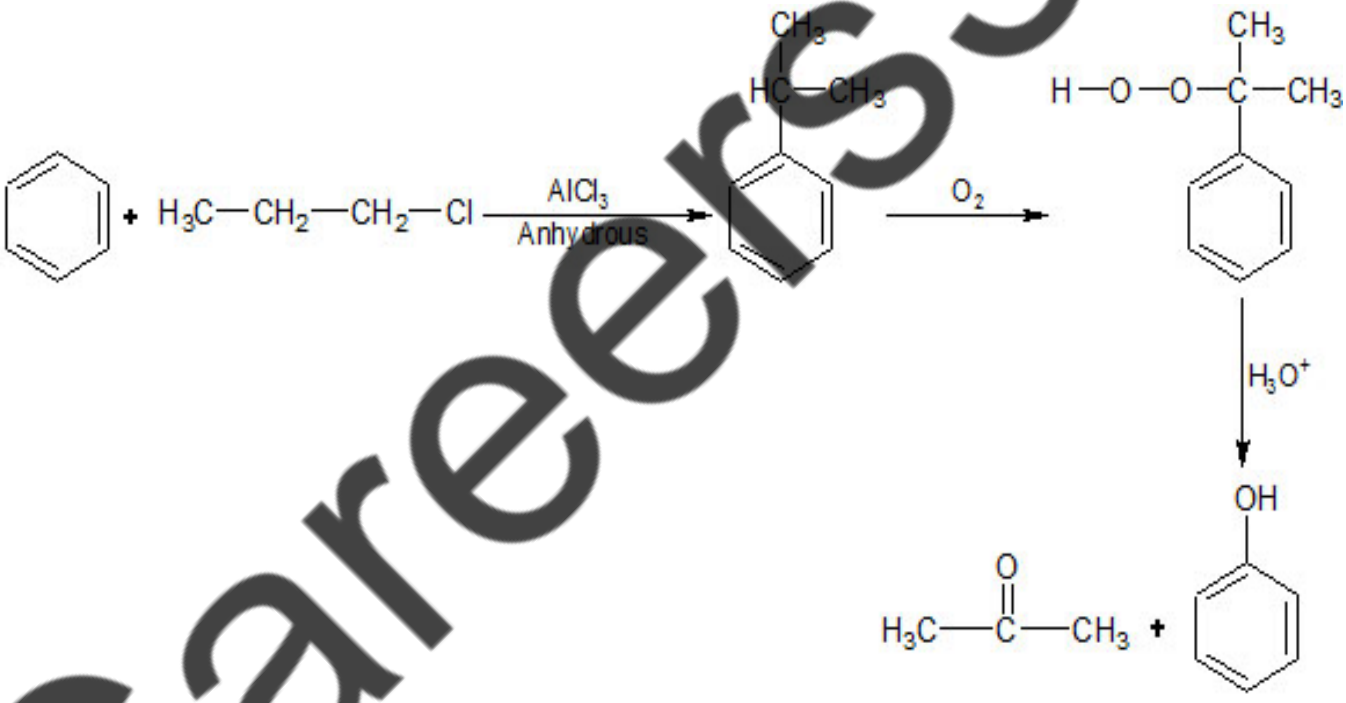
Careers360

Correct Answer:



Solution:
The reaction will be -

Due to Cl being a good leaving group, it will leave the alkane in the presence of anhydrous $AlCl_3$ and forms 1-degree carbonation then convert to 2-degree carbocation.



Q. 8 Which of the following compounds can form a zwitterion ?

Option 1:
Benzoic acid

Option 2:
Acetanilide

Careers360

Option 3:
Aniline

Option 4:
Glycine

Correct Answer:
Glycine

Solution:
Zwitter ion - A molecule with dipolar ends (an acidic end and a basic end).

In an aqueous solution,

$-NH_2$ the group undergoes ionization

$-NH_2$ accepts H^+ and becomes NH_3^+ and $-COOH$ losses H^+ and becomes $-COO^-$.

$$NH_2-CH_2-COOH \rightleftharpoons \underset{Zwitter\ ion}{H_3\overset{+}{N}-CH_2-COO^-}$$

Hence, the answer is the option (4).

Q. 9 Regarding cross-linked or network polymers, which of the following statements is incorrect ?

Option 1:
Examples are bakelite and melamine.

Option 2:
They are formed from bi- and tri-functional monomers.

Option 3:
They contain covalent bonds between various linear polymer chains.

Option 4:
They contain strong covalent bonds in their polymer chains.

Careers360

Correct Answer:
They contain strong covalent bonds in their polymer chains.

Solution:
Cross Linked / Network Polymers -

Rigid structure formed from bi-functional and tri-functional monomers forming strong covalent bonds.

- wherein

Bakellite, Melamine

Cross linked or network polymers are usually formed from bi-functional and tri-functional monomers and contains strong covalent bonds between various linear polymer chains .

Q. 10 Nitration of aniline in strong acidic medium also gives m-nitroaniline because

Option 1:
In absence of substituents nitro group always goes to m-position.

Option 2:
In electrophilic substitution reactions amino group is meta directive.

Option 3:
In spite of substituents nitro group always goes to only m-position.

Option 4:
In acidic (strong) medium aniline is present as anilinium ion.

Correct Answer:
In acidic (strong) medium aniline is present as anilinium ion.

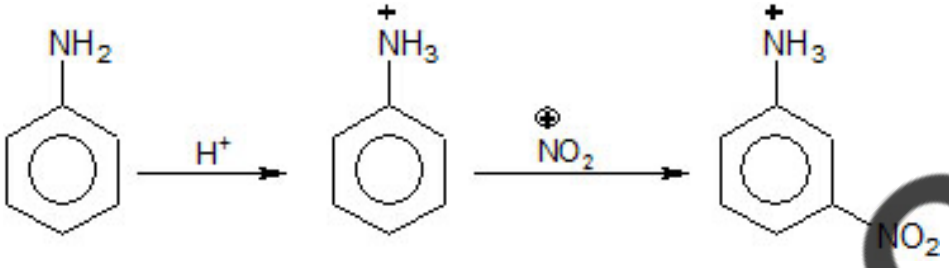
Solution:

Careers360

Reaction of Aniline with dil. HNO3 -

With dil. HNO_3 meta product is also formed in large amount because HN2 group is protonated and it becomes P - directing.

-



Q. 11 The difference between amylose and amylopectin is

Option 1:
Amylopectin have 1$\rightarrow$4 α-linkage and
1 $\rightarrow$ 6 β - linkage

Option 2:
Amylose have 1$\rightarrow$4 α-linkage and
1 $\rightarrow$ 6 β - linkage

Option 3:
Amylopectin have 1$\rightarrow$4 α-linkage and
1 $\rightarrow$ 6 α- linkage

Option 4:
Amylose is made up of glucose and galactose

Careers360

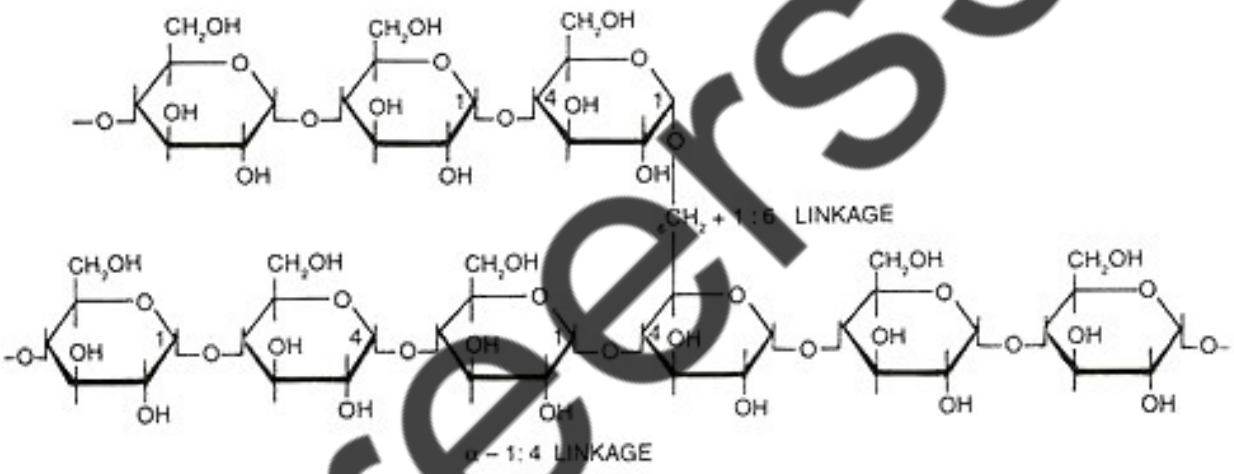
Correct Answer:
Amylopectin have 1$\longrightarrow$4 α-linkage and
1 $\longrightarrow$ 6 α- linkage

Solution:
Starch -

A polymer of $\alpha - D - glucose$ that consists of having two components , amylose (water soluble unbranched chain) and amylopectin (water insoluble having branched chain)

- wherein

Main storage polysaccharide of plants



Amylose : α , $1-4$ glycosidic linkage

Amylopectin : $\alpha, 1, 4$ linkage and $\alpha, 1, 6$ linkage

Amylopectin forms straight chain due to 1,4 linkage and branching occur due to 1, 6 - linkage.

Careers360

Q. 12 A mixture of 2.3 g formic acid and 4.5 g oxalic acid is treated with conc. H_2SO_4. The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be

Option 1:
2.8

Option 2:
3.0

Option 3:
1.4

Option 4:
4.4

Correct Answer:
2.8

Solution:
As we learnt in

Careers360

Number of Moles = given mass of substance/ molar mass of the substance

$$HCOOH \xrightarrow{H_2SO_4} CO + H_2O$$

$$\frac{2.3}{46} = \frac{1}{20} \qquad \frac{1}{20}\text{mole}$$

(H_2O is absorbed by H_2SO_4)

$$H_2C_2O_4 \xrightarrow{H_2SO_4} CO + CO_2 + H_2O$$

$$\frac{4.5}{90} = \frac{1}{20} \qquad \frac{1}{20}\text{mole}$$

(CO_2 is absorbed by KOH and H_2O by H_2SO_4)

Total moles of CO is 0.1 mole or 2.8 gm.

Hence, option number (1) is correct.

Q. 13 Which of the following oxides is most acidic in nature?

Option 1:
BaO

Option 2:
BeO

Option 3:
MgO

Option 4:
CaO

Correct Answer:
BeO

Careers360

Solution:
Elements of Group 2 of the periodic table Be, Mg, Ca, Sr, Ba, Ra

The basicity of oxides of alkaline earth metals increases down the group, this is because going down the extent of polarization decreases due to an increase in the size of cation hence, the ionic character increases down the group.

The acidic nature is directly proportional to the covalent character of oxide.

Their oxides are alkaline except for Be, BeO is amphoteric.

The correct answer is option 2.

Q. 14 Which oxide of nitrogen is not a common pollutant introduced into the atmosphere both due to natural and human activity ?

Option 1:
N_2O

Option 2:
NO_2

Option 3:
N_2O_5

Option 4:
NO

Correct Answer:
N_2O_5

Solution:
Nitrogen oxides -

Careers360

formed electro-photo chemically in air , by burning of fossil fuels and denitrifying bacteria.

- wherein

$$N_2\,(g) + O_2(g) \rightarrow 2NO\,(g)$$

$$2NO\,(g) + O_2(g) \rightarrow 2NO_2\,(g)$$

$$NO\,(g) + O_3(g) \rightarrow NO_2\,(g) + O_2$$

N_2O occur naturally in environment . In automobile engine , when fossil is burn N_2 and O_2 react to form NO and NO_2 .

Q. 15 The compound A on treatment with Na gives B, and with PCl_5 gives C. B and C react together to give diethyl ether. A, B and C are in the order

Option 1:
C_2H_5Cl, C_2H_5, C_2H_5OH

Option 2:
C_2H_5OH, C_2H_{5Cl}, C_2H_5ONa

Option 3:
C_2H_5OH, C_2H_6, C_2H_5Cl

Option 4:
C_2H_5OH, C_2H_5ONa, C_2H_5Cl

Correct Answer:
C_2H_5OH, C_2H_5ONa, C_2H_5Cl

Solution:

Careers360

$$ROH + Na \rightarrow 2RONa + H_2$$

Alcohols reaction with phosphorus trihalides -

Form alkyl halides

- wherein

$$ROH + PX_3 \rightarrow RX + H_3PO_3$$

X = Cl, Br, I

$$\underset{(A)}{C_2H_5OH} \xrightarrow{Na} \underset{(B)}{C_2H_5O^-Na^+}$$

$$\underset{(A)}{C_2H_5OH} \xrightarrow{PCl_5} \underset{(C)}{C_2H_5Cl}$$

$$\underset{(B)}{C_2H_5\overset{-}{O}\overset{+}{Na}} + \underset{(C)}{C_2H_5Cl} \longrightarrow \underset{\text{Riethyl ether}}{C_2H_5 - O - C_2H_5}$$

Q. 16 The compound C_7H_8 undergoes the following reactions. The product "C" is :

$$C_7H_8 \xrightarrow{3Cl_2/\Delta} A \xrightarrow{Br_2/Fe} B \xrightarrow{Zn/HCl} C$$

Option 1:
3-bromo-2,4,6-trichlorotoluene

Option 2:
o-bromotoluene

Option 3:
m-bromotoluene

Option 4:
p-bromotoluene

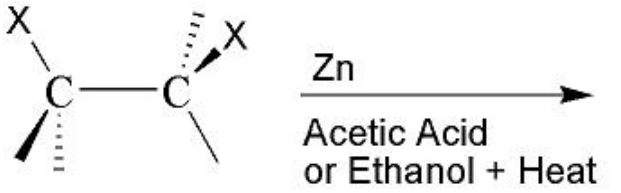
Careers360

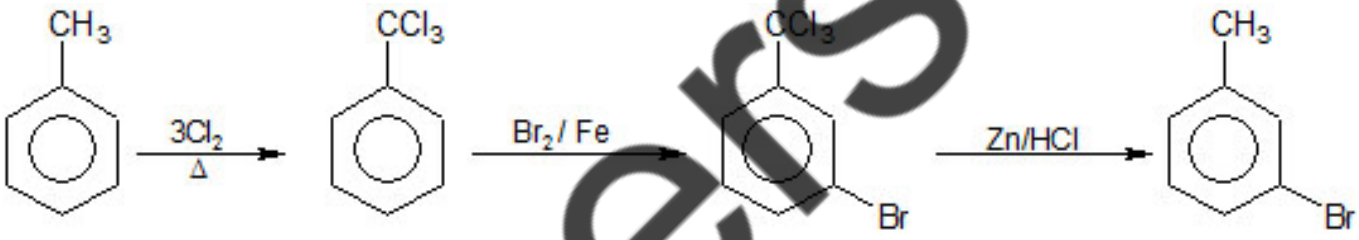
Correct Answer:
m-bromotoluene

Solution:
From vicinal dihalide -

Vicinal dihalide on treatment with Zn metal lose a molecule of $ZnCl_2$ to form alkene this reaction is known as Dehalogenation.

- wherein





Hence, the option number (3) is correct.

Q. 17 Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is

Option 1:
CH_3 — CH_3

Option 2:
CH_2= CH_2

Option 3:
$CH \equiv CH$

Careers360

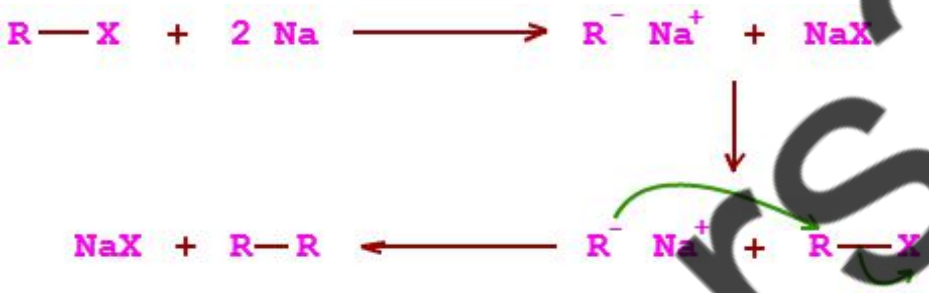
Option 4:
CH_4

Correct Answer:
CH_4

Solution:
Preparation of Alkane from alkyl halide (Wurtz Reaction) -

Alkyl halide on treatment with metal in dry ether (free from moisture) solution give higher alkane.

- wherein



$$CH_4 \xrightarrow[h\nu]{Br_2} CH_3Br \xrightarrow[ether]{Na} CH_3 - CH_3$$

(less than 4 C)

Q. 18 Which of the following molecules represents the order of hybridisation sp^2, sp^2, sp, sp from left to right atoms ?

Option 1:
$CH_2 = CH — CH = CH_2$

Option 2:
$CH_2 = CH — C \equiv CH$

Careers360

Option 3:

$CH \equiv C\text{-}C \equiv CH$

Option 4:

$CH_3 — CH = CH — CH_3$

Correct Answer:

$CH_2 = CH — C \equiv CH$

Solution:

Hybridization - The process of mixing atomic orbitals belonging to the same atoms of slightly different energies so that redistribution of energy takes place between them resulting in the formation of a new set of orbitals of equivalent energies and shapes is called hybridization.

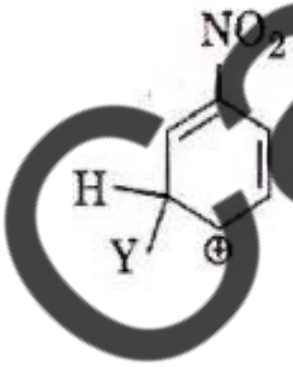
The new orbitals thus formed are called hybrid orbitals.

$$\underset{sp^2}{CH_2} = \underset{sp^2}{CH} - \underset{sp}{C} \equiv \underset{sp}{CH}$$

Hence, the answer is the option (2).

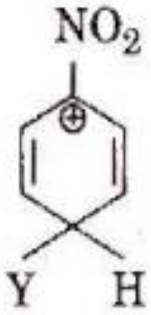
Q. 19 Which of the following carbocation is expected to be most stable ?

Option 1:

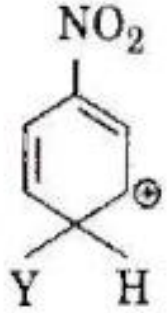


Careers360

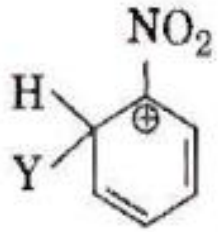
Option 2:



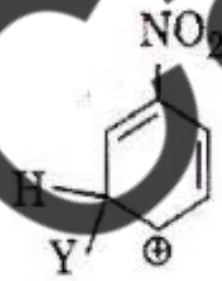
Option 3:



Option 4:



Correct Answer:



Solution:
Negative mesomeric effect -

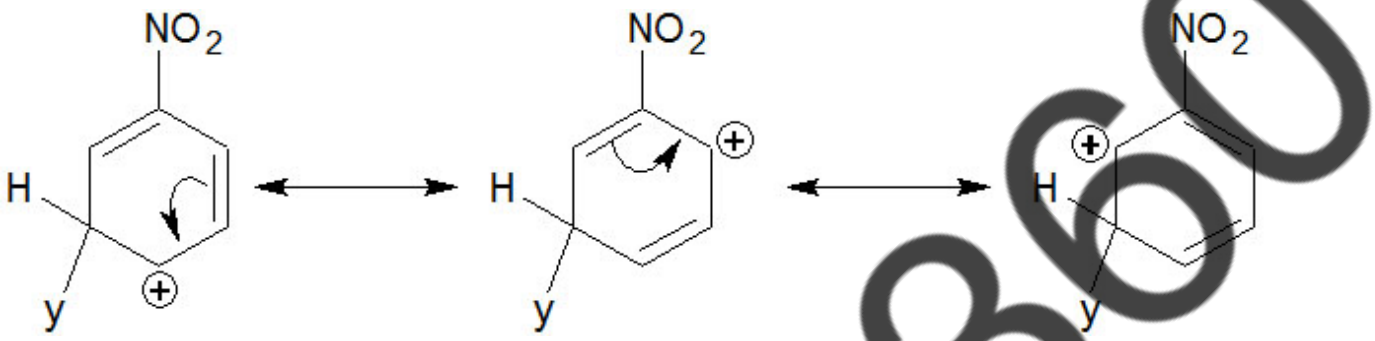
Decreases electron density at ortho and para position

Careers360

$-NO_2, -CHO, -CN, -COOH$

- wherein

Group showing -M are meta directing



More stable due to less e^- withdrawing effect of $-NO_2$

Q. 20 Which of the following is correct with respect to — I effect of the substituents ? (R = alkyl)

Option 1:
— NH_2 > — OR > — F

Option 2:
— NR_2 < — OR < — F

Option 3:
— NH_2 < — OR < — F

Option 4:
— NR_2 > — OR > — F

Correct Answer:
— NR_2 < — OR < — F

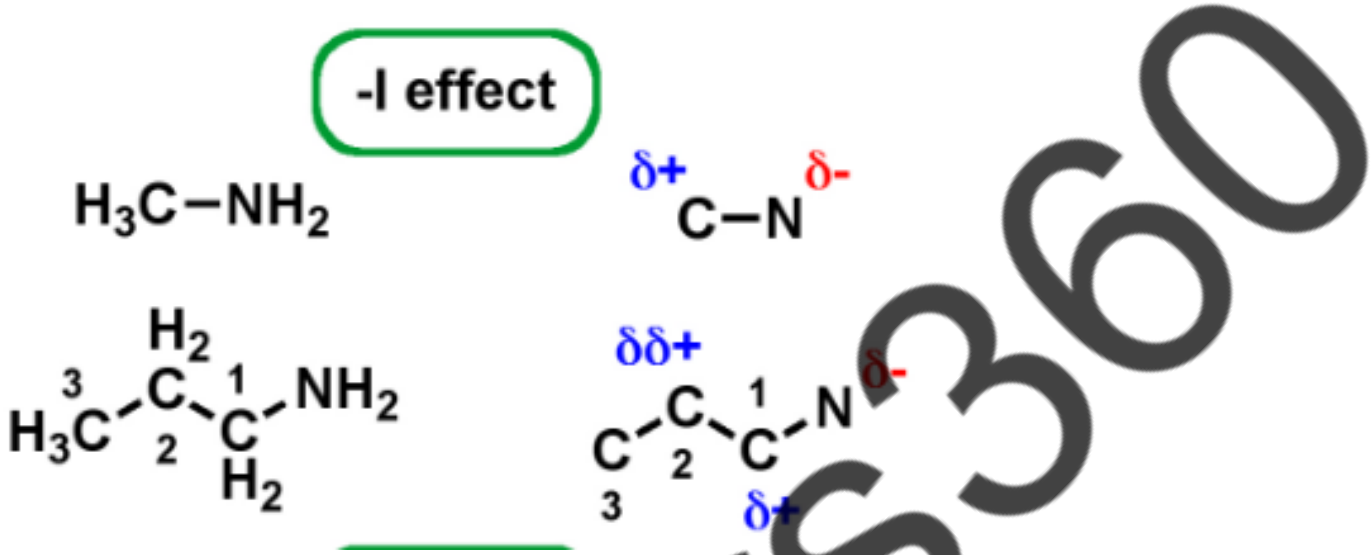
Solution:

Careers360

-I - effect (Negative inductive effect) -

Those atoms or group of atoms having more tendency to displace σ e - towards itself as compared to Hydrogen atom is called Negative Inductive effect

- wherein



Based on electronegativity

$-I\ effect \Rightarrow -\mathrm{NH_2} < -\mathrm{OR} < -\mathrm{F}$

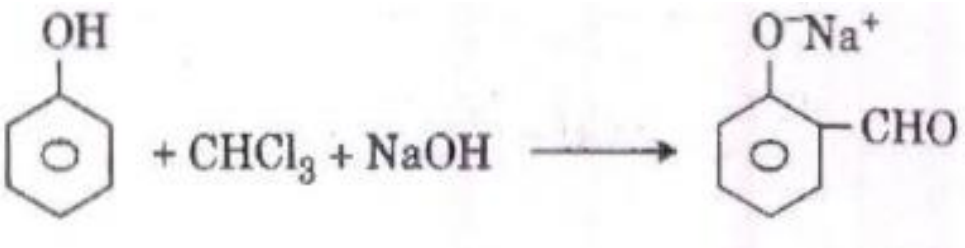
or

$-\mathrm{NR_2} < -\mathrm{OR} < -\mathrm{F}$

Option (2) and (3) both are correct.

Q. 21

In the reaction



Careers360

Option 1:
dichloromethyl anion ($\overset{\Theta}{C}HCl_2$)

Option 2:
formyl cation ($\overset{\oplus}{C}HO$)

Option 3:

dichloromethyl cation ($\overset{\oplus}{C}HCl_2$)

Option 4:
dichlorocarbene (:CCl_2)

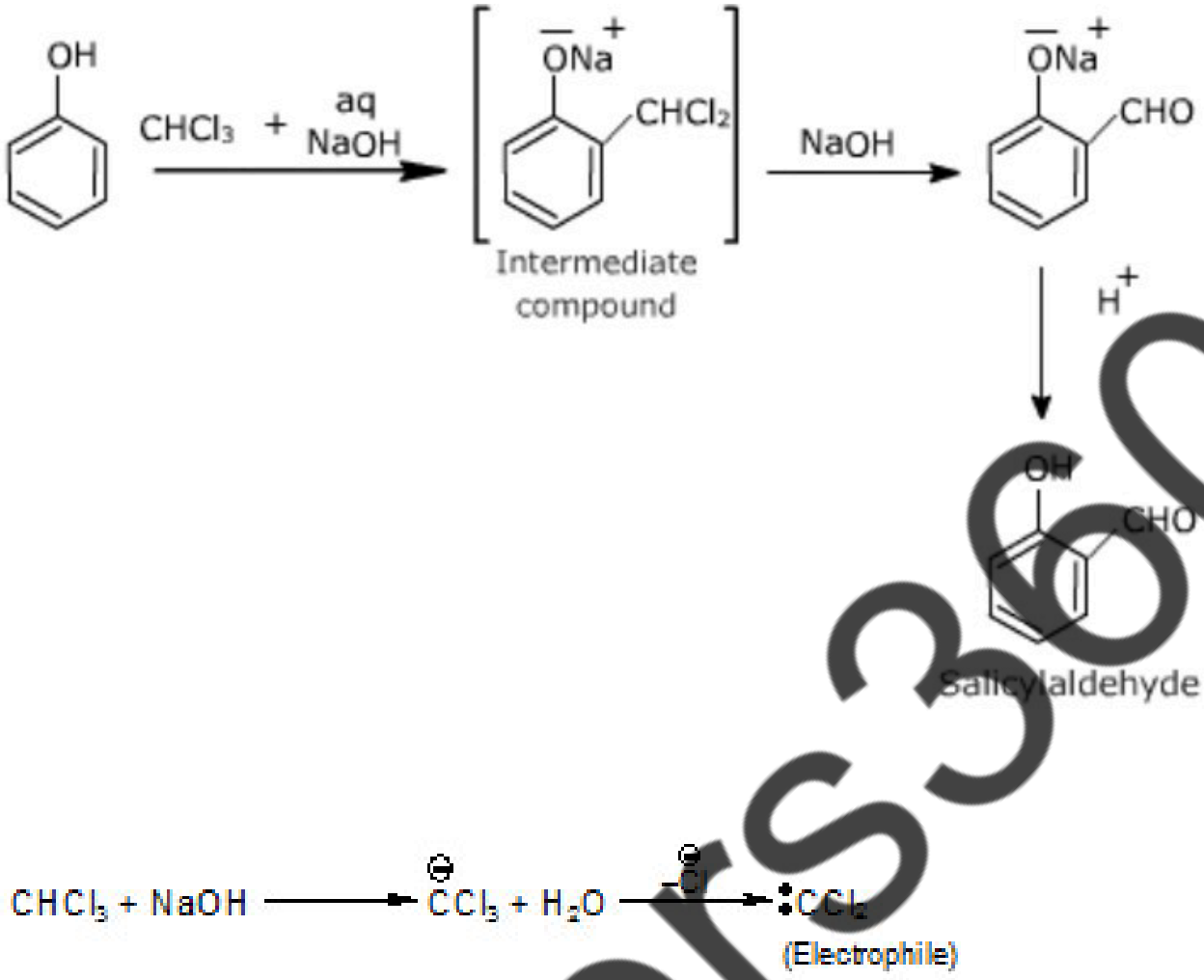
Correct Answer:
dichlorocarbene (:CCl_2)

Solution:
Reimer Tiemann's reaction -

The electrophile used is CCl_2 (Dichloromethylene) -CHO is introduced in the ortho position.

- wherein

Careers360



Q. 22 Carboxylic acids have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their:

Option 1:
more extensive association of carboxylic acid via van der Waals force of attraction

Option 2:
formation of carboxylate ion

Option 3:
formation of intramolecular H-bonding

Careers360

Option 4:
formation of intermolecular H-bonding

Correct Answer:
formation of intermolecular H-bonding

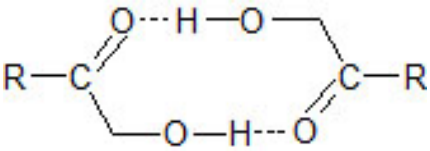
Solution:
Carboxylic acids -

An organic acid containing one or more carboxyl (-COOH) groups as a functional group.

- wherein

$C_nH_{2n+1}COOH \quad or \quad C_nH_{2n}O_2$

This is due to the more extensive association through intermolecular H-bond

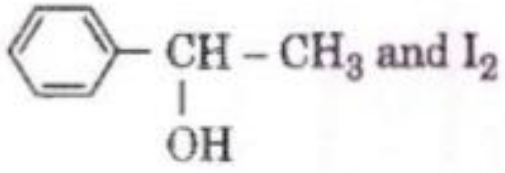


Hence, the option number (4) is correct.

Q. 23 Compound A, $C_8H_{10}O$, is found to react with NaOI produced by reacting Y with NaOH and yields a yellow precipitate with characteristic smell.

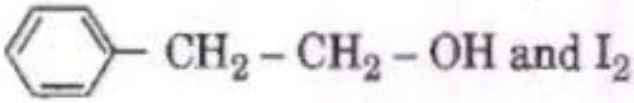
A and Y are respectively

Option 1:

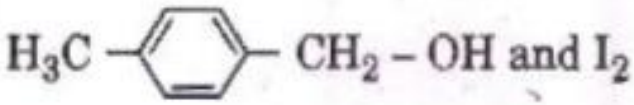


Careers360

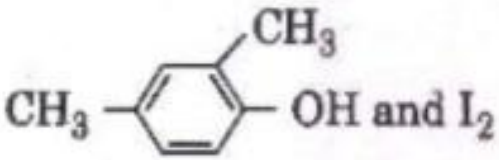
Option 2:



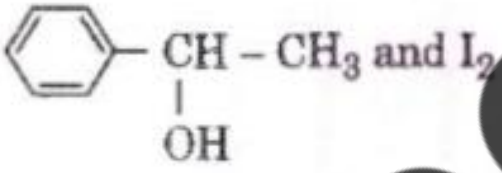
Option 3:



Option 4:



Correct Answer:



Solution:

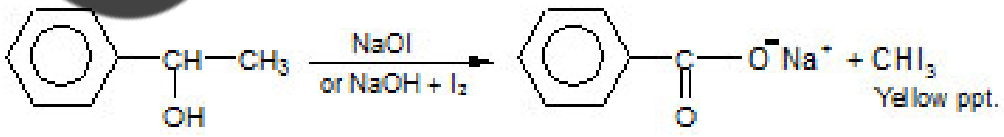
As we learnt in

Haloform reaction -

$CH_3COR \text{ is oxidised by } NaOI \text{ to form iodoform and } RCO_2Na$

- wherein

$C_2H_5COCH_3 + NaOH + I_2 \rightarrow C_2H_5CO_2Na + CHI_3$



Careers360

Q. 24 Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the correct code :

	Column I		Column II
a.	Co^{3+}	i.	$\sqrt{8}B.M.$
b.	Cr^{3+}	ii.	$\sqrt{35}B.M.$
c.	Fe^{3+}	iii.	$\sqrt{3}B.M.$
d.	Ni^{2+}	iv.	$\sqrt{24}B.M.$
		v.	$\sqrt{15}B.M.$

Option 1:

a	b	c	d
iv	i	ii	iii

Option 2:

a	b	c	d
i	ii	iii	iv

Option 3:

a	b	c	d
iv	v	ii	i

Careers360

Option 4:

a	b	c	d
iii	v	i	ii

Correct Answer:

a	b	c	d
iv	v	ii	i

Solution:

Magnetic Moments (spin only) - $\sqrt{n\left(n+2\right)}$

where n= number of the unpaired electrons.

Number of unpaired electrons and corresponding magnetic moments.

$1 \rightarrow 1.7$
$2 \rightarrow 2.8$
$3 \rightarrow 3.9$
$4 \rightarrow 4.9$
$5 \rightarrow 5.9$

$$Magnetic\ moment(\mu) = \sqrt{n(n+2)}\ B.M.$$

(a) $Co^{3+} \rightarrow n = 4$

$$\mu = \sqrt{4(4+2)} = \sqrt{24}\ B.M.$$

(b) $Cr^{3+} \rightarrow n = 3$

$$\mu = \sqrt{3(3+2)} = \sqrt{15}\ B.M.$$

(c) $Fe^{3+} \rightarrow n = 5$

$$\mu = \sqrt{5(5+2)} = \sqrt{35}\ B.M.$$

(d) $Ni^{2+} \rightarrow n = 2$

Careers360

$$\mu = \sqrt{2(2+2)} = \sqrt{8}\ B.M.$$

Hence, the answer is the option (3).

Q. 25 Which one of the following ions exhibits d-d transition and paramagnetism as well ?

Option 1:
MnO_4^-

Option 2:
$Cr_2O_7^{2-}$

Option 3:
CrO_4^{2-}

Option 4:
MnO_4^{2-}

Correct Answer:
MnO_4^{2-}

Solution:
MnO_4^{2-} has one unpaired electron so it can show a d-d transition and it is paramagnetic. Since it has one unpaired electron which can be shifted between the lower energy level and the higher energy level and thus a d-d transition exists.

Hence, the answer is the option (4).

Q. 26 Iron carbonyl, $Fe(CO)_5$ is

Careers360

Option 1:
trinuclear

Option 2:
mononuclear

Option 3:
tetranuclear

Option 4:
dinuclear

Correct Answer:
mononuclear

Solution:
Mononuclear Coordination entity - Contains a single central metal atom.

In $Fe(CO)_5$ has only one metal atom so it is a mononuclear complex.

Hence, the answer is the option (2).

Q. 27 The type of isomerism shown by the complex $[CoCl_2(en)_2]$ is

Option 1:
Ionization isomerism

Option 2:
Coordination isomerism

Option 3:
Geometrical isomerism

Option 4:
Linkage isomerism

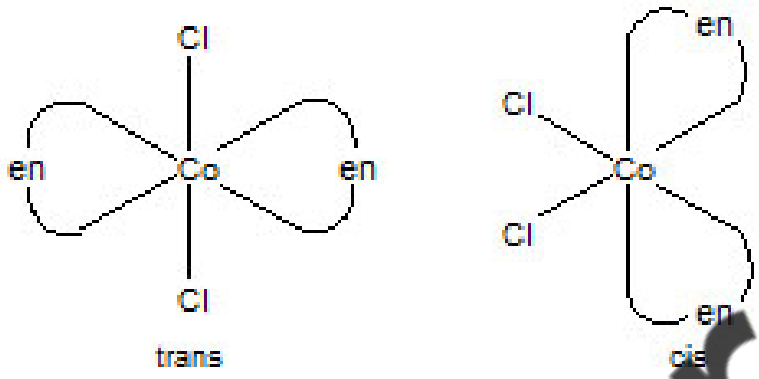
Careers360

Correct Answer:
Geometrical isomerism

Solution:
Geometrical Isomers -

This is due to the difference in the relative position of the ligands attached to the central metal atom with respect to each other.

$[CoCl_2(en)_2]$ shows geometrical isomerism.



The Correct answer is option 3.

Q. 28 The geometry and magnetic behaviour of the complex $[Ni(CO)_4]$ are:

Option 1:
square planar geometry and paramagnetic

Option 2:
tetrahedral geometry and diamagnetic

Option 3:
square planar geometry and diamagnetic

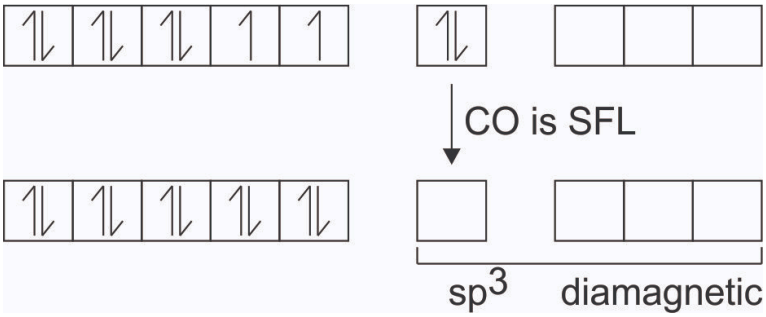
Option 4:
tetrahedral geometry and paramagnetic

Careers360

Correct Answer:
tetrahedral geometry and diamagnetic

Solution:

$Ni \Rightarrow 3d^8 4s^2$



Hence, option number (2) is correct.

Q. 29 The following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations :

$a. 60mL\ M/10\ HCl + 40mL\ M/10\ NaOH$
$b. 55mL\ M/10\ HCl + 45mL\ M/10\ NaOH$
$c. 75mL\ M/5\ HCl + 25mL\ M/5\ NaOH$
$d. 100mL\ M/10\ HCl + 100mL\ M/10\ NaOH$

pH of which one of them will be equal to 1 ?

Option 1:
d

Option 2:
a

Option 3:
b

Careers360

Option 4:

c

Correct Answer:

c

Solution:

The pH scale - Hydronium ion concentration in molarity is more conveniently expressed on a logarithmic scale known as the pH scale.

The p(H) of a solution is defined as the negative logarithm to base 10 of the activity of hydrogen ion

$$pH = -log[H^+]$$

As $N_1V_1 > N_2V_2$ so acid is left at the end of the reaction

For a

$$N_{mixture} = [H]^+ = \frac{N_1V_1 - N_2V_2}{V_1 + V_2} = \frac{\frac{1}{10} \times 60 - \frac{1}{10} \times 40}{60 + 40} = \frac{2}{100} = 2 \times 10^{-2}$$

$$pH = -log(2 \times 10^{-2}) = 1.7$$

For b

$$N_{mixture} = [H]^+ = \frac{N_1V_1 - N_2V_2}{V_1 + V_2} = \frac{\frac{1}{10} \times 55 - \frac{1}{10} \times 45}{55 + 45} = \frac{1}{100} = 10^{-2}$$

$$pH = -log(10^{-2}) = 2$$

For c

$$N_{mixture} = [H]^+ = \frac{N_1V_1 - N_2V_2}{V_1 + V_2} = \frac{\frac{1}{5} \times 75 - \frac{1}{5} \times 25}{75 + 25} = \frac{1}{10} = 10^{-1}$$

$$pH = -log10^{-1} = 1$$

For d

$$N_{mixture} = [H]^+ = \frac{N_1V_1 - N_2V_2}{V_1 + V_2} = \frac{\frac{1}{10} \times 100 - \frac{1}{10} \times 100}{100 + 100} = \frac{0}{200} = 0$$

$$pH = -log(10^{-7}) = 7$$

Careers360

Hence, the answer is the option (4).

Q. 30 On which of the following properties does the coagulating power of an ion depend?

Option 1:
Both magnitude and sign of the charge on the ion

Option 2:
Size of the ion alone

Option 3:
The magnitude of the charge on the ion alone

Option 4:
The sign of charge on the ion alone

Correct Answer:
Both magnitude and sign of the charge on the ion

Solution:
Properties of Colloidal Solutions -

Coagulation or Precipitation - The stability of the lyophobic sols is due to the presence of charge on colloidal particles. If somehow the charge is removed the particles will come nearer to each other and settle down under gravity.

According to Hardy Schulze's rule: The coagulating power of an ion depends on both the magnitude and sign of the charge on the ion.

Hence, option number (1) is correct.

Q. 31 Given van der Waals constant for NH_3, H_2, O_2, and CO_2 are respectively 4.17, 0.244, 1.36, and 3.59, which one of the following is most easily liquefied?

Careers360

Option 1:
O_2

Option 2:
H_2

Option 3:
NH_3

Option 4:
CO_2

Correct Answer:
NH_3

Solution:
As we know, real gases can be liquified by decreasing temperature or increasing pressure. The more the attraction between the particles of a certain gas, the more easily the gas can be liquified.

Vander Waal constant 'a' is a measure of attraction between the gases, the more attraction forces between the gases, more the ease of liquefication.\

hence, NH_3 is the most easily liquifiable gas.

Option 3 is correct.

Q. 32 The solubility of $BaSO_4$ in water is $2.42 \times 10^{-3} g L^{-1}$ at 298 K. The value of its solubility product (Ksp) will be
(Given molar mass of $BaSO_4 = 233 g mol^{-1}$)

Option 1:
$1.08 \times 10^{-14} mol^2 L^{-2}$

Careers360

Option 2:

$1.08 \times 10^{-12} mol^2 L^{-2}$

Option 3:

$1.08 \times 10^{-10} mol^2 L^{-2}$

Option 4:

$1.08 \times 10^{-8} mol^2 L^{-2}$

Correct Answer:

$1.08 \times 10^{-10} mol^2 L^{-2}$

Solution:

Solubility of $BaSO_4$ = $2.42 \times 10^{-3} g/L$

Molar mass of $BaSO_4$ = $233 g/mol$

$$\text{Molar solubility of } \mathrm{BaSO_4(s)} = \frac{2.42 \times 10^{-3}}{233}$$

$$\text{Molar solubility of } \mathrm{BaSO_4(s)} = 1.038 \times 10^{-5} \mathrm{mole/L}$$

$$\mathrm{K_{sp}} = \mathrm{s^2} = \left(1.038 \times 10^{-5}\right)^2 = 1.08 \times 10^{-10}\ \mathrm{mol^2/L^2}$$

Hence, option number (3) is correct.

Q. 33 In which case is the number of molecules of water maximum?

Option 1:

0.00224 L of water vapours at 1 atm and 273 K

Option 2:

0.18 g of water

Option 3:

18 mL of water

Careers360

Option 4:
10^{-3} mol of water

Correct Answer:
18 mL of water

Solution:
If the mass of water is maximum then the mole will be maximum hence number of molecules of water will be maximum.

Mass of solution = Density of Solution X Volume of Solution

A) 22.4 L of water vapors at 1 atm and 273 K = 18 g of water

0.00224 L of water vapors at 1 atm and 273 K $= 18 \times 10^{-1}$ g of water

B) 0.18 g of water

C) As density of water = 1 g/ml

and weight of water = volume of water

So 18 g = 18 ml

D) 1 mol of water =18 g

$10^{-3}\ \text{mol of water} = 18 \times 10^{-3}\ \text{g}$

So Option C has a maximum mass of water

which is 18 g of water which is 1 mole of water and contains molecules

Hence, the answer is the option (3).

Q. 34 The correct difference between first- and second-order reactions is that

Careers360

Option 1:
a first-order reaction can be catalyzed; a second-order reaction cannot be catalyzed.

Option 2:
the half-life of a first-order reaction does not depend on $[A]_o$; the half-life of a second-order reaction does depend on $[A]_o$.

Option 3:
the rate of a first-order reaction does not depend on reactant concentrations; the rate of a second-order reaction does depend on reactant concentrations

Option 4:
the rate of a first-order reaction does depend on reactant concentrations; the rate of a second-order reaction does not depend on reactant concentrations

Correct Answer:
the half-life of a first-order reaction does not depend on $[A]_o$; the half-life of a second-order reaction does depend on $[A]_o$.

Solution:
Both the first-order and second-order reactions can be catalyzed with the use of a suitable catalyst. For the first-order reaction, the half-life of a reaction is given by: $t_{\frac{1}{2}} = \frac{0.693}{k}$ while for the second-order reactions, it is $t_{\frac{1}{2}} = \frac{1}{kA_o}$.

In general, the half-life of an nth-order order reaction is $t_{\frac{1}{2}} = \frac{1}{[A_o]^{n-1}}$

Hence, For the first-order reaction, the half-life is independent of the initial concentration while for the second-order reaction, it is inversely proportional to it.

For the first-order reactions, $Rate = k\,[A]$, hence depends upon the concentration of reactant to the power 1 while for second-order reactions $Rate = k\,[A]^2$.

Hence, the answer is the option (2).

Careers360

Q. 35 Among CaH_2, BeH_2, BaH_2, the order of ionic character is

Option 1:
$BeH_2 < BaH_2 < CaH_2$

Option 2:
$CaH_2 < BeH_2 < BaH_2$

Option 3:
$BeH_2 < CaH_2 < BaH_2$

Option 4:
$BaH_2 < BeH_2 < CaH_2$

Correct Answer:
$BeH_2 < CaH_2 < BaH_2$

Solution:
Fajans Rules - It accounts for the covalent character in ionic compounds covalency is favored by

1. Small cation
2. Large anion and
3. Large charge on either ion

The smaller the size of the cation, the more will be its polarising power. Hence $BeCl_2$ will be least ionic.

Hence, the answer is the option (3).

Careers360

Q. 36 Consider the change in oxidation state of Bromine corresponding to different emf values as shown in the diagram below :

$$BrO_4^- \xrightarrow{1.82V} BrO_3^- \xrightarrow{1.5V} HBrO \xrightarrow{1.595V} Br_2 \xrightarrow{1.0652V} Br^-$$

Then the species undergoing disproportionation is

Option 1:
Br_2

Option 2:
BrO_4^-

Option 3:
BrO_3^-

Option 4:
$HBrO$

Correct Answer:
$HBrO$

Solution:
Calculate E^o_{cell} corresponding to each compound undergoing disproportionation reaction. The reaction for which E^o_{cell} comes out +ve is spontaneous.

$$HBrO \rightarrow Br_2 \;\; E^o = 1.595V \text{ (Cathode)}$$

$$HBrO \rightarrow BrO_3^- \;\; E^o = -1.5V \text{ (Anode)}$$

The reaction will be

$$2HBrO \rightarrow Br_2 + BrO_3^-$$

$$E^o_{cell} = E^o_{Cathode} - E^o_{Anode} = 1.595 - 1.5 = 0.095V$$

we know

Careers360

$E^{o}_{cell} > 0, \Delta G^{o} < 0, \text{spontaneous process}$

So, HBrO will go under disproportionation reaction.

Hence, the answer is the option (4).

Q. 37 For the redox reaction

$$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$$

the correct coefficients of the reactants for the balanced equation are

Option 1:

MnO_4^-	$C_2O_4^{2-}$	H^+
2	16	5

Option 2:

MnO_4^-	$C_2O_4^{2-}$	H^+
2	5	16

Option 3:

MnO_4^-	$C_2O_4^{2-}$	H^+
16	5	2

Careers360

Option 4:

MnO_4^-	$C_2O_4^{2-}$	H^+
5	16	2

Correct Answer:

MnO_4^-	$C_2O_4^{2-}$	H^+
2	5	16

Solution:
Balancing of Redox Reactions - Oxidation number method

1. Write the correct formula for each reactant and product.

2. Identify the atom whose oxidation number is changed.

3. Calculate the increase/decrease in oxidation number per atom and for the entire molecule/ion in which it occurs.

4. Add H^+/OH^- on the appropriate side so that the total ionic charges of reactants and products are equal.

5. Make the numbers of hydrogen atoms in the expression on the two sides equal by adding H_2O to the reactant or product

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

Hence, the answer is the option (2).

Careers360

Q. 38 Which one of the following conditions will favor maximum formation of the product in the reaction,

$$A_2(g) + B_2(g) \rightleftharpoons X_2(g) \Delta_r H = -X kJ\,?$$

Option 1:
High temperature and high pressure

Option 2:
Low temperature and low pressure

Option 3:
Low temperature and high pressure

Option 4:
High temperature and low pressure

Correct Answer:
Low temperature and high pressure

Solution:
Effect of Pressure Change - A pressure change obtained by changing the volume can affect the yield of the product in the case of a gaseous reaction where the total number of moles of gaseous reactant and the total number of moles of the gaseous product are different.

Effect of temperature change - When a temperature change occurs, the value of equilibrium constant K_c is changed.

For reaction $\Delta H = -ve \; and \; \Delta n_g = -ve$

$\therefore$ High pressure and low temperature favor product formation.

Hence, the answer is the option (3).

Careers360

Q. 39 When initial concentration of the reactant is doubled, the half-life period of a zero order reaction

Option 1:
is tripled

Option 2:
is doubled

Option 3:
is halved

Option 4:
remains unchanged

Correct Answer:
is doubled

Solution:
Zero order reaction means that the rate of the reaction is proportional to zeroth power of the concentration of reactants

$$R \rightarrow P$$

$$Rate = \frac{-d(R)}{dt} = K[R]^0 = K$$

$t_{\frac{1}{2}}$ = time required for half completion of the reaction $= \frac{A_o}{2k}$

$$\left(t_{\frac{1}{2}}\right)_{zero} = \frac{[A]_o}{2k}$$

$\therefore$ If $[A]_o$ is doubled then $t_{\frac{1}{2}}$ also doubled.

Hence, the answer is the option (2).

Careers360

Q. 40 The bond dissociation energies of X_2, Y_2 and XY are in the ratio of 1 : 0.5 : 1. ΔH for the formation of XY is -200 kJ mol^{-1}. The bond dissociation energy of X_2 will be

Option 1:
800 kJ mol^{-1}

Option 2:
100 kJ mol^{-1}

Option 3:
200 kJ mol^{-1}

Option 4:
400 kJ mol^{-1}

Correct Answer:
800 kJ mol^{-1}

Solution:
Bond dissociation enthalpy - It is the average enthalpy required to dissociate the said bond present in different gaseous compounds into free atoms in a gaseous state.

$$N_2 + Bond\,Energy \rightarrow 2N$$

Careers360

$$\frac{1}{2}X_2 + \frac{1}{2}Y_2 \rightarrow XY,\ \Delta H = -200\ KJ \cdot mol^{-1}$$

$$\Delta H = -200 = \Sigma Bond\ Energy\ Reactants - \Sigma Bond\ Energy\ Products$$

$$\Delta H = \frac{1}{2} \times B.E\ of\ X_2 + \frac{1}{2} \times B.E\ of\ Y_2 - 1 \times B.E\ of\ XY$$

$$\Delta H = \frac{1}{2} \times B.E\ of\ X_2 + \frac{1}{2} \times B.E\ of\ X_2 \times 0.5 - 1 \times B.E\ of\ X_2$$

$$-200 = \frac{3}{4} \times B.E\ of\ X_2 - B.E\ of\ X_2$$

$$-200 = -\frac{1}{4} \times B.E\ of\ X_2$$

$$\Rightarrow B.E\ of\ X_2 = 800\ KJ \cdot mol^{-1}$$

Hence, the answer is the option (1).

Q. 41 The correction factor 'a' to the ideal gas equation corresponds to

Option 1:
electric held present between the gas molecules

Option 2:
the volume of the gas molecules

Option 3:
density of the gas molecules

Option 4:
forces of attraction between the gas molecules

Careers360

Correct Answer:
forces of attraction between the gas molecules

Solution:
Vander Waal's constant 'a' is a measure of the intermolecular attraction of a gas

Vander Waal's constant (a) α force of attraction.

Hence, option 4 is correct.

Q. 42 Consider the following species :

CN^+, CN^-, NO, and CN

Which one of these will have the highest bond order?

Option 1:
CN^+

Option 2:
CN^-

Option 3:
NO

Option 4:
CN

Correct Answer:
CN^-

Solution:

Careers360

Bond Order - Bond order is defined as one-half the difference between the number of electrons present in the bonding and the antibonding orbitals.

$$Bond\,Order = \frac{N_b - N_a}{2}$$

Ion/Species	Total electron	Bond Order
NO	15	2.5
CN^-	14	3.0
CN^+	12	2.0
CN	13	2.5

Hence, the answer is the option (2).

Q. 43 Magnesium reacts with an element (X) to form an ionic compound. If the ground state electronic configuration of (X) is $1s^2\ 2s^2\ 2p^3$, the simplest formula for this compound is

Option 1:
Mg_2X

Option 2:
MgX_2

Option 3:
Mg_2X_3

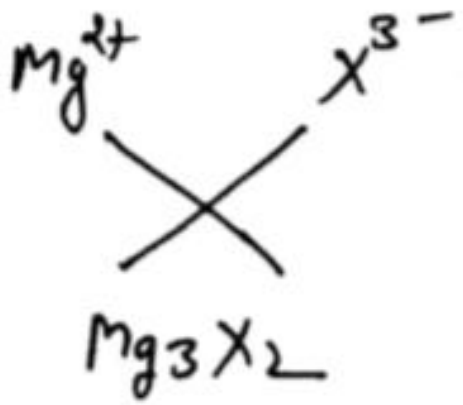
Option 4:
Mg_3X_2

Correct Answer:
Mg_3X_2

Solution:
Mg ion $\Rightarrow$ Mg^{2+} , X = N

Nitrogen ion = X^{3-}

Careers360



Option (4) is correct answer.

Q. 44 Iron exhibits bcc structure at room temperature. Above 900°C, it transforms to fcc structure. The ratio of density of iron at room temperature to that at 900°C (assuming molar mass and atomic radii of iron remains constant with temperature) is

Option 1:
$\frac{3\sqrt{3}}{4\sqrt{2}}$

Option 2:
$\frac{4\sqrt{3}}{3\sqrt{2}}$

Option 3:
$\frac{\sqrt{3}}{\sqrt{2}}$

Option 4:
$\frac{1}{2}$

Correct Answer:
$\frac{3\sqrt{3}}{4\sqrt{2}}$

Careers360

Solution:
As we learned in

The density of cubic unit cell -

$$d = \frac{zM}{a^3 N_A}$$

Where,

d = density of crystal

z = no. of effective constituent particles in one unit cell

M = molecular weight

a = edge length of the unit cell

$N_A = 6.022 \text{ X } 10^{23}$

Now, the Molar mass and atomic radii of iron remains constant with temperature, But edge length in bcc will be different from fcc. So need the ratio between the edge length and atomic radius for both structures due to radius is constant with temperature.

$$\text{For BCC}, Z_{bcc} = 2 \text{ and } 4r = \sqrt{3}a_{bcc}$$

$$d_{bcc} = \frac{Z_{bcc} M}{a_{bcc}^3 N_A}$$

$$d_{bcc} = \frac{2 \times M}{\left(\frac{4r}{\sqrt{3}}\right)^3 N_A}$$

$$\text{For FCC}, Z_{fcc} = 4 \text{ and } 4r = \sqrt{2}a_{fcc}$$

$$d_{fcc} = \frac{Z_{fcc} M}{a_{fcc}^3 N_A}$$

Careers360

$$d_{fcc} = \frac{4 \times M}{\left(\frac{4r}{\sqrt{2}}\right)^3 N_A}$$

Now the ratio will be -

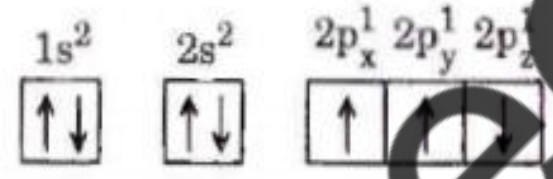
$$\frac{d_{bcc}}{d_{fcc}} = \frac{2 \times (\frac{4r}{\sqrt{2}})^3}{(\frac{4r}{\sqrt{3}})^3 \times 4}$$

$$\frac{d_{bcc}}{d_{fcc}} = \frac{3}{4}\sqrt{\frac{3}{2}}$$

Hence, the option number (1) is correct.

Q. 45 Which one is a wrong statement ?

Option 1:
The electronic configuration of N atom is



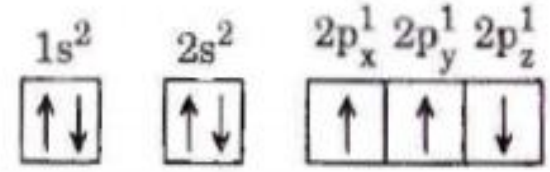
Option 2:
An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers.

Option 3:
Total orbital angular momentum of electron in 's' orbital is equal to zero.

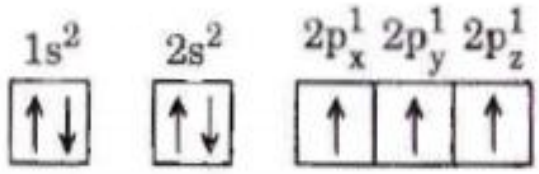
Option 4:
The value of m for d_{z^2} is zero.

Careers360

Correct Answer:
The electronic configuration of N atom is



Solution:
The correct configuration of N is



Therefore, the correct option is (1).

Biology

Q. 1 Oxygen is **not** produced during photosynthesis by

Option 1:
Cycas

Option 2:
Nostoc

Option 3:
Green sulphur bacteria

Option 4:
Chara

Correct Answer:
Green sulphur bacteria

Solution:

Careers360

As we learnt in

Photosynthesis -

Enzyme regulated anabolic process of manufacture of organic compoounds inside the chlorophyll containing cells.

- wherein

$$6CO_2 + 12H_2O \overset{Sunlight}{\rightarrow} C_6H_{12}O_6 + 6H_2O + 6O_2 \uparrow$$

Since Charan Cycas comes under plant kingdom so they perform oxygenic photosynthesis. Green sulphur bacteria utilises H_2S instead of H_2O and performs Anoxygenic Photosynthesis or do not evolve oxygen.

Q. 2 Double fertilization is

Option 1:
Fusion of two male gametes with one egg

Option 2:
Fusion of one male gamete with two polar nuclei

Option 3:
Fusion of two male gametes of a pollen tube with two different eggs

Option 4:
Syngamy and triple fusion

Correct Answer:
Syngamy and triple fusion

Solution:
As we learnt in

Primary Endosperm Cell (PEC) -

Careers360

Since two types of fusions, syngamy and triple fusion take place in an embryo sac the phenomenon is termed double fertilisation, an event unique to flowering plants. The central cell after triple fusion becomes the primary endosperm cell (PEC) and develops into the endosperm while the zygote develops into an embryo.

- There are two types of fusions, syngamy and triple fusion take place in an embryo sac the phenomenon is termed double fertilization.

Q. 3 Which one of the following plants shows a very close relationship with a species of moth , where none of the two can complete its life cycle without the other ?

Option 1:
Banana

Option 2:
Yucca

Option 3:
Hydrilla

Option 4:
Viola

Correct Answer:
Yucca

Solution:

As we learnt in

Mutualism -

This interaction confers benefits on both the interacting species.

Yucca have an obligate mutualism with a species of moth i.e. Pronuba.

Careers360

obligate mutualism means i.e., one organism cannot survive without the other.

Q. 4 Pollen grains can be stored for several years in liquid nitrogen having a temperature of

Option 1:
-196^{o}C

Option 2:
-80^{o}C

Option 3:
-120^{o}C

Option 4:
-160^{o}C

Correct Answer:
-196^{o}C

Solution:
As we learnt in

Storing Pollen grains -

It is possible to store pollen grains of a large number of species for years in liquid nitrogen (-1960C). Such stored pollen can be used as pollen banks

Pollen grains can be stored for several years in liquid nitrogen at –196°C.

This process is known as Cryo-preservation or cryo-conservation.

Hence, the correct option is (a).

Careers360

Q. 5 Which of the following elements is responsible for maintaining turgor in cells?

Option 1:
Potassium

Option 2:
Sodium

Option 3:
Magnesium

Option 4:
Calcium

Correct Answer:
Potassium

Solution:

-

Potassium is required for the maintenance of the turgidity of guard cells. In the guard cells, starch is converted into malic acid in the presence of light (during daytime).
Protons formed are used by the guard cells for the uptake of K+ ions. In this way, the concentration of K+ ions increases in guard cells. The potassium combines with malate to form potassium malate.
This increases the solute concentration of the guard cells. The guard cell withdraws water from the nearby cells and becomes turgid.

Careers360

Hence, the correct answer is potassium.

Q. 6 What is the role of NAD^+ in cellular respiration ?

Option 1:
It is a nucleotide source for ATP synthesis.

Option 2:
It functions as an electron carrier.

Option 3:
It functions as an enzyme.

Option 4:
It is the final electron acceptor for anaerobic respiration.

Correct Answer:
It functions as an electron carrier.

Solution:

-

At various chemical reactions, the NAD+ picks up an electron from glucose, at which point it becomes NADH. Then NADH, along with another molecule flavin adenine dinucleotide ($FADH_2$) will ultimately transport the electrons to the mitochondria, where the cell can harvest energy stored in the electrons.

Hence It functions as an electron carrier.

Careers360

Q. 7 In which of the following forms is iron absorbed by plants ?

Option 1:
Free element

Option 2:
Ferrous

Option 3:
Ferric

Option 4:
Both ferric and ferrous

Correct Answer:
Ferric

Solution:
As we learnt in

Iron Function -

Component of ferredoxin, cytochromes. Activates catalase, formation of chlorophyll.

- wherein

Deficiency - Interveinal chlorosis, green netting of citrus.

Iron is absorbed by plants in the form of ferric ions. Plants absorb iron in both form i.e. Fe++ and Fe+++. (Preferably Fe++)

Q. 8 Which of the following is commonly used as a vector for introducing a DNA fragment in human lymphocytes ?

Careers360

Option 1:
λ phage

Option 2:
Ti plasmid

Option 3:
Retrovirus

Option 4:
pBR 322

Correct Answer:
Retrovirus

Solution:
As we learnt in

Gene Therapy -

If a person is diagnosed to carry a defective gene leading to disability, one may like to get this gene replaced by a normal functional or may like to get this gene repaired called gene therapy.

- wherein

1st gene therapy in animal was carried out in milk, to correct the dwarfness by introducing growth hormone gene.

Gene therapy : Lymphocyte from blood of patient are grown in culture outside the body, a functional gene is introduced by using a retroviral vector, into these lymphocytes. Retrovirus is commonly used as vector for introducing a DNA fragment in human lymphocyte.

Careers360

Q. 9 Use of bioresources by multinational companies and organisations without authorisation from the concerned country and its people is called

Option 1:
Biodegradation

Option 2:
Biopiracy

Option 3:
Bio-infringement

Option 4:
Bioexploitation

Correct Answer:
Biopiracy

Solution:
As we learnt in

Biopiracy -

Use of bioresources by multi-national companies and other organisation without proper authorization from the countries and people concerned without compensatary payment.

-

Biopiracy is term used for or refer to the use of bioresources by multinational companies and other organisation without proper authorisation from the countries and people concerned with compensatory payment.

Q. 10 In India , the organisation responsible for assessing the safety of introducing genetically modified organisms for public use is

Careers360

Option 1:
Research Committee on Genetic Manipulation (RCGM)

Option 2:
Council for Scientific and Industrial Research (CSIR)

Option 3:
Indian Council of Medical Research (ICMR)

Option 4:
Genetic Engineering Appraisal Committee (GEAC)

Correct Answer:
Genetic Engineering Appraisal Committee (GEAC)

Solution:
As we learnt in

Genetically modified crops (GMOs) -

Are transgenic crops which contain and express one or more useful foreign genes or transgenes. The food which is prepared by GMC is called genetically modified food.

- wherein

Ex. BT Cotton, Flavre-Savre Tomato.

Transgenic plants also known as bioreactors or factories to produce specific products.

Indian Government has setup organisation such as GEAC (Genetic Engineering Appraisal Committee) which will make decisions regarding the validity of GM research and safety of introducing GM-organism for public services.

Q. 11 The correct order of steps in Polymerase Chain Reaction (PCR) is

Option 1:
Denaturation , Extension , Annealing

Careers360

Option 2:
Annealing , Extension , Denaturation

Option 3:
Extension , Denaturation , Annealing

Option 4:
Denaturation , Annealing , Extension

Correct Answer:
Denaturation , Annealing , Extension

Solution:
As we learnt in

PCR -

The polymerase chain Reaction (PCR) was invented by Kary Mullis (1985). In PCR, Template DNA, Oligonucleotides or primers, Taq polymerase and dNTP, are used to amplify DNA

- wherein

PCR is used to amplify (many copies) targeted sequence

Polymerase Chain Reaction (PCR), technique is used for making multiple copies of gene (or DNA) of interest in vitro. Each cycle has three steps

(i) Denaturation

(ii) Primer annealing

(iii)Extension of primer

Careers360

Q. 12 Select the correct match :

Option 1:
T.H. Morgan - Transduction

Option 2:
F_2 X Recessive parent - Dihybrid cross

Option 3:
Ribozyme - Nucleic acid

Option 4:
G. Mendel - Transformation

Correct Answer:
Ribozyme - Nucleic acid

Solution:
As we learnt in

Ribozymes -

The nucleic acids that behave like enzymes are called ribozymes.

-

Ribozyme is a catalytic RNA, which is nucleic acid.

Q. 13 A 'new' variety of rice was patented by a foreign company , though such varieties have been present in India for a long time . This is related to

Option 1:
Lerma Rojo

Option 2:
Sharbati Sonora

Careers360

Option 3:
Co - 667

Option 4:
Basmati

Correct Answer:
Basmati

Solution:
As we learnt in

Green Revolution -

The novel technological development of the Green Revolution was the production of novel wheat cultivars.

- wherein

Agronomists bred cultivars of maize, wheat, and rice that are generally referred to as HYVs or "high-yielding varieties”

Sharbati Sonora and Lerma Rojo are varieties of wheat. Basmati is the variety of rice on which got patented by an American company

through the US patent and trademark office that was actually been derived from Indian farmer’s varieties in 1997.

Q. 14 Which of the following pairs is **wrongly** matched ?

Option 1:
XO type sex determination : Grasshopper

Option 2:
ABO blood grouping : Co - dominance

Careers360

Option 3:
Starch synthesis in pea : Multiple alleles

Option 4:
T.H. Morgan : Linkage

Correct Answer:
Starch synthesis in pea : Multiple alleles

Solution:
As we learnt in

Alleles -

Genes which code for a pair of contrasting traits are known as alleles.

- wherein

They are slightly different forms of the same gene.

Starch synthesis in pea is controlled by pleiotropic gene. Pleiotropic gene has a multiple phenotypic effect because it can control expression of a number of characters.

Q. 15 Select the correct statement :

Option 1:
Spliceosomes take part in translation .

Option 2:
Punnett square was developed by a British scientist .

Option 3:
Franklin Stahl coined the term "Linkage" .

Option 4:
Transduction was discovered by S. Altman.

Careers360

Correct Answer:
Punnett square was developed by a British scientist .

Solution:
As we learnt in

Punnett square -

The production of gametes by the parents , the formation of the Zygotes , the F_1 and F_2 plants can be understod by a punnett square

- wherein

It was developed a british geneticist reginald C.Punnett.

Punnett square was developed by a British geneticist, Reginald C. Punnett.

– Franklin Stahl proved semi-conservative mode of replication.

– Transduction was discovered by Zinder and Laderberg.

Careers360

Q. 16 The experimental proof for semiconservative replication of DNA was first shown in a

Option 1:
Plant

Option 2:
Bacterium

Option 3:
Fungus

Option 4:
Virus

Correct Answer:
Bacterium

Solution:
As we learnt in

Observations of experiments done by meselsons and stahl -

The DNA that was extracted from the culture of E.coli bacteria one generation after the transfer from 15 N to 14 N medium had a hybrid or intermediate density or DNA extracted another generation (40 minutes) was composed of an equal amount of this hybrid DNA

-

Semi-conservative DNA replication was first shown in Bacterium Escherichia coli by Matthew Meselson and Franklin Stahl.

Q. 17 Which of the following flowers only once in its life-time ?

Option 1:
Mango

Option 2:
Jackfruit

Option 3:
Bamboo species

Option 4:
Papaya

Careers360

Correct Answer:
Bamboo species

Solution:
As we learnt in

Unusual flowering phenomenon -

Bamboo species flower only once in their life time, generally after 50-100 years, produce large number of fruits and die. Another plant, Strobilanthus kunthiana (neelakuranji), flowers once in 12 years. This plant flowered during September-October 2006.

-

Bamboo species are monocarpic i.e., it flowers generally only once in its life-time after 50-100 years. Jackfruit, papaya and mango are polycarpic i.e., produce flowers and fruits many times in their life-time.

Q. 18 Offsets are produced by

Option 1:
Parthenocarpy

Option 2:
Mitotic divisions

Option 3:
Meiotic divisions

Option 4:
Parthenogenesis

Correct Answer:
Mitotic divisions

Careers360

Solution:
As we learnt in

Vegetative reproduction -

In plants, the units of vegetative propagation such as runner, rhizome, sucker, tuber, offset, bulb are all capable of giving rise to new offspring. These structures are called vegetative propagules.

Offset is a vegetative part of a plant, formed by mitosis in the somatic cells. Meiotic divisions do not occur in somatic cells. Reproduction occurring through offset is therefore called vegetative propagation

---

Q. 19 Select the correct match :

Option 1:
Matthew Meselson and F. Stahl - *Pisum sativum*

Option 2:
Alfred Hershey and Martha Chase - TMV

Option 3:
Alec Jeffreys - *Streptococcus pneumoniae*

Option 4:
Francois Jacob and Jacques Monod - *Lac* operon

Correct Answer:
Francois Jacob and Jacques Monod - *Lac* operon

Solution:
As we learnt in

The lac operon -

In lac operon, a polycistronic structural gene is regulated by a common promoter and regulatory genes.

Careers360

- wherein

Such arrangements are very common in bacteria and is referred to as operon.

Francois Jacob and Jacque Monod proposed model of gene regulation known as operon model/lac operon.

Alec Jeffreys – DNA fingerprinting technique.

Matthew Meselson and F. Stahl – Semiconservative DNA replication in E. coli.

Alfred Hershey and Martha Chase – Proved DNA as genetic material not protein

---

Q. 20 Which of the following has proved helpful in preserving pollen as fossils ?

Option 1:
Oil content

Option 2:
Cellulosic intine

Option 3:
Pollenkitt

Option 4:
Sporopollenin

Correct Answer:
Sporopollenin

Solution:
As we learnt in

Structure of pollen grains -

Careers360

The hard outer layer called the exine is made up of sporopollenin which is one of the most organic material known. It can withstand high temperatures and strong acids and alkali. No enzyme that degrades sporopollenin is so far known. Pollen grains exine has prominent apertures called germ pores where sporopollenin is absent.

- wherein

Pollen grains are well preserved as fossils because of the presence of sporopollenin.

Sporopollenin cannot be degraded by enzyme; strong acids and alkali, therefore it is helpful in preserving pollen as fossil.

Pollenkitt – Help in insect pollination.

Cellulosic Intine – Inner sporoderm layer of pollen grain known as intine made up cellulose & pectin.

Oil content – No role is pollen preservation.

Q. 21 Natality refers to

Option 1:
Number of individuals leaving the habitat

Option 2:
Birth rate

Option 3:
Death rate

Option 4:
Number of individuals entering a habitat

Correct Answer:
Birth rate

Careers360

Solution:
Natality -

Refers to the number of births during a given period in the population that are added to the initial density.

-

Q. 22 World ozone day is celebrated on

Option 1:
16th September

Option 2:
21st April

Option 3:
5th June

Option 4:
22nd April

Correct Answer:
16th September

Solution:

As learnt in The Earth Summit -

World Ozone day is celebrated on 16th September.

Hence, the correct option is (a).

Careers360

Q. 23 Which of the following is a secondary pollutant ?

Option 1:
SO_2

Option 2:
CO_2

Option 3:
CO

Option 4:
O_3

Correct Answer:
O_3

Solution:

As we have already discussed in Description of Air Pollutants from Man made Sources -

Secondary Air pollutants :

Secondary air pollutants are poisonous substance formed from primary air pollutants. In bright sun light nitrogen, nitrogen oxides, hydrocarbons and O_2 interact to produce more powerful photochemical oxidants like ozone (O_3), peroxyacetyl nitrate (PAN), aldehydes, sulphuric acid, peroxides, etc.

- wherein

All these constitute photochemical smog, which retard photosynthesis in plants.

O_3 (ozone) is a secondary pollutant. These are formed by the reaction of primary pollutant.

Careers360

Hence, the correct option is (d).

Q. 24 Niche is

Option 1:
the range of temperature that the organism needs to live

Option 2:
the physical space where an organism lives

Option 3:
all the biological factors in the organism's environment

Option 4:
the functional role played by the organism where it lives

Correct Answer:
the functional role played by the organism where it lives

Solution:
As we learnt in

Importance of Species diversity to the Ecosystem -

Biodiversity is important because it helps in preventing the extinction of species. It allows the organisms to adapt according to the environment and provide wider range of material and food for survival. It is the biodiversity that provides vast genetic pools that helps in preserving existence of life on earth.

- wherein

Biodiversity helps in maintaining equilibrium in the environment and also defines the manner in which different organisms interact with each other. It helps in maintaining the balance by recycling and storage of nutrients, stabilizing climate, forming and protecting soil, combating pollution and thereby maintaining eco-balance. It is threatened by climatic change, overexploitation of natural resources,

Careers360

habitat loss and pollution. Loss of biodiversity weakens the connection between various species in the ecosystem.

Ecological niche was termed by J. Grinnel. It refers the functional role played by the organism where it lives.

Q. 25 What type of ecological pyramid would be obtained with the following data ?

Secondary consumer : 120 g

Primary consumer : 60 g

Primary producer : 10 g

Option 1:
Upright pyramid of numbers

Option 2:
Pyramid of energy

Option 3:
Inverted pyramid of biomass

Option 4:
Upright pyramid of biomass

Correct Answer:
Inverted pyramid of biomass

Solution:
As we learnt in

The pyramid of biomass -

Biomass refers to the total weight of living matter per unit area. In an ecosystem, the biomass decreases from the producer level to the consumer level.

Careers360

-

• The given data depicts the inverted pyramid of biomass, usually found in aquatic ecosystem.

• Pyramid of energy is always upright

• Upright pyramid of biomass and numbers are not possible, as the data depicts primary producer is less than primary consumer and this is less than secondary consumers.

Q. 26 In stratosphere , which of the following elements acts as a catalyst in degradation of ozone and release of molecular oxygen ?

Option 1:
Fe

Option 2:
Cl

Option 3:
Carbon

Option 4:
Oxygen

Correct Answer:
Cl

Solution:

As we have already discussed in Ozone Depletion in the Stratosphere -

- CFCs discharged in the lower part of the atmosphere move upward and reach the stratosphere.
- In the stratosphere, UV rays act on them, releasing Cl atoms.
- Cl degrades ozone releasing molecular oxygen, with these atoms acting merely as catalysts; Cl atoms are not consumed in the reaction.

Careers360

- Hence, whatever CFCs are added to the stratosphere, they have permanent and continuing effects on Ozone.

Hence, the correct option is (b).

Q. 27 The two functional groups characteristic of sugars are

Option 1:
Carbonyl and phosphate

Option 2:
Carbonyl and methyl

Option 3:
hydroxyl and methyl

Option 4:
Carbonyl and hydroxyl

Correct Answer:
Carbonyl and hydroxyl

Solution:

Sugar is basically a carbohydrate and carbohydrates are polyhydroxy aldehyde, ketone or their derivatives, which means they have carbonyl and hydroxyl groups. All sugars are characterized by two functional groups such as carbonyl (C=O) and hydroxyl (C−OH).

Q. 28 Which among the following is not a prokaryote ?

Careers360

Option 1:
Nostoc

Option 2:
Mycobacterium

Option 3:
Saccharomyces

Option 4:
Oscillatoria

Correct Answer:
Saccharomyces

Solution:
As we learnt in

Prokaryotic Cells -

These are mainly represented by bacteria, blue-green algae, mycoplasma and PPLO (Pleuro Pneumonia Like Organisms).

-

Saccharomyces are yeast which is an eukaryote (unicellular fungi)

Mycobacterium – a bacterium

Oscillatoria and Nostoc are cyanobacteria.

Q. 29 The Golgi complex participates in :

Option 1:
Respiration in bacteria

Careers360

Option 2:
Formation of secretory vesicles

Option 3:
Fatty acid breakdown

Option 4:
Activation of amino acid

Correct Answer:
Formation of secretory vesicles

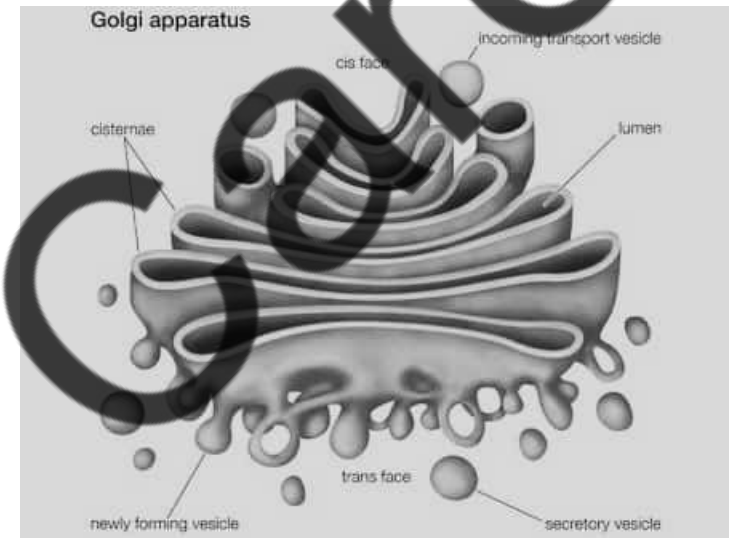
Solution:
As we learnt in

Golgi Apparatus -

Principally performs the function of packaging materials, to be delivered either to intracellular targets or secreted outside of the cell.

- wherein

It is the important site for the formation of glycoproteins and glycolipids.



Golgi complex, after processing releases secretory vesicles from their trans-face.

Careers360

Q. 30 Which of the following is **not** a product of light reaction of photosynthesis ?

Option 1:
NADPH

Option 2:
NADH

Option 3:
ATP

Option 4:
Oxygen

Correct Answer:
NADH

Solution:
As we learnt in

Glycolysis or EMP pathuray -

Occurs in cytoplasm, common to both aerobic and anaerobic respiration. One molecule of glusose is reduced to two molecules of pyruvate.

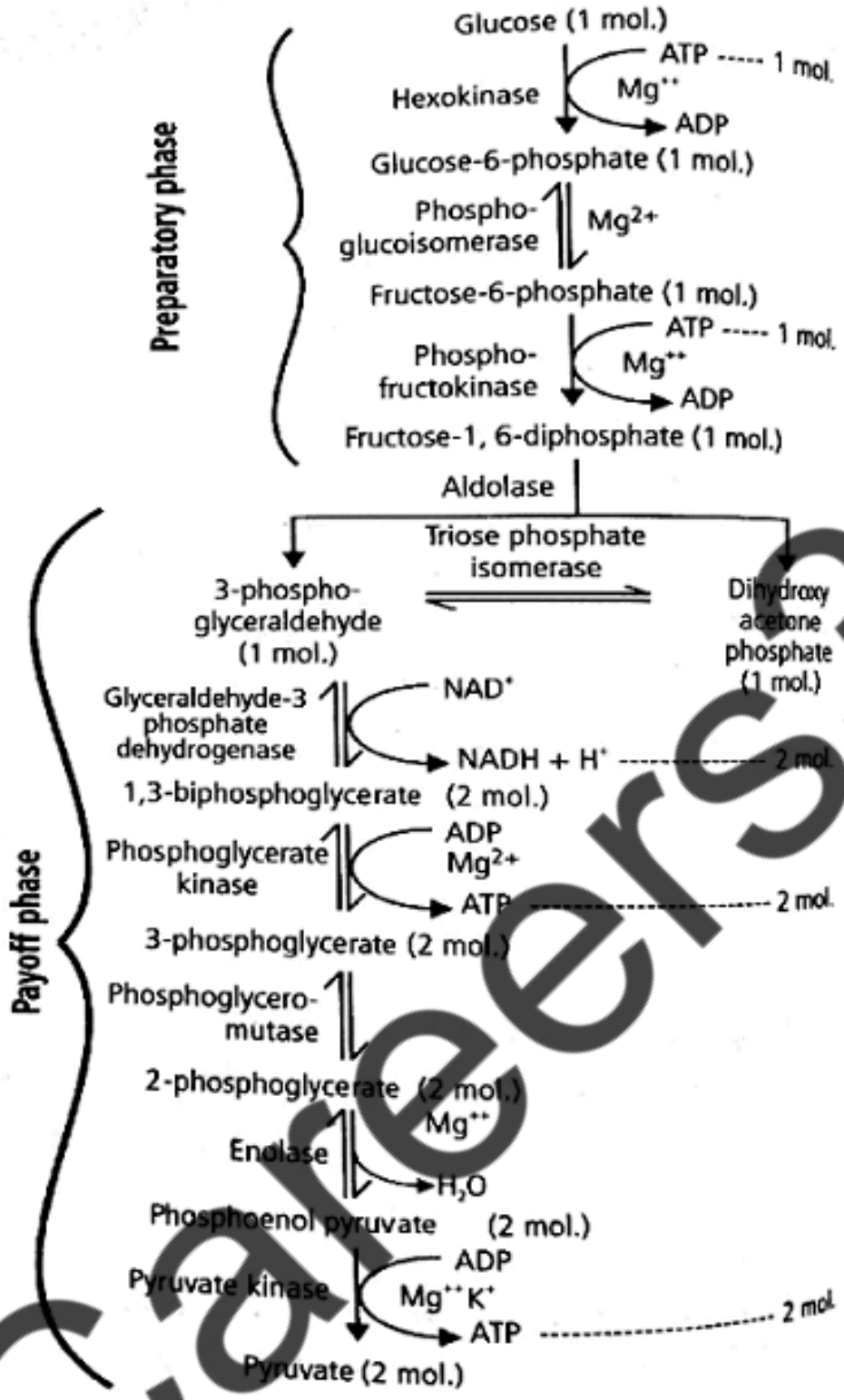
$Glucose + 2NAD^{+} + 2ADP + 2H_3PO_4 \longrightarrow 2Pyruvate + 2NADH + 2H^{+} + 2ATP$

Net gain of ATP = 2ATP + 6ATP (from 2NADH) = 8ATP

- wherein

Given by Embden, Mayerhof and Parnas

Careers360



NADH is a product of respiration process.

Q. 31 Which of the following is true for nucleolus ?

Careers360

Option 1:
It takes part in spindle formation.

Option 2:
It is a membrane - bound structure.

Option 3:
Larger nucleoli are present in dividing cells.

Option 4:
It is a site for active ribosomal RNA synthesis.

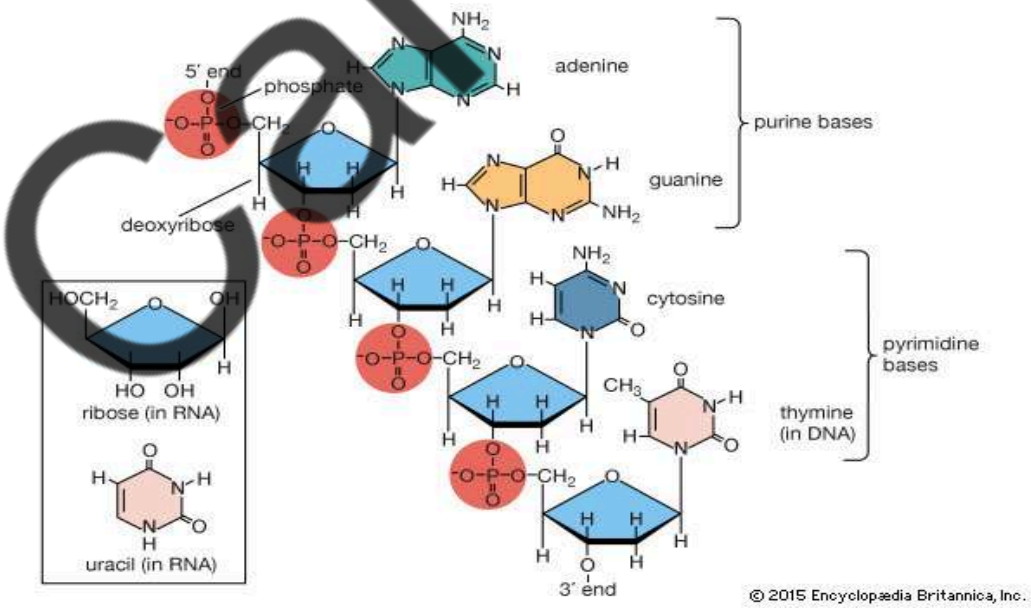
Correct Answer:
It is a site for active ribosomal RNA synthesis.

Solution:
As we learnt in

Nucleic Acids -

Nucleic Acids like DNA, RNA contains only Nucleotides and function as genetic material.

- wherein



Nucleolus is a non membranous structure and is a site of r-RNA synthesis.

Careers360

Q. 32 Stomatal movement is not affected by

Option 1:
O_2 concentration

Option 2:
Light

Option 3:
Temperature

Option 4:
CO_2 concentration

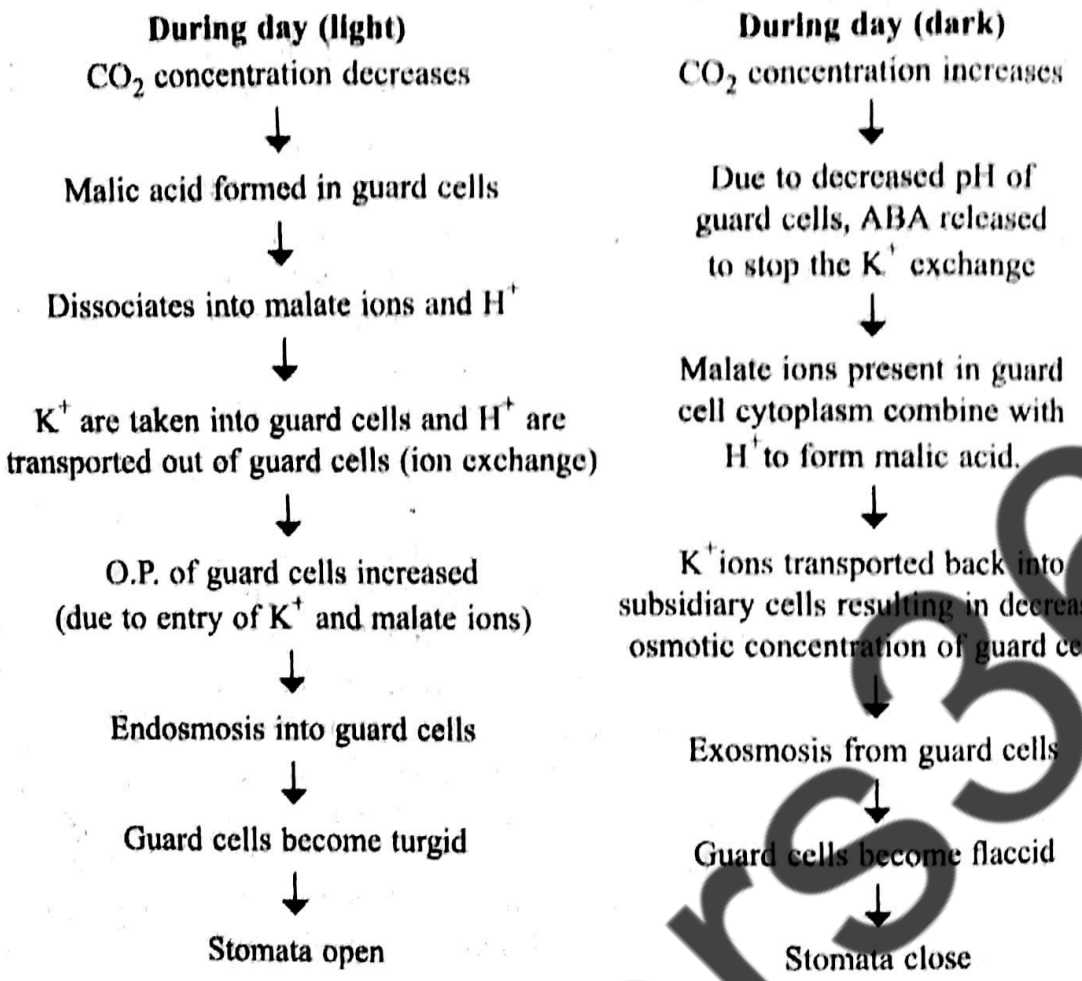
Correct Answer:
O_2 concentration

Solution:
As we learnt in

Active K+ transport theory -

This is the most accepted theory of stomatal movements.

Careers360



- wherein

Given by Fujino (1959) and modifies by Levitt (1974).

Stomata close when it is dark and dry, If carbon dioxide concentrations inside the leaf start to fall, the plant will open its stomata so that more CO_2 can enter.

Thus, Light, temperature and concentration of CO_2 affect opening and closing of stomata while they are not affected by O_2 concentration.

Careers360

Q. 33 The stage during which separation of paired homologous chromosomes begins is

Option 1:
Diakinesis

Option 2:
Diplotene

Option 3:
Pachytene

Option 4:
Zygotene

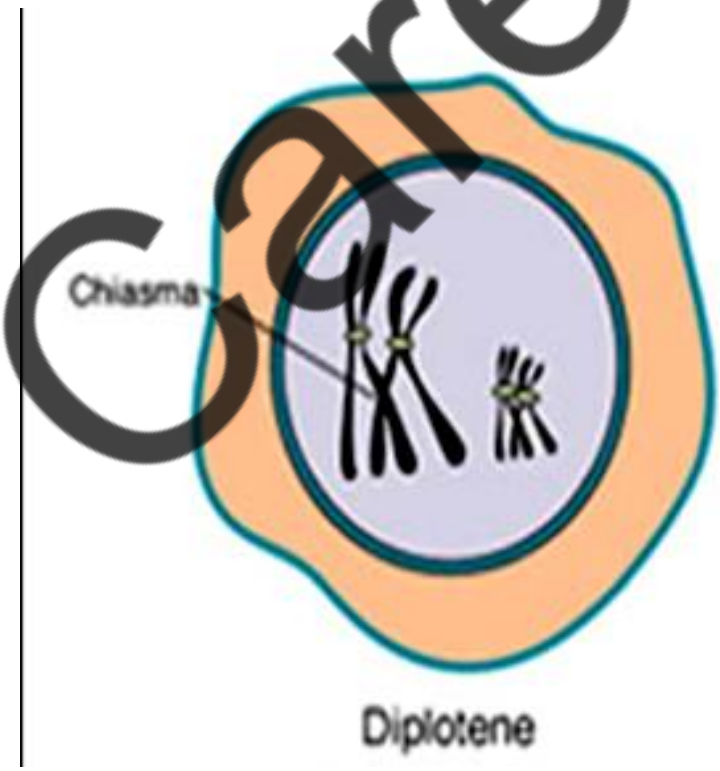
Correct Answer:
Diplotene

Solution:
As we learnt in

Diplotene -

The fourth stage of prophase I, marked by dissolution of synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate. The X-shaped structures are called chiasmata.

- wherein



Careers360

Synaptonemal complex disintegrates. Terminalisation begins at diplotene stage i.e. chiasmata start to shift towards end.

Q. 34 Stomata in grass leaf are

Option 1:
Rectangular

Option 2:
Kidney shaped

Option 3:
Dumb-bell shaped

Option 4:
Barrel shaped

Correct Answer:
Dumb-bell shaped

Solution:
As we learnt in

Stomata -

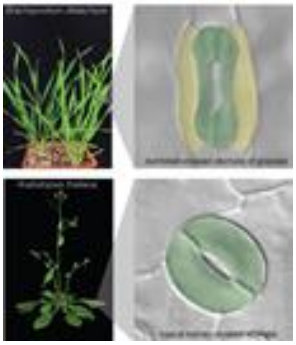
Tiny pores present on the epidermal surface of leaves, young stems and in some fruits.

- wherein

It is surrounded by two kidney/ bean shaped guard cells.

Grass being a monocot, has Dumb-bell shaped stomata in their leaves.

Careers360



Q. 35 Secondary xylem and phloem in dicot stem are produced by

Option 1:
Phellogen

Option 2:
Vascular cambium

Option 3:
Apical meristems

Option 4:
Axillary meristems

Correct Answer:
Vascular cambium

Solution:

The vascular cambium is the main growth layer in the stems and roots of many plants, specifically in dicots. It produces xylem on the inside and phloem on the outside.

Q. 36 Pneumatophores occur in

Careers360

Option 1:
Carnivororus plants

Option 2:
Free- floating hydrophytes

Option 3:
Halophytes

Option 4:
Submerged hydrophytes

Correct Answer:
Halophytes

Solution:
As we learnt in

Pneumatophores -

Respiratory roots or pneumatophores - In mangrove plants (marshy area), the tap root grows vertically upward into the air (negative geotropic) because of low oxygen.

- wherein

These roots have minute pores called pneumathodes or lenticel. Example: Rhizophora, Avicennia, Ceriops, Sonneratia.

Pneumatophores, an aerial root specialized for gaseous exchange, occur in Halophytes like mangroves and other swamp plants.

Q. 37 Casparian strips occur in

Option 1:
Cortex

Careers360

Option 2:
Pericycle

Option 3:
Epidermis

Option 4:
Endodermis

Correct Answer:
Endodermis

Solution:

Endodermis have casparian strip on radial and inner tangential wall. It is rich in suberin.

Q. 38 Plants having little or no secondary growth are

Option 1:
Conifers

Option 2:
Deciduous angiosperms

Option 3:
Grasses

Option 4:
Cycads

Correct Answer:
Grasses

Solution:
As we learnt in

Secondary Growth in Plants -

Careers360

By the activity of secondary lateral meristems, increase in the cicumference, growth of the plant organs, due to the formation of secondary tissue called secordary growth. Normally secondary growth take place in roots and stem of dicot and gymnosperms.

- wherein

Due to lack of cambium, secondary growth is absent in monocot - coconut, agave, smilax, dracaena, yucca, palm.

Grasses are monocots and monocots usually do not have secondary growth.

Q. 39 Sweet potato is modified

Option 1:
Tap root

Option 2:
Adventitious root

Option 3:
Stem

Option 4:
Rhizome

Correct Answer:
Adventitious root

Solution:
As we learnt in

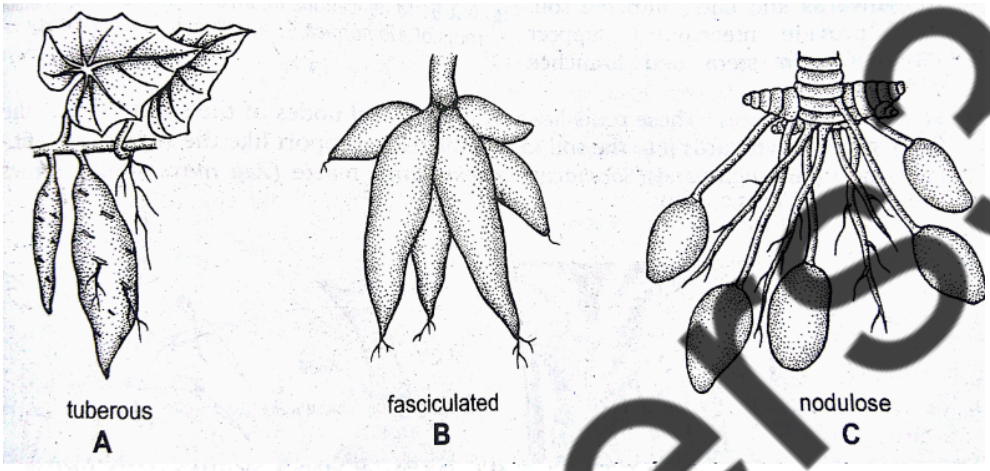
Storage Roots -

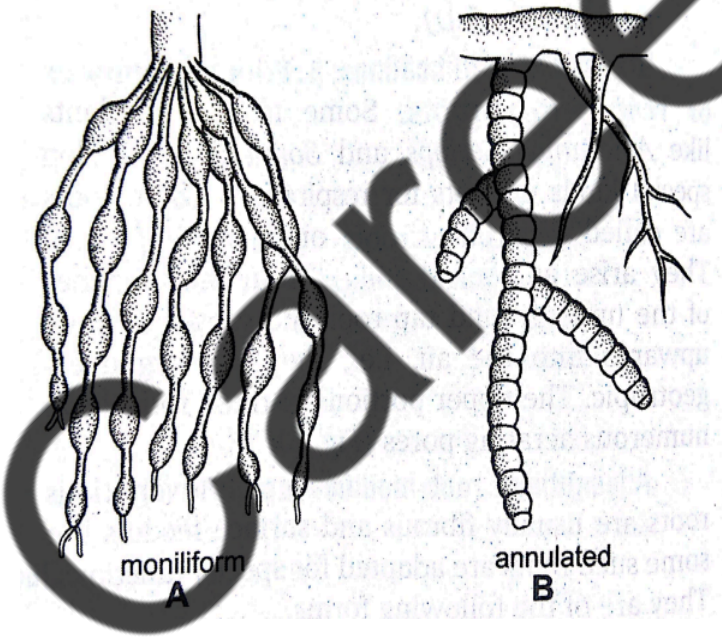
Storage roots -

Careers360

- Tuberous roots: A single root arises from the nodes of the stem. Example: sweet potato (Ipomoea batatas)
- Fasciculated roots: Cluster of tubercular roots. Example: Asparagus and Dahlia
- Nodulose roots: Tips of roots swell up. Example: Arrow-root (Maranta) and turmeric (Curcuma domestica)
- Beaded or moniliform roots: These roots swell up at regular intervals. Example: Momordica, mango ginger (Curcuma amada), Vitis, portulaca
- Annulated roots: Series of ring-like swelling. Example: Ipecac (Psychotria)

- wherein





Sweet potato is a modified adventitious root for storage of food Rhizomes are underground modified stem Tap root is primary root directly elongated from the radicle.

careers360

Q. 40 Which of the following statements is correct ?

Option 1:
Horsetails are gymnosperms .

Option 2:
Selaginella is heterosporous , while *Salvinia* is homosporous .

Option 3:
Ovules are not enclosed by ovary wall in gymnosperms .

Option 4:
Stems are usually unbranched in both *Cycas* and *Cedrus* .

Correct Answer:
Ovules are not enclosed by ovary wall in gymnosperms .

Solution:
As we learnt in

Seed -

Seed is fertilized or ripned ovule or a mature integumented megasporangium. Seed is characterstic of spermatophytes (gymnosperm & angiosperm).

-

Gymnosperms have naked ovule.

Q. 41 Select the wrong statement :

Option 1:
Pseudopodia are locomotory and feeding structures in sporozoans .

Careers360

Option 2:
Mushrooms belong to Basidiomycetes.

Option 3:
Cell wall is present in members of Fungi and Plantae.

Option 4:
Mitochondria are the powerhouse of the cell in all kingdoms except Monera.

Correct Answer:
Pseudopodia are locomotory and feeding structures in sporozoans .

Solution:
As we learnt in

Example of amoeboid movement in human body -

Some specialised cells in our body like macrophages and leucocytes in blood exhibit amoeboid movement.

- wherein

It is effected by pseudopodia formed by the streaming of protoplasm (as in Amoeba).
Pseudopodia are locomotory structures in sarcodines (Amoeboid)

Q. 42 After karyogamy followed by meiosis , spores are produced exogenously in

Option 1:
Agaricus

Option 2:
Alternaria

Option 3:
Neurospora

Careers360

Option 4:
Saccharomyces

Correct Answer:
Agaricus

Solution:
As we learnt in

Basidiomycetes / Club fungi -

1. Mycelium is septate and branched.
2. Sexual reproduction is somatogamy.
3. They decompose cellulose and lignin.
4. Basidiospores formed exogenously on sterigmata
5. Fruiting body is basidiocarp.

- wherein

Eg. Agaricus, Ustilago, Puccinia.

Deuteromycetes / Fungi imperfecti -

1. Mycelium is seperate and branched.
2. Sexual reproduction absent
3. Fruiting body absent.
4. Asexual reproduction by conidia.

- wherein

Eg. Alternaria, Trichoderma.

Ascomycetes / Sac fungi -

1. Mycelium is septate and branched.
2. Sexual reproduction is Isogamy or anisogamy.
3. Four to eight ascospores are formed in an ascus.
4. Fruiting body is Ascocarp.
5. Centrum is present.

- wherein

Careers360

Eg. Saccharomyces, cerevisiae, Baker's yeast, Penicillium.

In Agaricus (a genus of basidiomycetes), basidiospores or meiospores are produced exogenously.

-Neurospora (a genus of ascomycetes) produces ascospores as meiospores but endogenously inside the ascus.

-Alternaria (a genus of deuteromycetes) does not produce sexual spores.

-Saccharomyces (Unicellular ascomycetes) produces ascospores, endogenously.

Q. 43 Match the items given in Column I with those in Column II and select the correct option given below :

	Column I		Column II
a.	Herbarium	i.	It is a place having a collection of preserved plants and animals.
b.	Key	ii.	A list that enumerates methodically all the species found in an area with brief description aiding identification.
c.	Mueseum	iii.	Is a place where dried and pressed plant specimens mounted on sheets are kept.
d.	Catalogue	iv.	A booklet containing list of characters and their alternates which are helpful in identification of various taxa.

Careers360

Option 1:

a	b	c	d
ii	iv	iii	i

Option 2:

a	b	c	d
iii	ii	i	iv

Option 3:

a	b	c	d
i	iv	iii	ii

Option 4:

a	b	c	d
iii	iv	i	ii

Correct Answer:

a	b	c	d
iii	iv	i	ii

Careers360

Solution:

As we learnt in

Museum -

Collection of preserved plants and animals for study and reference.

- wherein

Prince of Wales Museum - Mumbai .

Herbarium -

It is a place where dried and pressed plant species, mounted on sheets are kept systematically according to a widely accepted system of classification.

- wherein

Largest herbarium of world is at kew.

Indented /Yoked Key -

Contains the sequence of choice between two or more characteristic.

-

Herbarium – Dried and pressed plant specimen

Key – Identification of various taxa

Museum – Plant and animal specimen are preserved

Catalogue – Alphabetical listing of species

Careers360

Q. 44 Winged pollen grains are present in

Option 1:
Mango

Option 2:
Cycas

Option 3:
Mustard

Option 4:
Pinus

Correct Answer:
Pinus

Solution:
As we learnt in

Pollen Grains -

The reduced male gametophytes which develop from microspores and are highly reduced and confined to only a limited number of cells .

-

In Pinus, winged pollen grains are present. It is extended outer exine on two lateral sides to form the wings of pollen. It is the characteristic feature, only in Pinus.

This is because Pinus performs anemophily i.e., pollination by wind.

Q. 45 Which one is **wrongly** matched ?

Option 1:
Gemma cups - *Marchantia*

Option 2:
Biflagellate zoospores - Brown algae

Option 3:
Uniflagellate gametes - *Polysiphonia*

Option 4:
Unicellular organism - *Chlorella*

Careers360

Correct Answer:
Uniflagellate gametes - *Polysiphonia*

Solution:
As we learnt in

Rhodophyceae (Red algae) -

Major pigments present are r - phycoerythrin, r - phycocyanin, chlorophyll a and d. Reserve foods are floridean starch and sugar flouridoside.

- wherein

Eg. Polysiphonia, Porphyra, Gelidium.

olysiphonia is a genus of red algae, where asexual spores and gametes are non-motile or non-flagellated.

Q. 46 Which of the following options correctly represents the lung conditions in asthma and emphysema , respectively ?

Option 1:
Increased respiratory surface ; Inflammation of bronchioles

Option 2:
Increased number of bronchioles ; Increased respiratory surface

Option 3:
Inflammation of bronchioles ; Decreased respiratory surface

Option 4:
Decreased respiratory surface ; Inflammation of bronchioles

Correct Answer:
Inflammation of bronchioles ; Decreased respiratory surface

Careers360

Solution:
As we learnt in

Hypothyroidism -

Hypothyroidism during pregnancy causes defective development and maturation of the growing baby leading to stunted growth (cretinism), mental retardation, low intelligence quotient, abnormal skin, deaf-mutism, etc.

-

Asthma is a difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles. Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased.

Q. 47 Match the items given in Column I with those in Column II and select the correct option given below :

	Column I		Column II
a.	Tricuspid valve	i.	Between left atrium and left ventricle
b.	Bicuspid valve	ii.	Between right ventricle and pulmonary artery
c.	Semilunar valve	iii.	Between right atrium and right ventricle

Careers360

Option 1:

a	b	c
i	ii	iii

Option 2:

a	b	c
i	iii	ii

Option 3:

a	b	c
iii	i	ii

Option 4:

a	b	c
ii	i	iii

Correct Answer:

a	b	c
iii	i	ii

Careers360

Solution:

As we learnt in

Structure of heart -

tricuspid valve

- wherein

each of these septa are provided with an opening through which the two chambers of the same side are connected. The opening between the right atrium and the right ventricle is guarded by a valve formed of three muscular flaps or cusps, the tricuspid valve.

Tricuspid valves are AV valve present between right atrium and right ventricle. Bicuspid valves are AV valve present between left atrium and left ventricle. Semilunar valves are present at the openings of aortic and pulmonary aorta.

Q. 48 Match the items given in Column I with those in Column II and select the **correct** option given below :

	Column I		Column II
a.	Tidal volume	i.	2500 - 3000 mL
b.	Inspiratory Reserve volume	ii.	1100 - 1200 mL
c.	Expiratory Reserve volume	iii.	500 - 550 mL
d.	Residual volume	iv.	1000 - 1100 mL

Option 1:

a	b	c	d
i	iv	ii	iii

Option 2:

a	b	c	d
iii	i	iv	ii

Careers360

Option 3:

a	b	c	d
iii	ii	i	iv

Option 4:

a	b	c	d
iv	iii	ii	i

Correct Answer:

a	b	c	d
iii	i	iv	ii

Solution:
As we learnt in

Tidal Volume (TV) -

Volume of air inspired or expired during a normal respiration.

- wherein

It is approx. 500 mL., i.e., a healthy man can inspire or expire approximately 6000 to 8000 mL of air per minute.

Inspiratory Reserve Volume (IRV) -

Additional volume of air, a person can inspire by a forcible inspiration.

- wherein

This averages 2500 mL to 3000 mL.

Careers360

Expiratory Reserve Volume (ERV) -

Additional volume of air, a person can expire by a forcible expiration.

- wherein

This averages 1000 mL to 1100 mL.

Residual Volume (RV) -

Volume of air remaining in the lungs even after a forcible expiration.

- wherein

This averages 1100 mL to 1200 mL.

Tidal volume is volume of air inspired or expired during normal respiration. It is approximately 500 mL.

Inspiratory reserve volume is additional volume of air a person can inspire by a forceful inspiration. It is around 2500 – 3000 mL.

Expiratory reserve volume is additional volume of air a person can be expired by a forceful expiration. This averages 1000 – 1100 mL.

Residual volume is volume of air remaining in lungs even after forceful expiration. This averages 1100 – 1200 mL.

Careers360

Q. 49 The transparent lens in the human eye is held in its place by

Option 1:
smooth muscles attached to the iris

Option 2:
ligaments attached to the iris

Option 3:
ligaments attached to the ciliary body

Option 4:
smooth muscles attached to the ciliary body

Correct Answer:
ligaments attached to the ciliary body

Solution:

- **As we have learnt in** :

Middle Layer or Vascular Coat:

- In front of the lens, the aperture surrounded by the iris is called the **pupil**. The diameter of the pupil is regulated by the muscle fibres of iris.
- The eyeball contains a transparent crystalline **lens** which is held in place by ligaments attached to the ciliary body.
- The ciliary body itself continues forward to form a pigmented and opaque structure called the **iris** which is the visible coloured portion of the eye.
- The choroid layer is thin over the posterior two-thirds of the eyeball, but it becomes thick in the anterior part to form the **ciliary body.**
- The middle layer, **choroid**, contains many blood vessels and looks bluish in colour.

Hence, The transparent lens in the human eye is held in its place by ligaments attached to the ciliary body.

Q. 50 Which of the following is an amino acid derived hormone ?

Option 1:
Estradiol

Careers360

Option 2:
Ecdyosone

Option 3:
Epinephrine

Option 4:
Estriol

Correct Answer:
Epinephrine

Solution:
As we learnt in

Hormones secreted by Adrenal Gland -

The adrenal medulla secretes two hormones called adrenaline or epinephrine and noradrenaline or norepinephrine.

- wherein

These are commonly called as catecholamines

Epinephrine is derived from tyrosine amino acid by the removal of carboxyl group. It is a catecholamine.

Q. 51 Which of the following hormones can play a significant role in osteoporosis ?

Option 1:
Estrogen and Parathyroid hormone

Option 2:
Progesterone and Aldosterone

Careers360

Option 3:
Aldosterone and Prolactin

Option 4:
Parathyroid hormone and Prolactin

Correct Answer:
Estrogen and Parathyroid hormone

Solution:
As we learnt in

Disorders Of Muscular And Skeletal System -

Osteoporosis

- wherein

Age-related disorder characterised by decreased bone mass and increased chances of fractures. Decreased levels of estrogen is a
common cause.

Estrogen promotes the activity of osteoblast and inhibits osteoclast. In an ageing female osteoporosis occurs due to deficiency of estrogen. Parathormone promotes mobilisation of calcium from bone into blood. Excessive activity of parathormone causes demineralisation leading to osteoporosis

---

Q. 52 Which of the following structures or regions is **incorrectly** paired with its function ?

Option 1:
Hypothalamus : production of releasing hormones and regulation of temperature, hunger and thirst

Careers360

Option 2:
Limbic system : consists of fibre tracts that interconnect different regions of brain ; controls movement

Option 3:
Medulla oblongata : controls respiration and cardiovascular reflexes.

Option 4:
Corpus callosum : band of fibers connecting left and right cerebral hemispheres.

Correct Answer:
Limbic system : consists of fibre tracts that interconnect different regions of brain ; controls movement

Solution:

Limbic System

- The limbic system is the collective name for the parts of the brain that controls emotions, motivation, and emotional association with memory.
- Certain components of the cerebrum and diencephalon constitute the limbic system.

Hence, the statement 'Limbic system: consists of fibre tracts that interconnect different regions of the brain ; controls movement' is incorrect that is option b

Careers360

Q. 53 The amnion of mammalian embryo is derived from

Option 1:
mesoderm and trophoblast

Option 2:
endoderm and mesoderm

Option 3:
ectoderm and mesoderm

Option 4:
ectoderm and endoderm

Correct Answer:
ectoderm and mesoderm

Solution:
As we learnt in

Eggs of aves -

Eggs are cleidoic, macrolecithal and mesolecithal.

- wherein

Extraembryonic membrane (amniota) present.

The extraembryonic or foetal membranes are amnion, chorion, allantois and Yolk sac. Amnion is formed from mesoderm on outer side and ectoderm on inner side.

Q. 54 Hormones secreted by the placenta to maintain pregnancy are

Option 1:
hCG , hPL , progestogens , estrogens

Option 2:
hCG , hPL , estrogens , relaxin , oxytocin

Careers360

Option 3:
hCG , hPL , progestogens , prolactin

Option 4:
hCG , progestogens , estrogens , glucocorticoids

Correct Answer:
hCG , hPL , progestogens , estrogens

Solution:
As we learnt in

Foetal ejection reflex. -

The signals for parturition originate from the fully developed foetus and the placenta which induce mild uterine contractions called foetal ejection reflex.

- wherein

This triggers release of oxytocin from the maternal pituitary. Oxytocin acts on the uterine muscle and causes stronger uterine contractions, which in turn stimulates further secretion of oxytocin.

Placenta releases human chorionic gonadotropic hormone (hCG) which stimulates the Corpus luteum during pregnancy to release estrogen and progesterone and also rescues corpus luteum from regression. Human placental lactogen (hPL) is involved in growth of breast. Progesterone maintains pregnancy, keeps the uterus silent by increasing uterine threshold to contractile stimuli.

Careers360

Q. 55 The difference between spermiogenesis and spermiation is

Option 1:
In spermiogensis spermatozoa from sertoli cells are released into the cavity of seminiferous tubules , while in spermiation spermatozoa are formed .

Option 2:
In spermiogenesis spermatozoa are formed , while in spermiation spermatids are formed .

Option 3:
In spermiogensis spermatids are formed , while in spermiation spermatozoa are formed.

Option 4:
In spermiogenesis spermatozoa are formed , while in spermiation spermatozoa are released from sertoli cells into the cavity of seminiferous tubules.

Correct Answer:
In spermiogenesis spermatozoa are formed , while in spermiation spermatozoa are released from sertoli cells into the cavity of seminiferous tubules.

Solution:
As we learnt in

Sperms -

The spermatids are transformed into spermatozoa (sperms) by the process called spermiogenesis

-

Spermiation -

After spermiogenesis, sperm heads become embedded in the Sertoli cells, and are finally released from the seminiferous tubules by the process called spermiation

-

Spermiogenesis is transformation of spermatids into spermatozoa whereas spermiation is the release of the sperms from sertoli cells into the lumen of seminiferous tubule.

Careers360

Q. 56 The contraceptive 'SAHELI'

Option 1:
is an IUD.

Option 2:
increases the concentration of estrogen and prevents ovulation in females.

Option 3:
blocks estrogen receptors in the uterus , preventing eggs from getting implanted.

Option 4:
is a post - coital contraceptive.

Correct Answer:
blocks estrogen receptors in the uterus , preventing eggs from getting implanted.

Solution:
As we learnt in

Saheli -

A new oral contraceptive for the females–was developed by scientists at **Central Drug Research Institute** (CDRI) in Lucknow

where in-

Saheli is the first non-steroidal, once a week pill. It contains **centchroman** and its functioning is based upon selective **Estrogen Receptor** modulation. Instead of using the hormone estrogen to prevent pregnancy, it contains a drug that blocks estrogen. Usually, increased **doses of estrogen** stop a woman's **ovaries from releasing an egg.**

Careers360

Q. 57 Ciliates differ from all other protozoans in

Option 1:
having pseudopodia for capturing prey

Option 2:
having a contractile vacuole for removing excess water

Option 3:
using flagella for locomotion

Option 4:
having two types of nuclei

Correct Answer:
having two types of nuclei

Solution:
As we learnt in

Ciliated Protozoan -

Actively moving due to the presence of cilia.

- wherein

Eg. Paramecium.

Ciliates differs from other protozoans in having two types of nuclei. eg. Paramoecium have two types of nuclei i.e. **macronucleus & micronucleus**.

Careers360

Q. 58 Identify the vertebrate group of animals characterized by crop and gizzard in its digestive system

Option 1:
Aves

Option 2:
Reptilia

Option 3:
Amphibia

Option 4:
Osteichthyes

Correct Answer:
Aves

Solution:
As we learnt in

Digestive system in aves -

Consist of additional chambers, Crop and gizzard. Three chambered cloaca.

-

The digestive tract of Aves has additional chambers in their digestive system as crop and Gizzard. Crop is concerned with storage of food grains. Gizzard is a masticatory organ in birds used to crush food grain.

Q. 59 Which of the following features is used to identify a male cockroach from a female cockroach ?

Option 1:
Forewings with darker tegmina

Careers360

Option 2:
Presence of caudal styles

Option 3:
Presence of a boat shaped sternum on the 9^{th} abdominal segment

Option 4:
Presence of anal cerci

Correct Answer:
Presence of caudal styles

Solution:
As we learnt in

Anal Styles -

In male, the 9th sternite (segment) bears a pair of spine like anal styles, which are absent in female and are used to differentiate between male and female.

- wherein

Additionally male cockroaches have long antennae than female.

Males bear a pair of short, thread like anal styles which are absent in females. Anal/caudal styles arise from 9th abdominal segment in male cockroach.

Q. 60 Which one of these animals is **not** a homeotherm ?

Option 1:
Camelus

Option 2:
Chelone

Careers360

Option 3:
Macropus

Option 4:
Psittacula

Correct Answer:
Chelone

Solution:
As we learnt in

Body temperature of chondriocytes -

Cold blooded or poikilothermous.

-

Homeotherm are animals that maintain constant body temperature, irrespective of surrounding temperature.

Birds and mammals are homeotherm. Chelone (Turtle) belongs to class reptilia which is Poikilotherm or cold blood.

Q. 61 Which of the following animals does **not** undergo metamorphosis ?

Option 1:
Moth

Option 2:
Tunicate

Option 3:
Earthworm

Option 4:
Starfish

Careers360

Correct Answer:
Earthworm

Solution:
As we learnt in

Retrogressive metamorphosis -

Change from better developed larva to less developed adult.

-

Metamorphosis refers to transformation of larva into adult.

Animal that perform metamorphosis are said to have indirect development.

In earthworm development is direct which means no larval stage and hence no metamorphosis.

Q. 62 Which of the following organisms are known as chief producers in the oceans ?

Option 1:
Cyanobacteria

Option 2:
Diatoms

Option 3:
Dinoflagellates

Option 4:
Euglenoids

Correct Answer:
Diatoms

Solution:

Careers360

As we learnt in

Diatomas / Chrysophytes -

1. May be present in marine, fresh water or terrestrial habitats.
2. Movement is brought by mucilage propulsion.

- wherein

1. Serve as sewage pollution indicator.
2. Cell walls are embedded with silica, which are indestructible and thus, forms diatomaceous earth or kieselguhr.

Diatoms are algae with distinctive, transparent cell walls made of silicon dioxide hydrated with a small amount of water ($SiO_2 + H_2O$). Diatoms are chief producers of the ocean.

Q. 63 Which one of the following population interactions is widely used in medical science for the production of antibiotics ?

Option 1:
Parasitism

Option 2:
Mutualism

Option 3:
Commensalism

Option 4:
Amensalism

Correct Answer:
Amensalism

Solution:

As discussed in Microbes in Industrial Products: Antibiotics -

Careers360

Antibiotics are chemicalsubstances, which are produced by some microbes and can kill or retard the growth of other (disease-causing) microbes.

- wherein

e.g., antibiotic Penicillin

Antibiotics are chemicals secreted by one microbial group (eg : Penicillium) which harm other microbes (eg : *Staphylococcus*). It has no effect on Penicillium or the organism which produces it.

Hence, the correct option is (d).

Q. 64 All of the following are included in 'Ex - situ conservation ' except

Option 1:
Botanical gardens

Option 2:
Sacred groves

Option 3:
Wildlife safari parks

Option 4:
Seed banks

Correct Answer:
Sacred groves

Solution:

Careers360

As we have already discussed in - In-situ Conservation of Biodiversity - Hotspots, National Parks and Sacred groves -

In-situ conservation is the on-site conservation or the conservation of genetic resources in natural populations of plant or animal species, such as forest genetic resources in natural populations of tree species.

Therefore , Sacred groves falls under 'In-situ conservation'.

Hence, the correct option is (b).

Q. 65 Match the items given in Column I with those in Column II and select the **correct** option given below :

	Column I		Column II
a.	Eutrophication	i.	UV - B radiation
b.	Sanitary landfill	ii.	Deforestation
c.	Snow blindness	iii.	Nutrient enrichment
d.	Jhum cultivation	iv.	Waste disposal

Option 1:

a	b	c	d
iii	iv	i	ii

Careers360

Option 2:

a	b	c	d
i	iii	iv	ii

Option 3:

a	b	c	d
ii	i	iii	iv

Option 4:

a	b	c	d
i	ii	iv	iii

Correct Answer:

a	b	c	d
iii	iv	i	ii

Solution:

As we learnt in

Causes of water pollution -

Eutrophication

- wherein

Rich growth of micro-organisms consumes most of the dissolved oxygen, so as to deprive other organisms. It generally occurs at the bottom layers of deep lakes. Addition of excessive plant nutrients intensifies eutrophication. It is harmful to fish and other aquatic life.

Careers360

Jhum cultivation -

Trees are axed for timber, firewood, cattle ranching and for several other purposes.

- wherein

Slash and burn agriculture, commonly called as Jhum cultivation in the north-eastern states of India, has also contributed to deforestation.

Sanitary landfills -

Sanitary landfills were adopted as the substitute for open-burning dumps

.

- wherein

In a sanitary landfill, wastes are dumped in a depression or trench after compaction, and covered with dirt everyday.

Effect of UV light -

UV radiation of wavelengths shorter than UV-B, are almost completely absorbed by Earth's atmosphere, given that the ozone layer is intact. But, UV-B damages DNA and mutation may occur

- wherein

It causes aging of skin, damage to skin cells and various types of skin cancers. In human eye,cornea absorbs UV-B radiation, and a high dose of UV-B causes inflammation of cornea, called snow-blindness, cataract, etc. Such exposure may permanently damage the cornea.

Eutrophication- Nutrient enrichment

Careers360

Sanitary landfill - Waste disposal

Snow blindness - UV-B radiation

Jhum cultivation - Deforestation

Q. 66 In a growing population of a country

Option 1:
reproductive and pre -reproductive individuals are equal in number.

Option 2:
reproductive individuals are less than the post - reproductive individuals.

Option 3:
pre - reproductive individuals are more than the reproductive individuals.

Option 4:
pre - reproductive individuals are less than the reproductive individuals .

Correct Answer:
pre - reproductive individuals are more than the reproductive individuals.

Solution:
As we learnt in

Population Growth -

The size of a population for any species is not a static parameter. It keeps changing in time, depending on various factors including food availability, predation pressure and adverse weather.

- wherein

It is these changes in population density that give us some idea of what is happening to the population – whether it is flourishing or declining.

Careers360

Whenever the pre-reproductive individuals or the younger population size is larger than the reproductive group, the population will be an increasing population.

Q. 67 Which part of poppy plant is used to obtain the drug "Smack "?

Option 1:
Roots

Option 2:
Latex

Option 3:
Flowers

Option 4:
Leaves

Correct Answer:
Latex

Solution:

As we have already discussed in Commonly Abused Drugs -

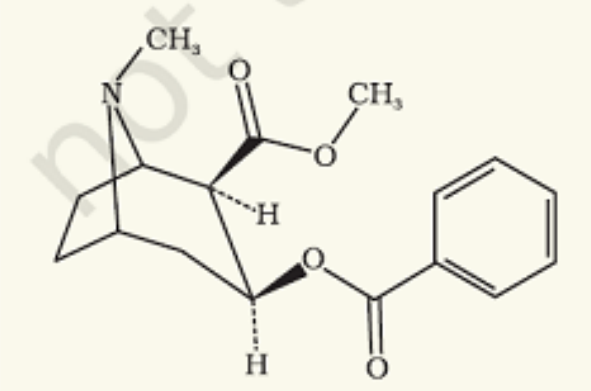
Heroin -

Commonly called smack is chemically diacetylmorphine which is a white, odorless, bitter crystalline compound.

- wherein

Careers360

This is obtained by acetylation of morphine, which is extracted from the latex of poppy plant Papaver somniferum
Generally taken by snorting and injection, heroin is a depressant and slows down body functions.



'Smack' also called as brown sugar/Heroin is formed by acetylation of morphine. It is obtained from the latex of unripe capsule of Poppy plant.

Hence, the correct option is (b).

Q. 68 All of the following are part of an operon except

Option 1:
an enhancer

Option 2:
structural genes

Option 3:
an operator

Option 4:
a promoter

Correct Answer:
an enhancer

Careers360

Solution:
As we learnt in

Haemophilia -

This is sex-linked recessive disease, which shows transmission from unaffected carrier female to some of the male progeny.

- wherein

In this, a simple cut will result in non-stop bleeding and no blood clotting will take place.

Operons are a part of prokaryotes and since Enhancer sequences are present in Eukaryote, operons are not a part of enhancer.

Q. 69 A woman has X - linked condition on one of her X chromosomes. This chromosme can be inherited by

Option 1:
Only grandchildren

Option 2:
Only sons

Option 3:
Only daughters

Option 4:
Both sons and daughters

Correct Answer:
Both sons and daughters

Solution:
As we learnt in

Haemophilia -

Careers360

This is sex-linked recessive disease, which shows transmission from unaffected carrier female to some of the male progeny.

- wherein

In this, a simple cut will result in non-stop bleeding and no blood clotting will take place.

- Woman is a carrier
- Both son & daughter inherit

X–chromosome

- Although only son be the diseased.

Q. 70 According to Hugo de Vries , the mechanism of evolution is

Option 1:
Phenotypic variations

Option 2:
Saltation

Option 3:
Multiple step mutations

Option 4:
Minor mutations

Correct Answer:
Saltation

Solution:
As we learnt in

Careers360

Discontinuous Distribution -

Descendants of a common ancestor in some cases inhabit different continents and also differ from each other.

- wherein

This is called Discontinuous Distribution.

Eg., aligators and mangolias, tulip and sassafras

As per mutation theory given by Hugo de Vries, the evolution is a discontinuous phenomenon or saltatory phenomenon/saltation.

Q. 71 AGGTATCGCAT is a sequence from the coding strand of a gene. What will be the corresponding sequence of the transcribed mRNA ?

Option 1:
ACCUAUGCGAU

Option 2:
UGGTUTCGCAT

Option 3:
AGGUAUCGCAU

Option 4:
UCCAUAGCGUA

Correct Answer:
AGGUAUCGCAU

Solution:
As we learnt in

DNA and RNA components -

Careers360

Cytosine is common for both DNA and RNA and Thymine is present in DNA.

- wherein

Uracil is present in RNA at the place of thymine.

Coding strand and mRNA has same nucleotide sequence except, 'T' – Thymine is replaced by 'U'–Uracil in mRNA.

Q. 72 Match the items given in Column I with those in Column II and select the correct option given below :

	Column I		Column II
a.	Proliferative Phase	i.	Breakdown of endometrial lining
b.	Secretory Phase	ii.	Follicular Phase
c.	Menstruation	iii.	Luteal Phase

Option 1:

a	b	c
ii	iii	i

Option 2:

a	b	c
i	iii	ii

Careers360

Option 3:

a	b	c
iii	ii	i

Option 4:

a	b	c
iii	i	ii

Correct Answer:

a	b	c
ii	iii	i

Solution:
As we learnt in

Follicular phase -

The menstrual phase is followed by the follicular phase. During this phase, the primary follicles in the ovary grow to become a fully mature Graafian follicle and simultaneously the endometrium of uterus regenerates through proliferation.

- wherein

(i) This phase usually includes cycle days 6-13 or 14 in a 28 days cycle.

(ii) The follicle stimulating hormone (FSH) secreted by the anterior lobe of the pituitary gland stimulates the ovarian follicle to secrete oestrogens.

(iii) Oestrogen stimulates the proliferation of the endometrium of the uterine wall.

(iv) The endometrium becomes thicker by rapid cell multiplication and this is accompanied by an increase in uterine glands & blood vessels

Careers360

Luteal phase -

The ovulation (ovulatory phase) is followed by the luteal phase during which the remaining parts of the Graafian follicle transform as the corpus luteum.

-

During proliferative phase, the follicles start developing, hence, called follicular phase. Secretory phase is also called as luteal phase mainly controlled by progesterone secreted by corpus luteum. Estrogen further thickens the endometrium maintained by progesterone.

Menstruation occurs due to decline in progesterone level and involves breakdown of overgrown endometrial lining.

Hence, the correctoption is (a).

Q. 73 Match the items given in Column I with those in Column II and select the **correct** option given below :

	Column I		Column II
a.	Glycosuria	i.	Accumulation of uric acid in joints
b.	Gout	ii.	Mass of crystallised salts within the kidney
c.	Renal calculi	iii.	Inflammation in glomeruli
d.	Glomerular nephritis	iv.	Presence of glucose in urine

Careers360

Option 1:

a	b	c	d
ii	iii	i	iv

Option 2:

a	b	c	d
i	ii	iii	iv

Option 3:

a	b	c	d
iii	ii	iv	i

Option 4:

a	b	c	d
iv	i	ii	iii

Correct Answer:

a	b	c	d
iv	i	ii	iii

Careers360

Solution:

Glycosuria denotes presence of glucose in the urine. This is observed when blood glucose level rises above 180 mg/100 ml of blood, this is called renal threshold value for glucose.

Gout is due to deposition of uric acid crystals in the joint.

Renal calculi are precipitates of calcium phosphate produced in the pelvis of the kidney.

Glomerular nephritis is the inflammatory condition of glomerulus characterised by proteinuria and haematuria.

Option 4 is the correct answer.

Q. 74 Match the items given in Column I with those in Column II and select the correct option given below :

	Column I (Function)		Column II (Part of Excretory System)
a.	Ultrafiltration	i.	Henle's loop
b.	Concentration of urine	ii.	Ureter
c.	Transport of urine	iii.	Urinary bladder
d.	Storage of urine	iv.	Malphigian corpuscle
		v.	Proximal convoluted tubule

Option 1:

a	b	c	d
v	iv	i	ii

Careers360

Option 2:

a	b	c	d
iv	i	ii	iii

Option 3:

a	b	c	d
iv	v	ii	iii

Option 4:

a	b	c	d
v	iv	i	iii

Correct Answer:

a	b	c	d
iv	i	ii	iii

Solution:

As we learnt in

Excretory Structures among Various Groups of Animals -

Insects like cockroaches

- wherein

Malpighian tubules are the excretory structures of most of the insects including cockroaches. Malpighian tubules help in the removal of nitrogenous wastes and osmoregulation.

Ultrafiltration refers to filtration of very fine particles having molecular weight less than 68,000 daltons through malpighian corpuscle. Concentration of urine refers to water absorption from glomerular filtrate as a result of hyperosmolarity in the medulla

Careers360

created by counter-current mechanism in Henle's loop. Urine is carried from kidney to bladder hrough ureter.

Urinary bladder is concerned with storage of urine.

Q. 75 Which of the following gastric cells indirectly help in erythropoiesis?

Option 1:
Goblet cells

Option 2:
Mucous cells

Option 3:
Chief cells

Option 4:
Parietal cells

Correct Answer:
Parietal cells

Solution:
As we learnt in

Gastric Glands -

Parietal or oxyntic cells which secrete HCl and intrinsic factor (factor essential for absorption of vitamin B12)

Parietal or oxyntic cell is a source of HCl and intrinsic factor. HCl converts iron present in diet from ferric to ferrous form so that it can be absorbed easily and used during erythropoiesis.

Intrinsic factor is essential for the absorption of vitamin B12 and its deficiency causes pernicious anaemia.

Careers360

Q. 76 Match the items given in Column I with those in Column II and select the correct option given below :

	Column I		Column II
a.	Fibrinogen	i.	Osmotic balance
b.	Globulin	ii.	Blood clotting
c.	Albumin	iii.	Defence mechanism

Option 1:

a	b	c
i	iii	ii

Option 2:

a	b	c
i	ii	iii

Option 3:

a	b	c
iii	ii	i

Careers360

Option 4:

a	b	c
ii	iii	i

Correct Answer:

a	b	c
ii	iii	i

Solution:
Fibrinogen is a blood plasma protein involved in blood clotting. Globulin is involved in defense mechanism and albumin is involved in osmotic balance.

Q. 77 Which of the following is an occupational respiratory disorder ?

Option 1:
Botulism

Option 2:
Silicosis

Option 3:
Anthracis

Option 4:
Emphysema

Correct Answer:
Silicosis

Solution:
As we learnt in

Careers360

DISORDERS OF RESPIRATORY SYSTEM -

Occupational Respiratory Disorders

- wherein

In certain industries, especially those involving grinding or stone-breaking, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation. Long exposure can give rise to inflammation leading to fibrosis (proliferation of fibrous tissues) and thus causing serious lung damage.
Silicosis is due to excess inhalation of silica dust in the workers involved grinding or stone breaking industries.

Long exposure can give rise to inflammation leading to fibrosis and thus causing serious lung damage.

Q. 78 Calcium is important in skeletal muscle contraction because it

Option 1:
detaches the myosin head from the actin filament.

Option 2:
activates the myosin ATPase by binding to it.

Option 3:
binds to troponin to remove the masking of active sites on actin for myosin.

Option 4:
prevents the formation of bonds between the myosin cross bridges and the actin filament.

Correct Answer:
binds to troponin to remove the masking of active sites on actin for myosin.

Solution:

Careers360

As we learnt in

Mechanism of Muscle Contraction -

Release of Calcium ions.

- wherein

A neural signal reaching the neuromuscular junction releases a neurotransmitter (Acetyl choline) which generates an action potential in the sarcolemma. This spreads through the muscle fibre and causes the release of calcium ions into the sarcoplasm.

Signal for contraction increase Ca++ level many folds in the sarcoplasm after which Ca++ binds with sub-unit of troponin (troponin "C") which is masking the active site on actin filament and displaces the sub-unit of troponin. Once the active site is exposed, head of the myosin attaches and initiate contraction by sliding the actin over myosin.

Q. 79 Nissl bodies are mainly composed of

Option 1:
Nucleic acids and SER

Option 2:
DNA and RNA

Option 3:
Proteins and lipids

Option 4:
Free ribosomes and RER

Correct Answer:
Free ribosomes and RER

Solution:

Careers360

-

Nissl body is also called as nissl substance. It is a large granular body present in neurons. Their main function is to synthesize and release proteins that are important for neuronal growth and regeneration of axons. It is made up of rough endoplasmic reticulum (RER) and free ribosomes.

Q. 80 Which of these statements **incorrect**?

Option 1:
Glycolysis operates as long as it is supplied with NAD that can pick up hydrogen atoms.

Option 2:
Glycolysis occurs in cytosol.

Option 3:
Enzymes of TCA cycle are present in mitochondrial matrix.

Option 4:
Oxidative phosphorylation takes place in outer mitochondrial membrane.

Correct Answer:
Oxidative phosphorylation takes place in outer mitochondrial membrane.

Solution:
As we learnt in

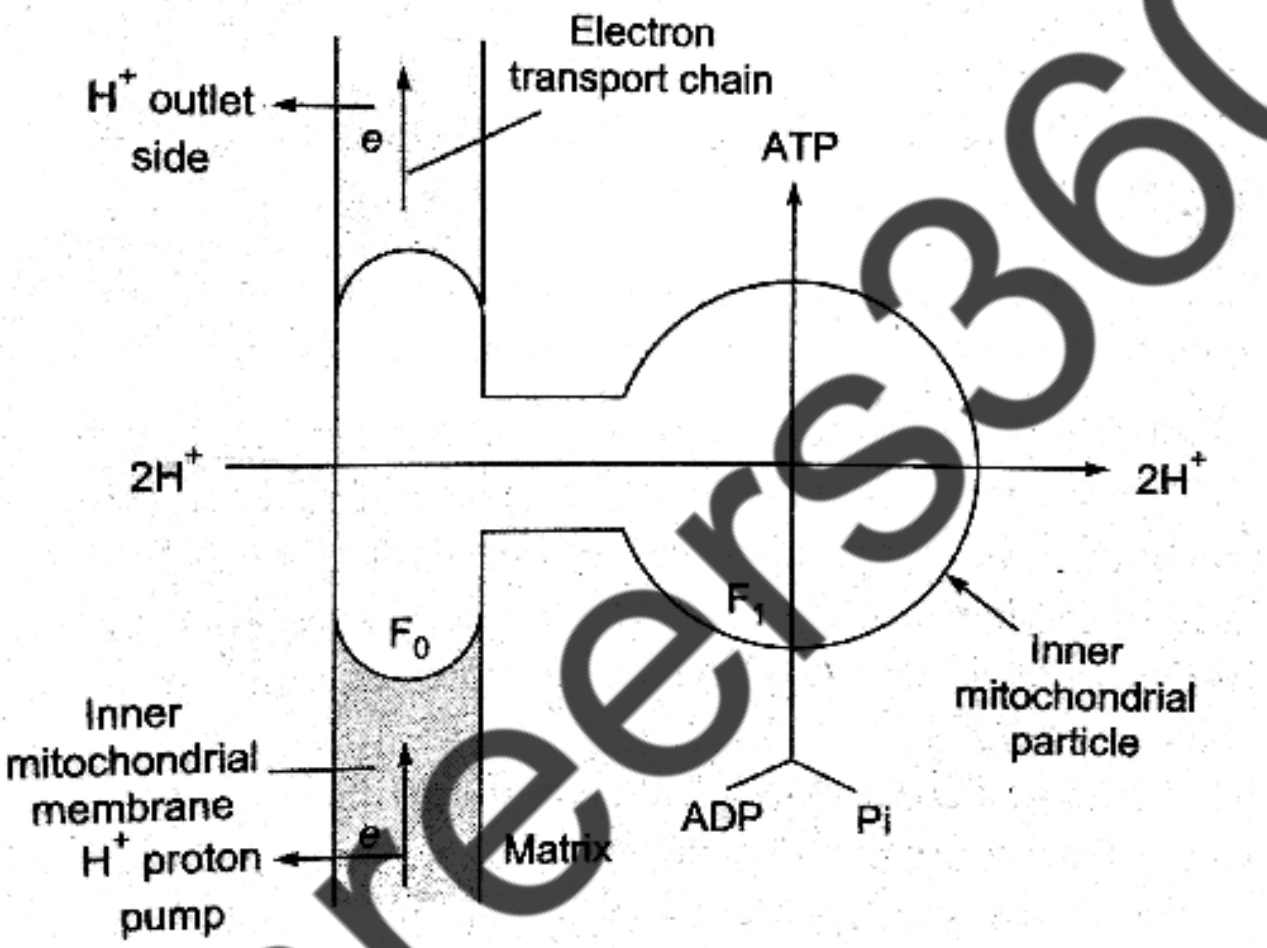
Oxidative phosphorylation -

Careers360

Synthesis of energy such as ATP molecules with the help of energy librated during oxidation of reduced coenzymes (NADH, FADH2) produced in respiration, is carried out in ETS and energy is generated through chemoiosmosis.

- wherein

Occours in the inner mitochondria membrane.



Oxidative phosphorylation is the process in which ATP is formed as a result of the transfer of electrons from NADH or $FADH_2$ to O_2 by a series of electron carriers. This process, which takes place in inner mitochondrial membrane, is the major source of ATP in aerobic organisms.

Careers360

Q. 81 Select the incorrect match :

Option 1:
Submetacentric chromosomes - L - shaped chromosomes

Option 2:
Allosomes - Sex chromosomes

Option 3:
Lampbrush chromosomes - Diplotene bivalents

Option 4:
Polytene chromosomes - oocytes of amphibians

Correct Answer:
Polytene chromosomes - oocytes of amphibians

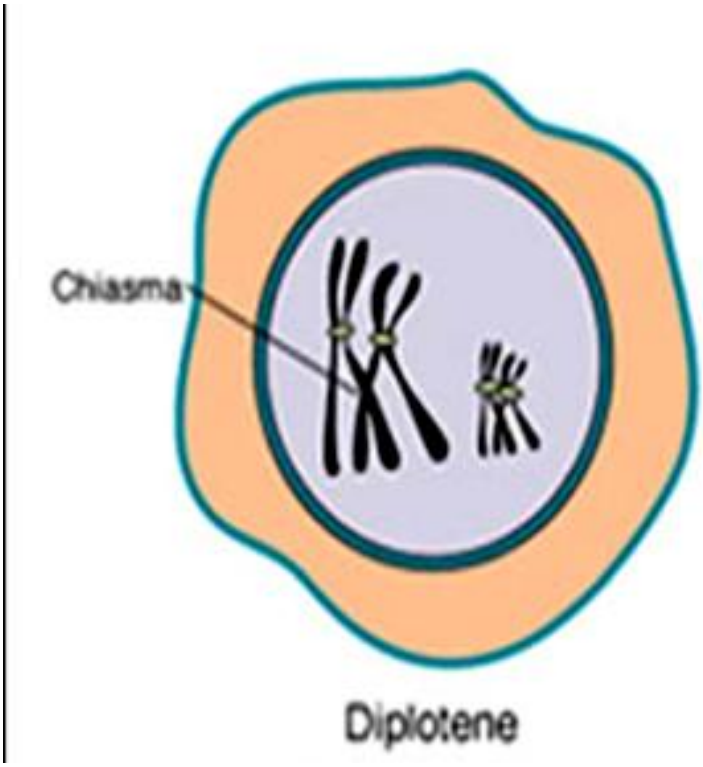
Solution:
As we learnt in

Diplotene -

The fourth stage of prophase I, marked by dissolution of synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate. The X-shaped structures are called chiasmata.

- wherein

Careers360



Types of chromosomes -

Based on the position of centromere, the chromosomes can be classified into four types:

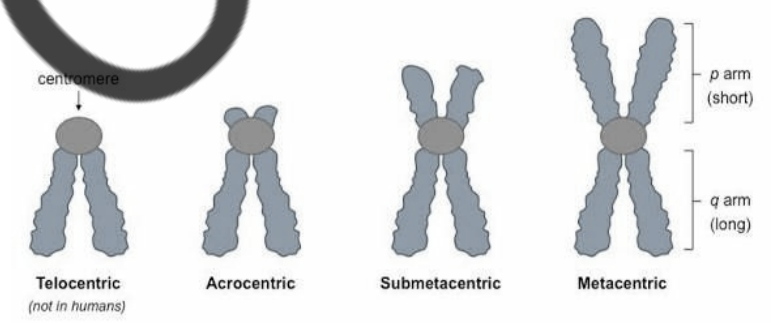
1) Metacentric chromosome- middle centromere forming two equal arms.

2) Sub-metacentric- centromere slightly away from the center.

3)Acrocentric- centromere is situated close to its end

4)Telocentric- terminal centromere

- wherein



Polytene chromosomes are found in salivary glands of insects of order Diptera.

Careers360

Q. 82 Which of the following terms describe human dentition ?

Option 1:
Pleurodont , Monophyodont , Homodont

Option 2:
Thecodont , Diphyodont , Heterodont

Option 3:
Thecodont , Diphyodont , Homodont

Option 4:
Pleurodont , Diphyodont , Heterodont

Correct Answer:
Thecodont , Diphyodont , Heterodont

Solution:
As we learnt in

Thecodont -

Each tooth is embedded in a socket of jaw bone . This type of attachment is called thecodont.

-Diphyodont -

Majority of mammals including human being forms two sets of teeth during their life, a set oftemporary milk or deciduous teeth replaced by a set of permanent or adult teeth

- wherein

This type of dentition is called diphyodont

In humans, dentition is

-Thecodont : Teeth are present in the sockets of the jaw bone called alveoli.

Careers360

- Diphyodont : Teeth erupts twice, temporary milk or deciduous teeth are replaced by a set of permanent or adult teeth.

- Heterodont dentition : Dentition consists of different types of teeth namely incisors, canine, premolars and molars.

Q. 83 Which of the following events does not occur in rough endoplasmic reticulum ?

Option 1:
Cleavage of signal peptide

Option 2:
Protein glycosylation

Option 3:
Protein folding

Option 4:
Phospholipid synthesis

Correct Answer:
Phospholipid synthesis

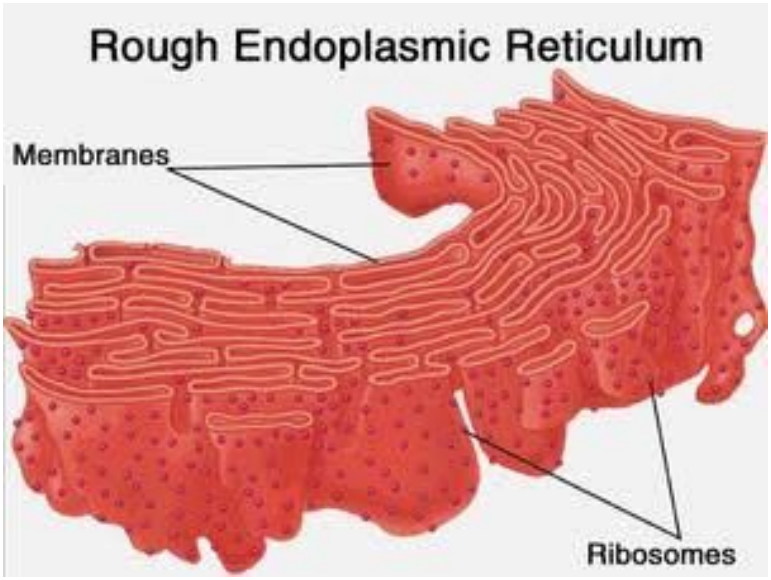
Solution:
As we learnt in

RER and SER -

The endoplasmic reticulum bearing ribosomes on their surface is called rough endoplasmic reticulum (RER), in the absence of ribosomes, they appear smooth and are called smooth endoplasmic reticulum (SER).

- wherein

Careers360



Phospholipid synthesis does not take place in RER. Smooth endoplasmic reticulum are involved in lipid synthesis.

Q. 84 Many ribosomes may associate with a single mRNA to form multiple copies of a polypeptide simultaneously . Such strings of ribosomes are termed as

Option 1:
Plastidome

Option 2:
Polyhedral bodies

Option 3:
Polysome

Option 4:
Nucleosome

Correct Answer:
Polysome

Solution:

Careers360

As we learnt in

Ribosomes -

It is a complex made up of RNA and protein and is the site for protein synthesis.

- wherein

Several ribosomes may attach to a single mRNA and form chains called polyribosomes or polysome.

The phenomenon of association of many ribosomes with single m-RNA leads to formation of polyribosomes or polysomes or ergasomes.

Q. 85 In which disease does mosquito transmitted pathogen cause chronic inflammation of lymphatic vessels ?

Option 1:
Ringworm disease

Option 2:
Ascariasis

Option 3:
Elephantiasis

Option 4:
Amoebiosis

Correct Answer:
Elephantiasis

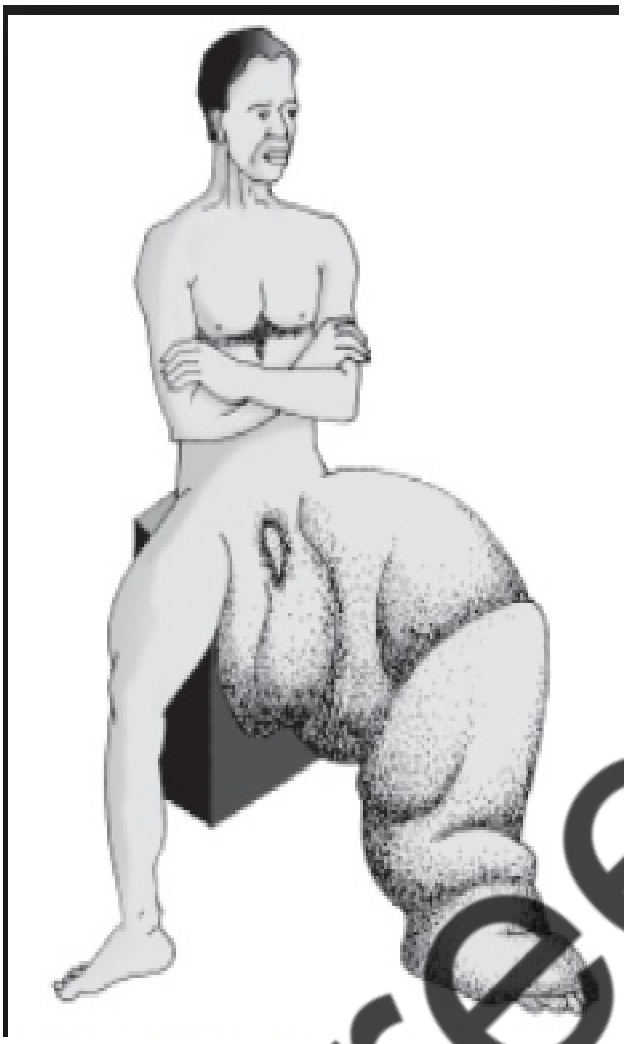
Solution:
As we learnt in

elephantiasis or filariasis -

Careers360

Wuchereria (W. bancrofti and W. malayi), the filarial worms that cause elephantiasis or filariasis

- wherein



Elephantiasis is caused by roundworm, Wuchereria bancrofti and it is transmitted by Culex mosquito.

Careers360

Q. 86 Which of the following is not an autoimmune disease ?

Option 1:
Alzheimer's disease

Option 2:
Rheumatoid arthritis

Option 3:
Psoriasis

Option 4:
Vitiligo

Correct Answer:
Alzheimer's disease

Solution:
As we learnt in

Auto-immune disease -

Sometimes, due to genetic and other unknown reasons, the body attack self-cells. This results in damage to the body and is called auto-immune disease.

- wherein

Rheumatoid arthritis which affects many people in our society is an auto-immune disease.

Alzheimer's disease is a chronic neurodegenerative disease. It is the cause of 60–70% of cases of dementia. The most common early symptom is difficulty in remembering recent events (short-term memory loss). Rheumatoid arthritis is an autoimmune disorder in which antibodies are produced against the synovial membrane and cartilage. Vitiligo causes white patches on skin also characterised as autoimmune disorder. Psoriasis is a skin disease that causes itchy or sore patches of thick red skin and is also autoimmune whereas Alzheimer's disease is due to deficiency of neurotransmitter acetylcholine.

Careers360

Q. 87 Among the following sets of examples for divergent evolution , select the **incorrect** option :

Option 1:
Brain of bat , man and cheetah

Option 2:
Heart of bat , man and cheetah

Option 3:
Forelimbs of man, bat and cheetah

Option 4:
Eye of octopus , bat and man

Correct Answer:
Eye of octopus , bat and man

Solution:
As we learnt in

Types of organic evolution -

1. Divergent Evolution
2. Co-Evaluation
3. Progressive
4. Microevolution
5. Macroevolution
6. Mega evolution
7. Parallel evolution
8. Convergent evolution

Divergent evolution occurs in the same structure, example - forelimbs, heart, brain of vertebrates which have developed along different directions due to adaptation to different needs whereas eye of octopus, bat and man are examples of analogous organs showing convergent evolution.

Careers360

Q. 88 Conversion of milk to curd improves its nutritional value by increasing the amount of

Option 1:
Vitamin B_{12}

Option 2:
Vitamin A

Option 3:
Vitamin D

Option 4:
Vitamin E

Correct Answer:
Vitamin B_{12}

Solution:

As discussed in Microbes in Household Products -

Micro-organisms such as *Lactobacillus* and others commonly called lactic acid bacteria (LAB) grow in milk and convert it to curd.

- wherein

During growth, the LAB produce acids that coagulate and partially digest the milk proteins. A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B12.

Curd has enriched presence of vitamins specially Vit-B12.

Hence, the correct option is (a).

Q. 89 The similarity of bone structure in the forelimbs of many vertebrates is an example of

Careers360

Option 1:
Convergent evolution

Option 2:
Analogy

Option 3:
Homology

Option 4:
Adaptive radiation

Correct Answer:
Homology

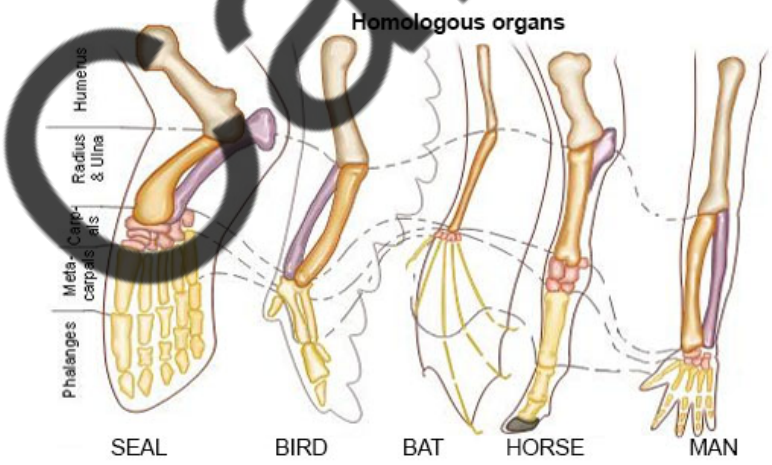
Solution:
As we learnt in

Homologous Organs -

Organs which look different and perform different functions, but have similar basic structure, similar topography and embryonic origin.

- wherein

These are called homologous organs. Eg: forelimbs of vertebrates like the seal, bat, bird wings, dragonfly, pterodactyl, eagle, humorous bat.



In homology, in different vertebrates, bones of forelimbs are similar but their forelimbs are adapted in different way as per their adaptation.

Careers360

Q. 90 Which of the following characteristics represent 'Inheritance of blood groups' in humans ?

a. Dominance

b. Co-dominance

c. Multiple allele

d. Incomplete dominance

e. Polygenic inheritance

Option 1:
b , d and e

Option 2:
a , b and c

Option 3:
b , c and e

Option 4:
a , c and e

Correct Answer:
a , b and c

Solution:
As we learnt in

Dominance in different alleles of gene (I) -

Since humans are diploid, each person possesses any two of the three genes I alleles.

Careers360

- wherein

I^A and I^B are completely dominant over i, but when I^A and I^B are present together, they both express their own types of sugar and none is dominant over other. This is called co-dominance and RBCs have both A and B types of sugar

Co-dominance -

In F1 generation, when the offspring resembles both parents, then it is called co-dominance.

- wherein

E.g., different types of red blood cells that determine ABO blood groups.

Dominance –(I^AI^O, I^BI^O)- the phenomenon whereby, in an individual containing two allelic forms of a gene, one is expressed to the exclusion of the other.

Codominance -(I^AI^B)- a form of dominance where in the alleles of a gene pair in a heterozygote are fully expressed. This results in offspring with a phenotype that is neither dominant nor recessive.

multiple allele- I^A, I^B & I^O - 3-different allelic forms of a gene.

Careers360